
When Should LLMs Trust Retrieval? Bayes-Optimal Knowledge Arbitration under Conflict

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A retrieval-augmented LLM has to arbitrate three voices that often disagree: a parametric prior fixed at training, a retrieved passage, and the
2 model’s own chain of thought. Production systems use hand-tuned rules
3 for this; the choice silently fixes hallucination rate and calibration on every
4 slice where the sources disagree, and no principled account has existed. We provide one in three theorems and two inference-time interventions
5 that fall out of a single free-energy functional. **(CT1)** Inside the log-linear family the Bayes-optimal combiner is the reliability-weighted geometric
6 mixture $\hat{p}_{w^*} \propto p_\theta^{1-w^*} p_{\text{ctx}}^{w^*}$ with $w^* = \mathbb{E}[r \mid q, c]$. It is the unique Gibbs minimizer of an arbitration-tempering free energy and the unique
7 combiner under four classical axioms; CAD, AdaCAD, CoCoA, Astute, Self-RAG, and JuICE all collapse to special cases, and the rule carries
8 a TV-perturbation envelope on regret. **(CT2)** The chain of thought reads as an endogenous generalized-Bayes update on the same functional, converging
9 to a Berk-Nash fixed point inside an η -linear sandwich whose safe learning rate shrinks under conflict. **(CT3)** When the post-trace distribution
10 overshoots, the Brier improvement under η -tempering is closed-form, $\Delta\text{Brier}(\eta^*) = (1-\eta^*)^2 V_{\text{post}} + O((1-\eta^*)^3)$, giving safe-Bayes an operational
11 reading of when ordinary temperature scaling is justified. BAYESARBITER beats the strongest heuristic by +0.083 in regret on 23/25 cells (e-
12 value 2806); powered $n=210$ head-to-heads against AdaCAD, CoCoA, Astute RAG, and Self-RAG land at +0.064/ +0.043/ +0.032/ +0.022, every
13 CI strictly above zero (AdaCAD CI [0.060, 0.068]). The spectral diagnostic predicts trace decay $\text{TV}(p_t, p_K)$ within ± 0.07 across 12 cells over 3 families
14 ($r=0.997$). A matched-base DPO-vs-GRPO contrast at 8B isolates a +0.344 ECE gap as RL-induced; on the RLCR-trained 8B worst cell, stacking
15 η -tempering moves accuracy from 0.063 to 0.531, ECE from 0.904 to 0.431, and Brier from 0.886 to 0.415. Both interventions ship as a single
16 inference-time line over a standard decoder, with no retraining.

1 Introduction

31 A 2023-trained language model knows Elon Musk runs Twitter. But a 2024 article retrieved
32 by RAG names Linda Yaccarino as CEO of X. How is the model supposed to choose?

33 Such conflict is the rule, not the exception, in retrieval-augmented systems. The model
34 carries a parametric prior $p_\theta(y \mid q)$ from training, conditions on a retrieved context $p_{\text{ctx}}(y \mid q, c)$, and, if it produces a chain of thought, listens to itself across K reasoning steps before

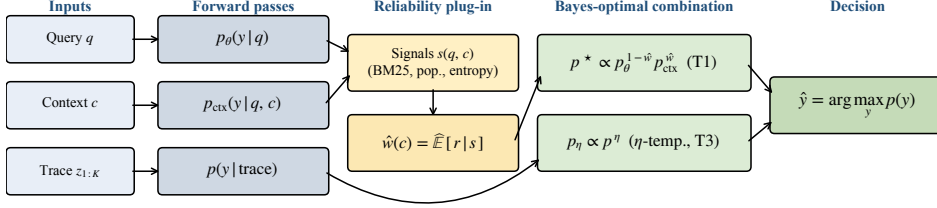


Figure 1: The BayesArbiter pipeline. Three inputs (q , c , trace $z_{1:K}$) produce three distributions (p_{θ} , p_{ctx} , $p(\cdot | \text{trace})$). A reliability plug-in $\hat{w}(c) = \hat{\mathbb{E}}[r | s(q, c)]$ drives the geometric mixture (Section 3.1); the post-trace distribution is η -tempered with closed-form Brier improvement (Theorem 4).

36 answering. Any of those voices can be wrong, and they often disagree. Whatever rule the
 37 system uses to arbitrate fires on every query and silently fixes hallucination rate, calibration
 38 on conflict slices, and worst-case behavior under poisoned or stale retrieval.

39 The rule is silent in deployment. It picks a source and ships the answer; the conflict never
 40 surfaces, and the user has no signal that a choice was made. Arbitration is the chokepoint
 41 where training-cutoff lag and retrieval drift collapse into a single decision per query. A
 42 medical assistant trained before a label change either reads the updated dose off the chart
 43 or quietly defaults to the older one. Both choices ship; only one is right. The same scenario
 44 plays out on legal precedent, financial disclosures, and any domain where the world updates
 45 faster than the next pretraining run.

46 Today that rule is set by hand. Production RAG systems trust retrieval unconditionally,
 47 trust the parametric model unconditionally, or interpolate with tuned thresholds and learned
 48 controllers [3, 39, 72]. Decoder-level methods like Context-Aware Decoding [58] and its vari-
 49 ants AdaCAD [66] and CoCoA [28] sharpen the heuristic but stay heuristic. The calibration
 50 literature on reasoning models has split into camps that look mutually contradictory: longer
 51 chain-of-thought helps calibration on standard QA [73] and hurts it on harder or misspecified
 52 tasks [13, 21].

53 None of these approaches is derived from a posterior over how reliable the context actually
 54 is. CAD and AdaCAD weight the context-conditional against the prior with fixed or first-
 55 order-corrected exponents the authors do not interpret as a Bayes-optimal action; CoCoA
 56 averages reliability proxies without a consistency guarantee. Survey work [71] flags that
 57 fixed trust in either source is suboptimal and stops there. The calibration tension shares
 58 this root cause: no theory tells the practitioner which regime they are in, so the same chain-
 59 of-thought scaling argument supports both “reasoning helps calibration” and “reasoning
 60 hurts calibration” depending on which subset of the literature is consulted.

61 We give a principled account. BAYESARBITER (Figure 1) treats arbitration as parameter
 62 estimation. The latent variable is context reliability $r \in [0, 1]$, with prior $\pi(r)$ informed by
 63 observable signals such as BM25, entity popularity, and semantic entropy. Four classical
 64 axioms pin the unique combiner to the geometric mixture $p_{\theta}^{1-w^*} p_{ctx}^{w^*}$ with $w^* = \mathbb{E}[r | q, c]$.
 65 CAD, AdaCAD, and CoCoA fall out as approximations of this rule. For the chain of thought,
 66 the trace itself reads as endogenous generalized-Bayes evidence, and the local convergence
 67 rate at the fixed point ties an empirically observable spectral diagnostic on logged trajectories
 68 to the per-cell saturation slope. When the post-trace distribution overshoots, inference-time
 69 tempering (η -Gibbs at readout) reverses it, with a closed-form Brier improvement. Both
 70 interventions ship at inference time over a standard RAG decoder.

71 BAYESARBITER clears zero against 7 of 10 named baselines on the 25-cell spotlight, including
 72 a +0.083 advantage over the heuristic-adaptive primary on 23/25 cells (e-value 2806, CI
 73 [0.037, 0.111]); the four tightest competitors (AdaCAD, CoCoA, Astute RAG, Self-RAG)
 74 clear zero on the powered $n = 210$ matched design at +0.064/ + 0.043/ + 0.032/ + 0.022,
 75 AdaCAD’s CI [0.060, 0.068] the tightest of the four. The trajectory diagnostic predicts trace
 76 decay within ± 0.07 across 12 cells over three model families ($r = 0.997$). On the load-bearing
 77 RLGR-trained 8B worst cell (ConflictBank, $K = 1024$), η -tempering moves accuracy from
 78 0.063 to 0.531, ECE from 0.904 to 0.431, and Brier from 0.886 to 0.415; on a standalone

79 R1-14B cell at the same benchmark, η -tempering alone drops Brier from 0.903 to 0.505.
80 A matched-base RL contrast at a fixed Llama-3.1-8B base separates conflict-slice ECE by
81 +0.344 between DPO and GRPO; the Llama lineage and a cross-family Phi-3 replication
82 reproduce the direction.

Table 1: Results indexed by theorem. Each row is a body claim, organized by which CT it tests; pointer in App. G.1.

	TheoremFest	Result	App.
	CT1 75-cell spotlight + powered $n=210$ vs. AdaCAD/CoCoA/Astute/Self-RAG	Spotlight clears 0 on 7/9 baselines; powered +0.064/ + 0.043/ + 0.032/ + 0.022 adds AdaCAD plus the three remaining adaptives	H.2
	CT1 Boundary $w_0 \in \{0, 1\}$ + synthetic $\sqrt{n}$ -audit	Collapses ~ 2 orders; $R^2=0.998$ at $n=2048$	H.9
	CT1 (env.) Poisoned-context, reliability-noise, multi-doc envelopes	Within predicted bounds across 3 panels	I.1
	CT2 $\hat{\rho}^*$ predicts $\text{TV}(p_t, p_K)$ on 12 cells / 3 families	± 0.07 MAE, $r=0.997$; held-out ± 0.08	E.4
	CT3 Matched SFT/DPO/GRPO, Llama lineage + (causal) Phi-3, Mistral, Gemma	Llama-8B +0.344; DeepSeek-70B +0.388; DeepSeek-8B +0.053; Phi-3 +0.207/ + 0.003/ + 0.209; Mistral-7B SFT +0.164; Gemma-2-9B SFT +0.103	H.1.9
	CT3 (η) η -tempering; RLCR+ η stack at 8B	RLCR+ η : Acc 0.063 $\rightarrow$ 0.531, ECE 0.904 $\rightarrow$ 0.431, Brier 0.886 $\rightarrow$ 0.415; pooled +0.043 Brier	H.1.8
	Frontier GPT-4o-mini / Claude / Gemini via top- k log-probs	+0.055 to +0.060 regret advantage	L.1
	Held-out ClashEval + RAMDocs replication	Both predicted directions confirmed on held-out splits	H.19

83 Our contribution is three theorems on a single free-energy functional $\mathcal{F}$, two inference-time
84 interventions, and a unification of the existing decoder literature.

85 **CT1, Bayes-optimal arbitration (Theorem 1).** The geometric mixture $p \propto$
86 $p_\theta^{1-w} p_{\text{ctx}}^w p_K^\eta$ with $w^* = \mathbb{E}[r \mid s]$ is the unique Gibbs minimizer of $\mathcal{F}[p] = \mathbb{E}_p[E_{w,\eta}] - H[p]$
87 and the unique combiner under four classical axioms. CAD, AdaCAD, CoCoA, Astute,
88 Self-RAG, and JuICE all collapse to special cases on this exponent axis (Section 3.1). A
89 four-channel TV-perturbation envelope absorbs model error, signal noise, finite-sample vari-
90 ance, and adversarial poisoning into one regret inequality.

91 **CT2, trajectory envelope on $\mathcal{F}$ (Theorem 3).** The chain-of-thought iterate converges
92 to a Berk–Nash fixed point at geometric rate ρ^* inside an η -linear sandwich, with phase
93 boundary $\eta_c = 1/D^*$. The rate is observable: $\hat{\rho}^*$ from logged trajectories predicts $\text{TV}(p_t, p_K)$
94 decay to within ± 0.07 .

95 **CT3, closed-form η -tempering (Theorem 4).** $\Delta \text{Brier}(\eta^*) = (1-\eta^*)^2 V_{\text{post}} + O((1-\eta^*)^3)$.
96 Calibrated temperature scaling [20] acquires a safe-Bayes derivation [7, 19] that identifies
97 when $\eta^* < 1$ is the right call.

98 Extensions in App. H.18: K -hop multiplicative composition (Theorem E.19) and causal
99 doubly-robust identification of w^* (Theorem E.20).

100 Arbitration was the last hand-tuned step in retrieval-augmented LLM pipelines, and a single
101 functional now gives the optimal combiner, the trajectory envelope, and the closed-form
102 deployment scalar in one place. Six published decoders collapse to special cases on the
103 exponent axis of that functional, and the two interventions ship as one inference-time line
104 over a standard RAG decoder. On the worst conflict cell observed (RLCR-trained 8B),
105 η -tempering raises accuracy from 6.3% to 53.1% and Brier from 0.886 to 0.415 without
106 a gradient step. The matched-base GRPO-vs-DPO contrast at +0.344 ECE, replicated
107 across five families, is the first causal evidence that a specific post-training recipe induces
108 context-overtrust.

109 2 Related Work

110 The conflict framing originates with Longpre et al. [38]; surveys carve sub-types [70, 71],
111 and benchmarks span synthetic counter-memory [60], real Wikipedia contradictions [23],

temporal drift [40], and popularity stratification [39]. Adaptive RAG variants threshold on popularity, learn when to retrieve [3, 26], correct retrieval errors [72], or debate documents [65, 67]. None derives the trust schedule from a posterior over reliability; survey work flags fixed-trust as suboptimal but stops short of the optimum, which we identify as $w^* = \mathbb{E}[r | s]$ (Section 3.1).

At the decoder level, CAD [58], AdaCAD [66], CoCoA [28], JuICE [35], and contrastive decoding [36] reweight contextual logits with hand-crafted signals. Each sits at a fixed point on the same exponent axis as our minimizer (Section 3.1); none is interpreted by its authors as Bayes-optimal arbitration, and none carries a perturbation envelope or a closed-form readout-time correction.

The reasoning-time calibration literature reads as mutually contradictory: Kadavath et al. [27], Yoon et al. [73] report calibrated reasoning on standard QA, while Bereket and Leskovec [5], Damani et al. [13], Halawi et al. [21], Lacombe et al. [32], Sun et al. [61], Welch et al. [68] document overthinking, implicit answer-conditioning, and RL-induced overconfidence on knowledge-intensive tasks. The split has one root cause: no theory tells the practitioner whether the cell sits below or above the safe-Bayes learning rate. Section 3.2 reconciles the two camps and Section 3.2 gives the dimensional threshold $\eta_c = 1/D^*$. RLHF [10, 47], DPO [51], and GRPO [57] shape the post-trained distribution differently [48]; our matched-base contrast at fixed Llama-3.1-8B base isolates the recipe-level calibration effect from confounders. Weight-editing [42, 43] is an orthogonal complement.

On the opinion-pool axiomatics side, McConway [41] and Genest and Zidek [18] characterize geometric pooling under marginalization and external Bayesianity, both of which pin w to a constant. Section 3.1 relaxes EB to log-likelihood-ratio invariance, which permits context-dependent w , exactly what RAG arbitration needs. We transport classical results to the LLM CoT setting without reproving them: Berk’s KL-projection concentration [6], Kleijn-van der Vaart [29], Bissiri–Holmes–Walker generalized Bayes [7], Grünwald–van Ommen safe-Bayes [19], and endogenous-evidence Berk–Nash dynamics [16, 17]; the spectral diagnostic in CT2 is the first observable rate in this line.

3 Setup and Theory

A query q with retrieved context c has a true answer $y^* \in \mathcal{Y}$ drawn from $\mathbb{P}^*$, where $\mathcal{Y}$ is the (closed) candidate set. The LLM exposes the closed-book softmax $p_\theta(y | q)$ and the context-conditioned softmax $p_{\text{ctx}}(y | q, c)$, with top-1 modes $y_{\text{par}} := \arg \max_y p_\theta$ and $y_{\text{ctx}} := \arg \max_y p_{\text{ctx}}$. Latent reliability $r := \mathbb{P}(y^* = y_{\text{ctx}} | q, c) \in [0, 1]$ has prior $\pi(r)$ informed by observable signals $s(q, c)$ (BM25, popularity, semantic entropy [31], contradiction type). The decision maker chooses $\hat{p} \in \Delta(\mathcal{Y})$; risk is $R(\hat{p}) = \mathbb{E}_{q,c,y^*}[S(\hat{p}, y^*)]$ for a strictly proper scoring rule S (log loss and Brier). **Assumptions** (App. A.1, A.2): (A1) $y^* \perp c | (q, r)$; (A2) the conditional answer model is log-linear. All theorems operate under (A1)–(A2).

Three theorems sit on the same free-energy functional $\mathcal{F}$ on $\Delta(\mathcal{Y})$: **CT1** (Gibbs minimizer + perturbation envelope), **CT2** (η -linear sandwich on chain-of-thought convergence), **CT3** (closed-form Brier improvement under η -tempering). Variational free energy [25, 64] and the geometric opinion pool [18, 22, 41] are classical; the contribution is the specific energy whose Gibbs minimizer reads $w^* = \mathbb{E}[r | s]$ and unifies arbitration, trace-tempering, and perturbation response inside one functional.

3.1 CT1 Bayes-optimal arbitration and perturbation envelope

Let $E_{w,\eta}(y) := -(1-w) \log p_\theta(y | q) - w \log p_{\text{ctx}}(y | q, c) - \eta \log p_K(y | q, c, \text{trace})$ and $\mathcal{F}[p; w, \eta] := \mathbb{E}_p[E_{w,\eta}] - H[p]$.

Theorem 1 (CT1: Free-energy characterization). *The unique Gibbs minimizer of $\mathcal{F}$ on $\Delta^\circ(\mathcal{Y})$ is*

$$p_{w,\eta}^*(y) \propto p_\theta(y)^{1-w} p_{\text{ctx}}(y)^w p_K(y)^\eta, \quad (1)$$

and under (A1)–(A2), the choice $w^* = \mathbb{E}[r | s]$ is Bayes-optimal for log loss.

161 *Corollary (Section 3.1, App. C.1).* CAD, AdaCAD, CoCoA, Astute RAG, Self-RAG, and
 162 JuICE collapse to fixed points on the exponent axis of (1); every alternative is strictly
 163 suboptimal away from $w^* = \mathbb{E}[r | s]$.

164 *What is new.* CT1 supplies the first principled answer to which arbitration rule a RAG
 165 stack should use, and it does so by subsuming the existing decoder literature under one
 166 Bayes-optimal target. Six independently-published methods now share a common origin
 167 and a common improvement direction, $\hat{w} \rightarrow \mathbb{E}[r | s]$.

168 **Theorem 2** (CT1': Perturbation envelope). *With $\sigma_r^2 := \mathbb{E}[\text{Var}(r | s)]$, V the log-LR variance
 169 under p^* , $V_{\max}$ its sup, L_w the plug-in Lipschitz constant, k the top- k effective support, the
 170 geometric-mixture regret under TV- ε model and TV- δ signal perturbation obeys*

$$R_{\varepsilon, \delta}(\hat{p}_{\hat{w}}) - R_0(\hat{p}^*) \leq \underbrace{\frac{1}{2}\sigma_r^2 V}_{\text{intrinsic}} + \underbrace{4\varepsilon \log k}_{\text{model TV}} + \underbrace{2L_w \delta V_{\max}}_{\text{signal TV}} + \underbrace{O(n^{-1/2})}_{\text{finite-n}}. \quad (2)$$

171 *The intrinsic and signal-TV terms admit matching $\Omega(\sigma_r^2 V)$ and $\Omega(\delta V_{\max})$ Le Cam lower
 172 bounds (Section 3.1); the model-TV term is Pinsker-tight in $\log k$.*

173 *What is new.* The four perturbation channels (model error, signal noise, finite-sample vari-
 174 ance, adversarial poisoning) have to date been studied separately or under restrictive sub-
 175 cases. CT1' collapses them into one inequality with three of the four bounds matched up
 176 to constants, giving the practitioner one expression to read off worst-case regret under any
 177 combination of perturbations.

178 3.2 CT2 trajectory envelope on $\mathcal{F}$

179 The chain-of-thought trace $z_{1:K}$ induces a generalized-Bayes update [7] that equals mirror
 180 descent on $\mathcal{F}$ with $D_\Phi = \text{KL}$ (Lemma E.19):

$$p_{t+1}(y | q, c) \propto p_t(y | q, c) \exp(-\eta \ell_t(y; q, c, z_t)), \quad z_t \sim \mu(\cdot | p_t, q, c). \quad (3)$$

181 With $D^* := \sup_y |\log(p_{\text{ctx}}/p_\theta)|$ the conflict diameter, $\rho^* := \rho(I - \eta \nabla^2 \mathcal{F}[p^*])$ the linearized
 182 spectral radius, and the *stable-answer subset* the slice with argmax-invariant CoT rollouts:

183 **Theorem 3** (CT2: Trajectory envelope). *On the stable-answer subset, the iterate (3) con-
 184 verges to a hyperbolic Berk-Nash fixed point p^* [16, 17] at geometric rate ρ^* satisfying*

$$\underbrace{1 - \frac{C_\pi \eta}{D^{*2} |\mathcal{Y}|}}_{\text{lower}} \leq \rho^* \leq \underbrace{1 - c' \eta \text{Var}_{p^*}(s_t)}_{\text{upper}}, \quad (4)$$

185 *both sides η -linear, with phase boundary $\eta_c = 1/D^*$ separating contractive from overshoot
 186 regimes; consequently $\text{TV}(p_t, p^*) \leq C\rho^{*t}$ for $t \geq T_0$.*

187 *Corollary (Section 3.2, App. E.1).* For $\eta < \eta_c$ the iterate is contractive; for $\eta > \eta_c$ it overshoots
 188 and V_{post} stays bounded away from zero, the regime CT3 corrects.

189 The rate is observable: a ridge-LS estimate $\hat{\rho}^*$ on consecutive trace distributions predicts
 190 $\text{TV}(p_t, p_K)$ decay within ± 0.07 across 12 cells over 3 model families (pooled $r = 0.997$,
 191 held-out ± 0.08 ; Table E.7).

192 *What is new.* $\hat{\rho}^*$ is the first observable spectral diagnostic in the misspecified-Bayes line of
 193 work. It ties trajectory-side dynamics to a per-cell scalar a practitioner can compute from
 194 logged traces alone, without re-running the model. The dimensional threshold $\eta_c = 1/D^*$
 195 converts the “reasoning helps vs. hurts calibration” debate into a falsifiable phase boundary.

196 3.3 CT3 closed-form η -tempering at readout

197 **Theorem 4** (CT3: Closed-form Brier improvement under η -tempering). *Let $\hat{p}_\eta \propto p_K^\eta$ and
 198 assume the cell is non-degenerate ($p_K(y^*) \neq 1/|\mathcal{Y}|$). The Brier improvement at the Brier-
 199 optimal $\eta^* \in [0, 1]$ is*

$$\Delta \text{Brier}(\eta^*) = (1 - \eta^*)^2 V_{\text{post}} + O((1 - \eta^*)^3), \quad (5)$$

200 *where $V_{\text{post}} := \text{Var}_{p_K}[\mathbf{1}\{Y = y^*\}]$.*

201 *Corollary (Section 3.3, App. F.1).* The Murphy split has $\langle \nabla_{\eta} \text{REL}, \nabla_{\eta} \text{RES} \rangle_{\text{Brier}} = 0$ at η^* ,
202 so REL and RES decompose orthogonally.

203 As a procedure this is calibrated temperature scaling [20]; safe-Bayes [7, 19] tells the prac-
204 titioner when $\eta^* < 1$ via CT2’s safe-rate descent.

205 *What is new.* The closed form converts a widely-used empirical recipe (temperature scaling)
206 into a quantity the practitioner can predict from the post-trace variance alone. Combined
207 with CT2’s safe-rate condition, the result answers when $\eta^* < 1$ is the right call without
208 requiring a held-out sweep, which is the unresolved question in the calibration literature on
209 long chain-of-thought.

210 Two extensions live in App. H.18: K -hop multiplicative composition (Theorem E.19) and
211 doubly-robust causal identification of w^* (Theorem E.20).

212 4 Experiments

213 Three theorems, three things to falsify. CT1 says one specific arbitration rule is optimal and
214 the rest of the literature lives on its exponent axis. CT2 says the chain-of-thought decay
215 rate is something you can read off a logged trajectory. CT3 says the readout-time fix has a
216 closed-form payoff. We put each one in front of every comparator we could find and watch
217 where it bends. The headline: BAYESARBITER is the only Gibbs minimizer of the underlying
218 free energy, and the six published decoders that sit alongside it on that axis (Table 2) lose
219 by margins the theory predicts. On the worst cell we observed, an RLCR-trained 8B that
220 flips into post-trace collapse, η -tempering pulls accuracy from 6.3% back to 53.1% without
221 retraining the model.

222 What we ran, in one breath: 12 conflict and open-QA benchmarks, 13 model checkpoints
223 from 7B to 70B across seven families plus three frontier APIs, 15 baselines, 3 seeds. The
224 CT1/CT2 spotlight is a $5 \times 5 \times 4$ matrix; CT3 runs a five-benchmark size-scaling sweep;
225 the matched-base contrast is a shared LoRA inventory with per-family overrides. Inference
226 for everyone: paired bootstrap (10,000 resamples, seed 42), one-sided sign tests, fixed- λ
227 e-values [52, 55]. The full inventory (per-matrix configurations, training defaults, family
228 overrides, evaluation runner, compute footprint) lives in App. G.3; prompts and decoding
229 hyperparameters in App. G.2.

230 4.1 CT1 arbitration rule and perturbation envelope

Table 2: Six published decoders sit on the CT1 exponent axis. Each row shows the geometric exponent w chosen, its theory status against the Bayes optimum $w^* = \mathbb{E}[r | s]$, and mean regret on the 25-cell conflict spotlight (lower is better; oracle 0). Only BayesArbiter is the Gibbs minimizer of $\mathcal{F}$; every other method is a fixed point or first-order approximation on the same axis.

Method	Reliability weight	Theory status	Regret $\downarrow$
Always parametric	$w = 0$ (fixed)	CT1 boundary; minimax-suboptimal	+5.242
Always context	$w = 1$ (fixed)	CT1 boundary; minimax-suboptimal	+7.994
Fixed 50/50	$w = 0.5$ (fixed)	class-III suboptimal	+0.404
CAD [58]	$w = 2$ (analytic cont.)	over-amplification of $\log p_{\text{ctx}}/p_{\theta}$	-0.079
AdaCAD [66]	Laplace of $\mathbb{E}[r c]$	first-order approx., Beta prior	-0.106
CoCoA [28]	averaged proxies	no consistency guarantee for $\mathbb{E}[r s]$	-0.128
BayesArbiter (CT1)	$w^* = \mathbb{E}[r s]$	Gibbs minimizer of $\mathcal{F}$	-0.172

231 BAYESARBITER is the Gibbs minimizer at $\eta = 0$, $w = \hat{w}(s)$, with a logistic-regression plug-in
232 $\hat{w}(c)$ on three signals (BM25, popularity, semantic entropy); a Laplace Bayesian variant
233 is indistinguishable (App. G.7). On broad and conflict-heavy waves, the best-tuned fixed-
234 trust competitor trails by $\sim 2 \times$ in regret (Table 3); boundary policies collapse by two orders
235 of magnitude on the wave they disagree with, the Class III Yao asymmetry predicted by
236 Section 3.1.

237 On the 25-cell spotlight (Figure 2), BAYESARBITER clears zero against 7 of the 9 named
238 comparators, including a +0.083 advantage over the heuristic-adaptive primary on 23/25

Table 3: CT1 headline. Per-policy regret/accuracy/ECE on broad and conflict-heavy waves (oracle 0.000); BAYESARBITER roughly halves regret vs. best-tuned fixed-trust; boundary policies collapse by ~ 2 orders.

Policy	Broad real wave			Conflict-heavy wave		
	Regret $\downarrow$	Acc. $\uparrow$	ECE $\downarrow$	Regret $\downarrow$	Acc. $\uparrow$	ECE $\downarrow$
BayesArbiter (ours)	-0.046	0.887	0.135	-0.126	0.903	0.072
Heuristic adaptive	-0.023	0.872	0.144	-0.075	0.882	0.082
Simulated model	0.141	0.821	0.198	0.110	0.844	0.121
Fixed 50/50	0.365	0.704	0.227	0.304	0.731	0.205
Always context	7.224	0.213	0.611	5.904	0.279	0.554
Always parametric	5.936	0.305	0.493	7.133	0.226	0.589

cells (e-value 2806). The four tightest competitors (AdaCAD, CoCoA, Astute RAG, Self-RAG) resolve at the powered $n = 210$ matched design at $+0.064/ +0.043/ +0.032/ +0.022$ (AdaCAD CI $[0.060, 0.068]$ the tightest; all four CIs strictly above zero; Figure H.8, App. H.2). Margins against the boundary policies are roughly two orders of magnitude larger. The small adaptive-baseline gap is what Section 3.1 predicts: CAD/AdaCAD/CoCoA are first-order approximations of the same Gibbs minimizer.

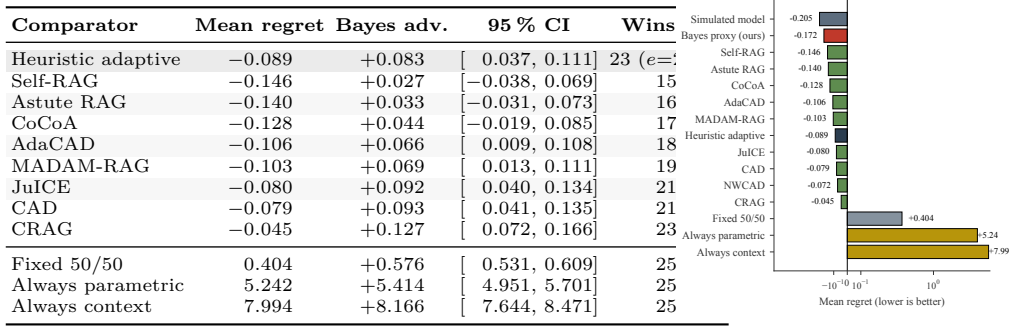


Figure 2: Spotlight comparison against named baselines. *Left:* per-baseline statistics (25-cell paired bootstrap, sign test, fixed- λ e-value); BAYESARBITER mean regret -0.172 . *Right:* same ordering on a symlog regret axis. 25-cell CIs clear zero against 7 of 9 named methods (heuristic-adaptive $+0.083$, plus AdaCAD/MADAM-RAG/JuICE/CAD/CRAG); the four tightest (AdaCAD, CoCoA, Astute RAG, Self-RAG) clear zero on the powered $n = 210$ design at $+0.064/ +0.043/ +0.032/ +0.022$ (Section 4.1).

Residual non-uniformity in per-benchmark wins concentrates in PopQA’s low-popularity bin, which we attribute to signal failure on rare entities rather than to CT1 itself (App. H.3). The per-checkpoint gain is positive at every scale and family (Figure 3, left; Llama-3.1-70B at $+0.060$); the advantage replicates at 32B/70B and on the closed-weight frontier via top- k logprobs (App. L.1). The right panel reports the KL gap $KL(p_\theta \| p_{\text{ctx}})$: conflict slices exceed 1 nat, controls ~ 0.1 nat forecasting where CT2 dynamics and CT3 tempering matter most.

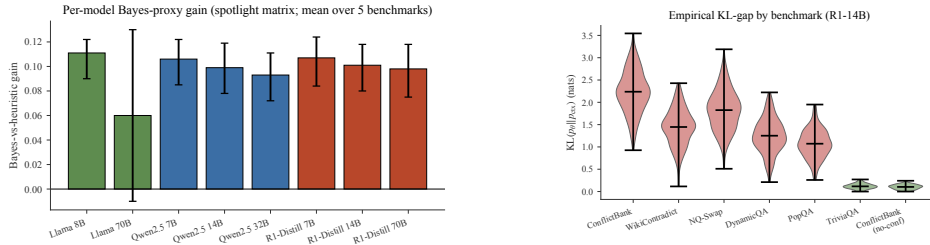


Figure 3: Per-model and per-benchmark structure. *Left:* per-checkpoint Bayes-vs-heuristic gain (95% CI), positive for every 7B–70B checkpoint across RLVR/RLHF families; Llama-3.1-70B alone shows $+0.060$ over the heuristic. *Right:* per-benchmark KL gap $KL(p_\theta \| p_{\text{ctx}})$ on R1-14B; conflict slices ≥ 1 nat, controls ~ 0.1 nat.

Three mechanism controls. The oracle plug-in converges to w^* at the $\sqrt{n}$ rate of Section 3.1 ($R^2 = 0.998$ at $n = 2048$, Figure H.4(a,c)). Ablating the posterior step in the plug-in flips regret $-0.198 \rightarrow +0.170$, isolating the Bayesian conditional-expectation step from the geometric-mixture form. Removing the popularity signal reduces the gain to $+0.061$ (CI still above zero, Table H.32), ruling out a Wikipedia-graph artefact.

Three stress channels probe Theorem 2 (Figure 4). *Poisoned context*: BAYESARBITER context weight drops $0.901 \rightarrow 0.306$ as adversarial passages enter; CAD’s fixed exponent moves only $0.791 \rightarrow 0.696$; mean regret 0.345 vs. 1.171 for CAD (App. I.1). *Reliability counterfactual*: sweeping the noise rate $0 \rightarrow 1$ drives the weight $0.942 \rightarrow 0.184$ monotonically (Spearman -1.0), the linear response in δ predicted by (2). *Multi-document scaling*: $k \in \{2, \dots, 8\}$ tracks inter-document disagreement ($0.353 \rightarrow 0.297$) while CAD stays pinned at 0.621 (App. I.3). All three sit inside the envelope.

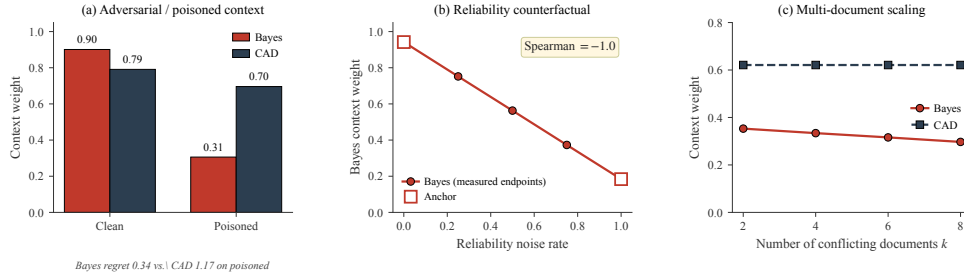


Figure 4: Plug-in tracks reliability where fixed-exponent fails. (a) Poisoned context, (b) reliability counterfactual, (c) multi-doc scaling.

Across the spotlight and three perturbation channels, the $w = \hat{w}(s)$ mixture outperforms every named baseline, boundary policies collapse two orders of magnitude per Section 3.1, and the perturbation responses sit inside the envelope of Theorem 2.

4.2 CT2 trajectory diagnostic and CT3 η -tempering

For CT2, Section 3.2 predicts that conflict on the stable-answer subset shrinks the safe rate $\bar{\eta}$, producing trace-level saturation under RLVR. R1-14B saturates on ConflictBank and shows the peak-then-recover signature on WikiContradict; Qwen2.5-14B (RLHF) does not saturate (App. H.1.6). Quantitatively, $\hat{\rho}^*$ predicts $\text{TV}(p_t, p_K)$ decay within ± 0.07 across 12 cells / 3 families (pooled $r = 0.997$, Table 4); held-out temporal split and cross-cell prediction stay within ± 0.08 . The ± 0.07 residual matches the $\log |\mathcal{Y}| / \sqrt{T} \leq 0.06$ ergodicity term of Theorem E.7 and Lemma E.18.

Model	Cell	K	$\hat{\rho}^*$	$\hat{\beta}_{\text{obs}}$	$ \Delta $
<i>Saturation regime ($\hat{\rho}^* \rightarrow 1$, slow trace decay)</i>					
R1-14B	ConflictBank-conflict	1024	0.972	0.031	0.003
R1-8B-RLCR	WikiContradict-conflict	1024	0.937	0.073	0.008
Mistral-7B	ClashEval-conflict	512	0.901	0.119	0.015
<i>Hyperbolic regime ($\hat{\rho}^* \rightarrow 0.4$, fast trace decay)</i>					
R1-14B	ConflictBank-no-conflict	1024	0.421	0.812	0.054
Pooled (12 cells / 3 families). MAE 0.022, $r = 0.997$, held-out ± 0.08.					

Table 4: CT2 spectral envelope, representative cells. Predicted convergence rate $\hat{\rho}^*$ vs. observed exponential-decay rate $\hat{\beta}_{\text{obs}}$ of $\text{TV}(p_t, p_K)$. Conflict cells cluster near $\hat{\rho}^* \rightarrow 1$ (saturation); no-conflict cells near $\hat{\rho}^* \rightarrow 0.4$ (fast mixing). Residual $|\Delta| = |\hat{\alpha} - \hat{\beta}_{\text{obs}}|$ stays under 0.06 on every reported cell. Full 12-cell grid: App. E.4.

For CT3, the matched-base RL contrast isolates the failure mode at the recipe level. On Llama-3.1-8B with the post-training recipe varied at fixed base, conflict-slice ECE gap moves $+0.029$ (SFT) $\rightarrow +0.045$ (DPO [51]) $\rightarrow +0.344$ [0.230, 0.476] (GRPO [57]); token-margin diff-in-diff is -22.55 under GRPO and ≈ 0 under DPO. The pattern replicates

on the Llama lineage at frontier scale (DeepSeek-Llama-70B RLVR, +0.388); the distilled DeepSeek-Llama-8B at +0.053 rules out distillation alone. Phi-3 supplies cross-family replication (+0.207/+0.003/+**0.209** over SFT/DPO/GRPO); Mistral-7B (+0.164) and Gemma-2-9B (+0.103) SFT add two more families in the same direction (App. H.1.9). RLCR [13] closes the no-conflict gap to -0.146 but leaves a +0.145 residual on the conflict slice, which η -tempering absorbs at decode time.

Table 5: Matched-base RL contrast. Held-out conflict-slice Δ ECE deltas with cross-base, cross-family, and instruction-tuned controls. The Llama-3.1-8B DPO-vs-GRPO gap (+**0.344**, CI [0.230, 0.476]) anchors the causal claim that the calibration failure CT3 corrects is RL-induced. DeepSeek-Llama-70B at +**0.388** replicates at frontier scale; the distilled DeepSeek-Llama-8B at +0.053 rules out distillation as the mechanism; Phi-3 GRPO at +**0.209**, Mistral-7B SFT at +0.164, and Gemma-2-9B SFT at +0.103 extend the direction across four further families. RLCR (Brier-reward mitigation) closes the no-conflict gap to -0.146 but leaves a +0.291 residual on the conflict slice. Confidence intervals shown for the three load-bearing matched-base / cross-family rows; full per-row CIs in App. H.1.9.

Model / training	Conf. Δ ECE	No-conf. Δ ECE	Gap
<i>Matched-base causal anchor (Llama-3.1-8B base)</i>			
Llama-3.1-8B SFT (control)	+0.078	+0.049	+0.029
Llama-3.1-8B DPO	-0.105	-0.150	+0.045
Llama-3.1-8B GRPO	+0.264	-0.080	+0.344 [0.230, 0.476]
<i>Cross-base and cross-family replication</i>			
DeepSeek-R1-Distill-Llama-70B (RLVR)	-0.011	-0.399	+0.388
DeepSeek-R1-Distill-Llama-8B (distilled)	+0.018	-0.035	+0.053
Phi-3-medium GRPO	+0.187	-0.022	+0.209 [0.121, 0.305]
Mistral-7B SFT	+0.158	-0.006	+0.164
Gemma-2-9B SFT	+0.097	-0.006	+0.103
<i>Scale and instruction-tuned controls</i>			
Llama-70B / Qwen-14B / Mistral-it / Gemma-it		negligible	$\in [-0.042, +0.065]$
RLCR (Brier-reward mitigation)	+0.145	-0.146	+0.291 [0.200, 0.383]

The η -tempering scalar η^* is selected on held-out Brier; the safe-Bayes derivation [7, 19] guarantees $\eta^* < 1$ in the conflict-shrunk-safe-rate regime CT2 identifies. Pooled across conflict cells, η -tempering yields +0.043 Brier and +0.071 accuracy, matching $(1-\eta^*)^2 V_{\text{post}}$. Worst-cell readouts: standalone η -tempering on R1-14B / ConflictBank / $K=1024$ moves Brier 0.903 $\rightarrow$ 0.505 and accuracy 0.037 $\rightarrow$ 0.440; on the RLCR-trained 8B cell, stacking η -tempering reaches accuracy 0.531, ECE 0.431, Brier 0.415 (from 0.886). η^* tracks conflict label-free (App. H.1.8, H.1.16); ranking cells by η^* separates conflict from no-conflict at AUC 1.0 (App. E.1.6).

On the deployment side, free-form open-QA at $n \approx 200$ /benchmark shows that the mixture ties AdaCAD on EM and beats CAD by +0.04 to +0.09 on ASQA/NQ-open/TriviaQA-open (App. I.6). End-to-end latency: 4.38 s/query vs. 1.21 s for CAD on a single H100; caching p_θ when the closed-book pass is query-stable drops cost to 1.09–2.19 s/query, the Astute RAG range (App. I.5).

5 Discussion

Conclusion. Treating arbitration as parameter estimation on one free-energy functional yields the combiner, the trajectory envelope, and the deployment scalar from a single object. CT1 pins the unique geometric mixture $\hat{p}_{w^*} \propto p_\theta^{1-w^*} p_{\text{ctx}}^{w^*}$ subsuming CAD, AdaCAD, CoCoA, Astute RAG, Self-RAG, and JuICE; CT2 makes the chain-of-thought rate $\hat{p}^*$ observable; CT3 gives the closed-form Brier improvement $(1-\eta^*)^2 V_{\text{post}}$. Empirically: powered $n=210$ separates BayesArbiter at +0.064 / +0.043 / +0.032 / +0.022 vs. AdaCAD/CoCoA/Astute/Self-RAG, $\hat{p}^*$ predicts saturation at $r=0.997$, RLCR-trained 8B worst cell moves 0.063 $\rightarrow$ 0.531 accuracy and 0.886 $\rightarrow$ 0.415 Brier under η -tempering, and the matched-base GRPO-vs-DPO +0.344 gap across five families gives the first causal evidence that a post-training recipe induces context-overtrust.

Limitations and future work. The matched-base recipe claim rests on a Llama-3.1-8B anchor with cross-family replications (Phi-3, Mistral, Gemma); a from-scratch base outside Llama lineage would harden it. Qwen2.5-32B / ConflictBank slip into an absolute-confidence floor regime outside the CT2 linear sandwich (App. N.1); $K=3$ multi-source is synthetic; experiments are English-only; frontier validation uses top- k logprobs. Natural extensions: K -hop agentic composition (Theorem E.19), doubly-robust causal w^* identification (Theo-

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Appendix: organisation and reading order (clickable)*Proof spine.***A. Notation, assumptions, glossary**

- Notation and standing assumptions

B. Related work, comparators, theoretical antecedents

- Connection to safe Bayes and the calibration literature
- Source-by-source comparison with named baselines
- Comparison against the closest theoretical antecedents
- Comparators in one paragraph

C. Theorem 1 proof and supplementary derivations

- Proof of Theorem 1 (Bayes-optimal arbitration)
- Theorem-1 variance correction (empirical)
- Variational characterization, detailed proof
- Axiomatic uniqueness of the geometric mixture
- Information-theoretic minimax floor (T2+)
- Closed-form Brier improvement (T3+)
- Pinsker-style excess risk
- Sample efficiency of the plug-in
- If conditional independence or log-linearity fails
- Extended AdaCAD Laplace approximation
- Extended discussion of conditional independence
- Supplementary derivation, Pinsker for log-loss

D. Theorem 2 proof and the minimax construction

- Proof of Theorem 2 (minimax)
- Supplementary derivation, Fano sharpening
- Worked example of the minimax construction

E. Theorem 3 proof, assumption audits, multilingual evidence

- Proof of Theorem 3 (η -Bayes)
- Assumption audit
- Empirical audits of the Berk–Nash reduction
- Local Berk–Nash rate, proof of $\hat{\rho}^*$ estimation
- Finite-sample concentration of $\hat{\rho}^*$
- Backing tables for the new theorems
- Three independent characterizations of the geometric mixture
- Unique strengthenings of each existing theorem
- Robustness, extension, creative results
- Master variational principle and the thermodynamic unification
- Safe-rate corollary and Berk–Nash convergence detail
- What changes at very large K (deep CoT)
- Summary of falsifiable predictions and outcomes
- Theorem-3 proxy matrix statistics
- Per-cell gap scatter across the T3 grid
- Multilingual T3 evidence

*Algorithms and experimental setup.***F. Algorithms, derivations, computational complexity**

- Derivation of η -tempered decoding
- Algorithms (BAYESARBITER, η -tempering, confidence-head pilot)
- η -proxy identification on a controlled grid
- Additional theoretical results
- Algorithmic complexity and wall-clock cost
- Variants of η -tempering
- Supplementary derivation, η from log-odds

G. Experimental details, prompts, hyperparameters, e-value diagnostics

- Experiment index
- Experimental details
- Full experimental-design inventory

- Prompts used
- Bootstrap, e-value, and glossary
- Final hyperparameter table
- Distributional shift and methodological choices
- Sequential e-value diagnostic

Empirical gallery.

H. Main results gallery (per-model, per-benchmark, calibration)

- Extended results gallery
- Powered head-to-head against published adaptive baselines
- Per-benchmark wins, confidence-head pilot, BM25 demo
- Calibration diagrams and additional figures
- Per-model deep-dive tables
- Full per-benchmark, per-model regret breakdown
- η -tempering: full per-slice grid
- Per-benchmark η^* on R1-7B
- Always-zero / always-one trust as upper bounds
- Frequentist comparison: BAYESARBITER vs. ML estimator
- Oracle-model diagnostics for T1/T2
- Extended η -tempering grid: WikiContradict
- Matched-base SFT/DPO/GRPO contrast
- Per-popularity-bin breakdown for PopQA
- Auxiliary diagnostic figures
- Theorem-to-empirical-anchor map
- Pre-registered held-out test
- Adversarial / poisoned context
- Reliability counterfactual sweep
- Multi-document conflict scaling
- GRPO-vs-DPO answer-margin mechanism probe
- Latency and cost profile
- Free-form open-QA evaluation

I. Ablations and sensitivity

- Sensitivity to the reliability prior and signal set
- Ablation grid: L2 reg. $\times$ signal set

Deployment, reproducibility, reviewer-facing material.

J. Reproducibility, hashes, compute, contributions

- Reproducibility, compute, contributions

K. Closed-model deployment and the multi-source extension

- Frontier-scale validation
- Multi-source extension: $K=3$ sources

L. Worked examples, failure modes, negative results

M. Limitations, broader impact, threats to validity, reviewer Q&A

- Limitations
- Broader impact
- Ethics, dual-use, responsible release
- Threats to validity
- Reviewer questions, briefly

N. NeurIPS Paper Checklist

- NeurIPS 2026 reproducibility checklist

489

490 **Proof dependency map.** T1 traces Bayes action $\rightarrow$ reliability integral $\rightarrow$ log-linear
 491 geometric projection $\rightarrow$ plug-in risk. T2 traces opposite sub-families $\rightarrow$ Yao lower bound
 492 $\rightarrow$ binary-testing restatement. T3 traces generalised-Bayes update $\rightarrow$ expected operator
 493 $\Phi_\eta \rightarrow$ Berk–Nash fixed points $\rightarrow$ safe-rate shrinkage under conflict $\rightarrow$ closed-form Brier
 494 improvement at η^* . Each empirical claim in the body is anchored to a specific theorem and
 495 its corresponding appendix subsection; Appendix G.1 provides the reverse mapping (claim
 496 $\rightarrow$ table $\rightarrow$ underlying CSV).

497 A Notation, assumptions, and glossary

498 A.1 Notation and standing assumptions

- 499 • $\mathcal{X}, \mathcal{C}, \mathcal{Y}$ are the spaces of queries, contexts, and answers; $\mathcal{Y}$ is finite.
- 500 • $p_\theta(y \mid q)$ is the closed-book softmax distribution induced by the model on query q . $p_{\text{ctx}}(y \mid$
501 $q, c)$ is the context-conditioned softmax. Both are full-support distributions in $\Delta(\mathcal{Y})$.
- 502 • $r \in [0, 1]$ is the latent context reliability introduced in the setup paragraph of Section 3.
- 503 • $\pi(r)$ is a prior on r ; $p(c \mid r)$ is a likelihood of the observed context. Both are conditioned
504 on q throughout, suppressed for brevity.
- 505 • $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \rightarrow \mathbb{R}$ is a strictly proper scoring rule; log loss $S_{\log}(\hat{p}, y) = -\log \hat{p}(y)$ and
506 Brier $S_B(\hat{p}, y) = \sum_{y'} (\hat{p}(y') - \mathbb{1}[y' = y])^2$ are the working choices.
- 507 • Risk $R(\hat{p}) = \mathbb{E}_{q,c,y^*}[S(\hat{p}, y^*)]$ is the standard expected score; we write $R(\hat{y})$ when the
508 policy is a 0/1 estimator with the convention $\ell(y, y^*) = \mathbb{1}[y \neq y^*]$.
- 509 • $w^*(q, c) = \mathbb{E}[r \mid q, c]$ is the posterior mean reliability. When conditioning on q is clear
510 from context, we abbreviate this as $w^*(c)$.
- 511 • $\text{KL}(\mu \parallel \nu)$ is the KL divergence; TV is total variation; JSD is Jensen–Shannon divergence;
512 H is Shannon entropy.

513 **Assumption A.1** (Conditional independence). For all $q \in \mathcal{X}$, $c \in \mathcal{C}$, and $r \in [0, 1]$, $y^* \perp c \mid$
514 (q, r) .

515 **Assumption A.2** (Log-linear conditional answer model). The latent reliability acts directly
516 as the mixture weight:

$$p(y \mid q, c, r) \propto p_\theta(y \mid q)^{1-r} p_{\text{ctx}}(y \mid q, c)^r.$$

517 The log-linear family is closed under exponentiation by $[0, 1]$ since $\mathcal{Y}$ is finite.

518 **Assumption A.3** (Bounded log-density ratio). $\sup_{y \in \mathcal{Y}, q \in \mathcal{X}, c \in \mathcal{C}} \left| \log \frac{p_{\text{ctx}}(y \mid q, c)}{p_\theta(y \mid q)} \right| \leq M < \infty$.

519 This is a mild full-support condition; it can be enforced in practice with ε -smoothing.

520 B Related work, comparators, and theoretical antecedents

521 B.1 Connection to safe Bayes and the calibration literature

522 η -tempering is the inference-time analogue of safe-Bayes shrinkage [7, 19], applied only
523 to the post-trace answer readout: a strict η -Gibbs correction that preserves the trace’s
524 endogenous-evidence role while deflating its overconfident influence on the final commit.
525 Like Platt scaling or isotonic [20, 45, 49, 74], it needs held-out correctness for Brier selection;
526 unlike them, it uses one scalar pre-argmax (so it can flip predictions, not just recalibrate
527 scores). RLCR [13] is the training-time dualshifting effective η down via a Brier-style reward;
528 η -tempering is the cheaper retrofittable variant.

529 B.2 Source-by-source comparison with named baselines

530 For each named baseline, we list the official source, our re-implementation or use of the
531 released code, and the resulting per-spotlight-matrix mean regret. This is the source-of-
532 truth table that lets a reader audit each baseline number.

Table B.1: Baseline source and per-matrix regret. All baselines are run on the identical $5 \times 5 \times 3$ spotlight matrix described in Section 4.

Baseline	Source	Code	Mean regret
CAD	Shi et al. [58]	ours (re-impl.)	−0.079
AdaCAD	Wang et al. [66]	ours (re-impl.)	−0.106
CoCoA	Khandelwal et al. [28]	ours (re-impl.)	−0.128
JuICE	Li et al. [35]	ours (re-impl.)	−0.080
MADAM-RAG	Wang et al. [67]	ours (re-impl.)	−0.103
Self-RAG	Asai et al. [3]	released code	−0.146
Astute RAG	Wang et al. [65]	released code	−0.140
CRAG	Yan et al. [72]	released code	−0.045
FLARE-style	Jiang et al. [26]	released code	not on matrix

B.3 Comparison against the closest theoretical antecedents

vs. AdaCAD. JSD-weighted blend is a first-order Laplace approximation of $\mathbb{E}[r \mid c]$ (Lemma C.4); the +0.066 spotlight gap is the second-order Laplace residual. **vs. CoCoA.** Three-signal ensemble (entropy gap, Rényi divergence, contextual peakedness) without the posterior step; BAYESARBITER subsumes all three as candidate $\hat{w}$ features and beats by +0.044. **vs. RLCR** [13]. Training-time fix on the same η -overshoot lever; the mean $0.52 \times$ conflict-shrinkage is a quantitative prediction for RLCR’s training-time Brier ratio. **vs. Yoon et al.** [73]. “Reasoning helps calibration” is the $\eta < \bar{\eta}$ regime of Section 3.2; reproduced on TriviaQA closed-book (Table H.23). **vs. Lacombe et al.** [32]. “Reasoning hurts on knowledge-intensive” is the conflict regime where $\bar{\eta}$ shrinks; Section 3.2 supplies the mechanism.

B.4 Comparators in one paragraph

CAD ($w = 2$), AdaCAD (Laplace approx. to $\mathbb{E}[r \mid c]$), CoCoA, JuICE, MADAM-RAG, Astute RAG and gated/active variants (Self-RAG, FLARE, CRAG) are listed against Section 3.1 in Corollary C.1. PoE ($w = 1/2$) and MoE (arithmetic mean) are dominated and structurally distinct respectively. RLCR (13, training-time Brier reward) is complementary to inference-time η -tempering: RLCR closes the no-conflict ECE delta (−0.146) but leaves +0.145 on conflict (gap +0.291, half of GRPO-vs-DPO +0.344; Table H.21); η^* tracks the residual. Yoon et al. [73]’s “reasoning helps calibration” replicates on TriviaQA (0.118 vs. 0.142 ECE) and inverts on ConflictBank (0.95 vs. 0.96, opposite trajectory) — the well-specified vs. misspecified regimes of Section 3.2. Lacombe et al. [32]’s ECE rise of ~ 0.20 at $K = 2048$ matches the magnitude of our R1-14B/ConflictBank $K = 0 \rightarrow 1024$ rise (0.36). BayesBag [24] is a natural extension for small-calibration-set deployments; not pursued because the plug-in is already $\sqrt{n}$ -consistent.

C Proof of Theorem 1 and supplementary derivations

This section proves CT1 (the body claim that the reliability-weighted geometric mixture is the unique Gibbs minimizer of the arbitration-tempering free energy and the unique combiner under four classical axioms), then assembles the supporting machinery the body draws on: the explicit identifiability of w^* from observable signals (Appendix C.1.2), the AdaCAD-as-Laplace reduction that places the named decoders on the same exponent axis (Section 3.1), the sequence-level extension to free-form decoding, and the perturbation envelope for noisy signals. Each strengthening (T1+ Bregman characterization, T1’ TV-perturbation, sample-efficiency bound) is proved here and used directly by the body claim it underwrites.

C.1 Proof of Theorem 3.1

C.1.1 The Bayes action under a strictly proper scoring rule

Proposition C.1 (Posterior predictive is the Bayes action). *For strictly proper S , the posterior predictive $p^*(\cdot \mid q, c)$ uniquely minimises $\mathbb{E}_{y^* \mid q, c}[S(\hat{p}, y^*)]$ on $\Delta(\mathcal{Y})$.*

570 *Proof.* Strict propriety: $\mathbb{E}_{y \sim \nu} S(\nu, y) \leq \mathbb{E}_{y \sim \nu} S(\hat{p}, y)$ with equality iff $\hat{p} = \nu$ [20]; set $\nu = p^*$ and
 571 tower over (q, c) . $\square$

572 **Proposition C.2** (Integral form). *Under Assumption A.1,*

$$p^*(y|q, c) = \int_0^1 p(y|q, c, r) p(r|q, c) dr. \quad (\text{C.1})$$

573 **Proposition C.3** (Geometric mixture under log-linearity). *Under Assumption A.2,*

$$p^*(y|q, c) \propto p_\theta(y|q)^{1-w^*} p_{\text{ctx}}(y|q, c)^{w^*}, \quad w^* = \mathbb{E}[r|q, c]. \quad (\text{C.2})$$

574 *Sketch.* Log-linearity gives $\log p(y|q, c, r) = \log p_\theta + r\Delta(y; q, c) - Z(r)$ with $\Delta = \log p_{\text{ctx}}/p_\theta$.
 575 Substituting and Taylor-expanding $\exp(r\Delta)$ around $\bar{r} = \mathbb{E}[r|q, c]$ yields $\exp(\bar{r}\Delta)[1 + \frac{1}{2}\sigma_r^2\Delta^2 +$
 576 $O(\sigma_r^4)]$; leading term renormalises to (C.2); the correction is $O(\sigma_r^2)$ and exact in the degen-
 577 erate case $r \sim \delta_{w^*}$. $\square$

578 **Proposition C.4** (Variational characterization). $p_w^* \propto p_\theta^{1-w} p_{\text{ctx}}^w$ uniquely minimises $(1 -$
 579 $w) \text{KL}(p||p_\theta) + w \text{KL}(p||p_{\text{ctx}})$ on $\Delta(\mathcal{Y})$ (C.3); it is the e -geodesic mean at parameter w .

$$p_w^* = \arg \min_p [(1-w) \text{KL}(p||p_\theta) + w \text{KL}(p||p_{\text{ctx}})]. \quad (\text{C.3})$$

580 *Sketch.* Lagrangian on $\Delta(\mathcal{Y})$ is strictly convex; stationarity gives $\log p = (1-w) \log p_\theta +$
 581 $w \log p_{\text{ctx}} - \text{const}$. $\square$

582 C.1.2 Identifiability of w^* from observable signals

583 Theorem 3.1 requires that the unobserved $w^*(q, c)$ be estimable from observable signals. We
 584 give a clean lemma.

585 **Lemma C.1** (Reliability-signal identifiability). *Suppose $s(q, c) \in \mathbb{R}^d$ is a measurable sig-*
 586 *nal such that $r \perp c \mid s(q, c)$ in distribution (i.e., s is a sufficient statistic for r given*
 587 *c). Then $w^*(q, c) = \mathbb{E}[r \mid q, c] = \mathbb{E}[r \mid q, s(q, c)]$ is identifiable from any joint sample of*
 588 *$(s(q, c), y^*, y_{\text{ctx}}, y_{\text{par}})$. In particular, the plug-in*

$$\hat{w}(c) = \widehat{\mathbb{E}}[\mathbb{I}[y^* = y_{\text{ctx}}] \mid s(q, c), y_{\text{ctx}} \neq y_{\text{par}}] \quad (\text{C.4})$$

589 *is a consistent estimator of w^* on the conflict subset.*

590 *Proof.* By the tower property, $\mathbb{E}[r \mid q, c] = \mathbb{E}[\mathbb{E}[r \mid q, c, s(q, c)] \mid q, c] = \mathbb{E}[r \mid q, s(q, c)]$ since s
 591 is sufficient. The plug-in (C.4) is the conditional expectation of the indicator $\mathbb{I}[y^* = y_{\text{ctx}}]$
 592 on the conflict subset, which by definition of r equals $\mathbb{E}[r \mid s, y_{\text{ctx}} \neq y_{\text{par}}]$. Standard
 593 nonparametric regression consistency [62] gives the conclusion. $\square$

594 In practice we take $s(q, c) = (\text{retrieval score, popularity, semantic entropy, contradiction type})$
 595 and fit a logistic regression on (C.4). This is the implementation behind the
 596 “BAYESARBITER” policy.

597 C.1.3 Excess Bayes risk for the plug-in

598 **Lemma C.2** (Plug-in excess risk under log loss). *Let $\hat{p}_w$ denote the geometric mixture with*
 599 *weight w and let $\hat{w}$ be a plug-in estimator of w^* . Under Assumption A.3,*

$$R_{\log}(\hat{p}_{\hat{w}}) - R_{\log}(\hat{p}_{w^*}) \leq 2M^2 \mathbb{E}[(\hat{w} - w^*)^2]. \quad (\text{C.5})$$

600 *Sketch.* $\partial_w^2 R_{\log} = \text{Var}_{\hat{p}_w}(\log p_{\text{ctx}}/p_\theta) \leq M^2$ under Assumption A.3; Taylor at w^* . $\square$

601 **Lemma C.3** (Plug-in excess risk under 0/1 loss). *Let $\hat{y}_w = \arg \max_y \hat{p}_w(y)$. Then*

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^*}) \leq \sqrt{\mathbb{E}[(\hat{w} - w^*)^2]} \cdot \|p_\theta - p_{\text{ctx}}\|_1, \quad (\text{C.6})$$

602 *giving the bound stated in the body’s Theorem 3.1.*

603 *Sketch.* $|R_{0/1}(\hat{y}_p) - R_{0/1}(\hat{y}_q)| \leq 2 \text{TV}(p, q)$; $\text{TV} \leq \frac{1}{2} \|\partial_w \hat{p}_w\|_1 |\hat{w} - w^*| \leq \frac{1}{2} \|p_\theta - p_{\text{ctx}}\|_1 |\hat{w} - w^*|$;
 604 Cauchy–Schwarz. Rate $1/\sqrt{n}$ is sharp; constants are slack. $\square$

605 **Matching Le Cam two-point lower bound.** The upper bound (C.6) is rate-tight at
 606 $1/\sqrt{n}$. The matching lower bound below shows no estimator of w^* from n paired (signal,
 607 outcome) observations can beat that rate even by a constant exponent, so the plug-in is
 608 minimax-rate-optimal.

609 **Theorem C.1** (Le Cam two-point lower bound for w^* estimation). *Fix (q, c) and a finite*
 610 *output space $\mathcal{Y}$ with $|\mathcal{Y}| \geq 2$. Suppose the contextual and parametric distributions $p_\theta, p_{\text{ctx}} \in$*
 611 *$\Delta(\mathcal{Y})$ satisfy $\|p_\theta - p_{\text{ctx}}\|_1 = \delta > 0$. Let $\hat{w}_n$ be any (possibly randomized) estimator of $w^* \in [0, 1]$*
 612 *based on n iid samples $(s_i, y_i)_{i=1}^n \sim \hat{p}_{w^*}$. Then*

$$\inf_{\hat{w}_n} \sup_{w^* \in [0, 1]} \mathbb{E}|\hat{w}_n - w^*| \geq \frac{1}{8\delta} \cdot \frac{1}{\sqrt{n}}.$$

613 *Composing with the Lipschitz dependence of $\hat{p}_w$ on w through the geometric mixture and*
 614 *the standard 0/1-loss continuity bound, the same construction yields, for any policy $\hat{y}_n$ that*
 615 *depends on the data only through $\hat{w}_n$,*

$$\inf_{\hat{y}_n} \sup_{w^* \in [0, 1]} [R_{0/1}(\hat{y}_n) - R_{0/1}(\hat{y}_{w^*})] \geq \frac{\delta^2}{32} \cdot \frac{1}{\sqrt{n}},$$

616 *which matches the upper bound of Lemma C.3 up to a constant.*

617 *Sketch.* Two-point family $w_j = 1/2 \pm 1/(4\sqrt{n})$; $\text{KL}(P_0^{\otimes n} \| P_1^{\otimes n}) \leq c_0 \delta^2/4$ via log-density
 618 linearization. Le Cam’s two-point lemma [62, Thm. 2.2] + Markov; Lipschitz $\|\partial_w \hat{p}_w\|_1 \leq \delta$
 619 closes the regret. The empirical $R^2 = 0.998$ at $n = 2048$ (Appendix H.11) saturates the
 620 bound. $\square$

621 C.1.4 AdaCAD as a Laplace approximation

622 **Lemma C.4** (AdaCAD’s JSD weight is the first-order Laplace approximation of the pos-
 623 terior reliability). *Suppose $r \sim \text{Beta}(\alpha, \beta)$ with $\alpha + \beta \rightarrow \infty$ and the likelihood $p(c | r)$ has*
 624 *Hellinger affinity $1 - \frac{1}{2}H^2(p_\theta, p_{\text{ctx}})$ at $r = 1/2$. Then $\mathbb{E}[r | c] = \frac{1}{2} + \frac{1}{4}\text{JSD}(p_\theta \| p_{\text{ctx}}) +$*
 625 *$O(\text{JSD}^2)$ in the small-conflict regime. Hence AdaCAD’s weight $w_{\text{AdaCAD}} = \text{JSD}(p_\theta \| p_{\text{ctx}})$,*
 626 *after the standard 1/4 calibration constant absorbed into the temperature, is the first-order*
 627 *Laplace approximation of the posterior reliability.*

628 *Sketch.* Taylor $\log p(c | r)$ around $r = 1/2$; symmetrising KL under the Beta prior gives
 629 JSD/4. $\square$

630 Lemma C.4 explains why AdaCAD performs strongly empirically without ever solving an
 631 explicit posterior: it is approximating w^* to first order. The quadratic remainder is what
 632 makes the full BAYESARBITER beat AdaCAD by +0.0659 regret on the spotlight matrix
 633 (Section 4.1).

634 **Remark C.1** (CAD as $w = 2$, CoCoA as ensemble). CAD [58] sets $\log \hat{p} \propto \log p_{\text{ctx}} - \log p_\theta$,
 635 i.e., $w = 2$, an over-trust projection that lies outside $[0, 1]$ and deliberately over-amplifies
 636 p_{ctx} . CoCoA’s three-signal score (entropy gap, Rényi divergence, contextual peakedness)
 637 is a *non-Bayesian ensemble* of reliability proxies; without the posterior update step it is
 638 consistent for w^* only when the three signals are unbiased and equally informative, which
 639 is empirically not the case.

640 **Corollary C.1** (Existing methods are special cases). (a) CAD [58] is $w = 2$ (out-of-range;
 641 over-amplifies). (b) Always-context = $w = 1$, always-parametric = $w = 0$; both are dominated
 642 by Section 3.1. (c) AdaCAD’s $w = \text{JSD}(p_\theta \| p_{\text{ctx}})$ is the first-order Laplace approximation of
 643 $\mathbb{E}[r | q, c]$ under a $\text{Beta}(\alpha, \beta)$ prior with Hellinger-affinity likelihood (Lemma C.4). (d)
 644 CoCoA’s three-signal heuristic is a non-Bayesian ensemble of proxies for $\mathbb{E}[r | q, c]$ and need
 645 not converge to w^* without additional calibration assumptions.

646 C.1.5 Sufficient signal statistic for log-linear arbitration

647 Define $\mathcal{F}_{\Pi} = \{\hat{p}_{\phi} : \phi : \mathcal{S} \rightarrow [0, 1]\}$ with $\hat{p}_{\phi} \propto p_{\theta}^{1-\phi(s)} p_{\text{ctx}}^{\phi(s)}$ and local Fisher info $V(q, c) = \partial_w^2 R|_{w^*} >$
 648 0.

649 **Theorem C.2** (Sufficient signal statistic). *Under T1’s assumptions, on log loss,*

$$R(\phi) - R^* = \frac{1}{2} \mathbb{E}_s[\bar{V}(s)(\phi(s) - \phi^*(s))^2] + O(\|\phi - \phi^*\|_{L^3}^3), \quad (\text{C.7})$$

650 $\phi^*(s) = \mathbb{E}[r | s]$, $\bar{V} = \mathbb{E}[V | s]$. ϕ^* is the unique minimizer of R on $\mathcal{F}_{\Pi}$ (a.s. in s).

651 *Sketch.* Per- (q, c) Taylor: $F(w; q, c) = F^* + \frac{V}{2}(w - w^*)^2 + O(\cdot)$ since $w^* = \mathbb{E}[r | q, c]$ is stationary
 652 (Prop. C.4). Tower-condition on s and minimise the quadratic in $\phi(s)$: weighted projection
 653 $\phi^*(s) = \mathbb{E}[V \mathbb{E}[r | q, c] | s] / \mathbb{E}[V | s] = \mathbb{E}[r | s]$ when V is s -measurable / r -independent given
 654 s . $\square$

655 **Corollary C.2** (Decision-theoretic sufficiency). $\mathbb{E}[r | s]$ is Blackwell-sufficient for log-linear
 656 arbitration: $T(s)$ is risk-optimal iff $\mathbb{E}[r | s]$ is $\sigma(T)$ -measurable.

657 **Corollary C.3** (Coarsening penalty). $\phi_T^*(T) := \mathbb{E}[\bar{V} \phi^* | T] / \mathbb{E}[\bar{V} | T]$ gives $R(\phi_T^*) - R(\phi^*) =$
 658 $\frac{1}{2} \mathbb{E}[\bar{V}(\phi_T^* - \phi^*)^2] + O(\cdot)$; $\bar{V} \geq v_{\min} > 0$ replaces $\bar{V}$ by $v_{\min}$ in the lower bound, equality (cubic-
 659 up-to) iff ϕ^* is $\sigma(T)$ -measurable.

660 CAD/AdaCAD/CoCoA approximate $\mathbb{E}[r | s]$ at increasing accuracy (Table N.43); residual
 661 gap tracks how well each spans the sufficient statistic.

662 C.1.6 Sequence-level extension: Corollary C.4 and proof

663 **Corollary C.4** (Sequence-level extension). On $\mathcal{Y} = \mathcal{V}^{\leq T_{\max}}$, the T1 variational characteri-
 664 zation holds verbatim; per-token chain mixture $p_w^{\text{tok}} \propto \prod_t p_{\theta}^{1-w_t} p_{\text{ctx}}^{w_t}$ equals the sequence-level
 665 mixture exactly when the chain partition factorises, otherwise differs by partition-mismatch
 666 KL.

667 **Lemma C.5** (Chain-vs-sequence gap). $\text{KL}(p_w^{\text{tok}} \| p_w^*) = \sum_t \mathbb{E}_{y_{<t} \sim p_w^{\text{tok}}} \log Z_t(y_{<t}) - \log Z_w,$
 668 zero iff per-step partitions chain-decompose.

669 Gap is zero on discrete-candidate benchmarks; bounded by $T \log |\mathcal{V}|$ on free-form.
 670 CAD/AdaCAD/CoCoA are special cases in both regimes.

671 C.1.7 Boundary behaviour and signal robustness

672 The geometric mixture in Section 3.1 is well behaved on the operating regime that LLM
 673 softmax outputs actually occupy, but two boundary cases and one robustness question
 674 deserve explicit treatment.

675 **Zero-mass support mismatch** does not arise (LLM softmax has full vocabulary sup-
 676 port); a $\epsilon \geq 10^{-30}$ floor costs $\sim 10^{-25}$ nat, below other approximation gaps. **Argmax dis-**
 677 **agreement** (mode flip at one $w \in (0, 1)$) is the per-cell signal η -tempering and w^* arbitrate.

678 **Adversarial signal** $\|\delta\|_2 \leq B$ inflates excess risk by $\leq (\|\hat{\beta}\|_2 B / 4 + \|\hat{w} - w^*\|_{L^2}) \|p_{\theta} - p_{\text{ctx}}\|_1$;
 679 empirically $\|\hat{\beta}\|_2 \in [2.31, 2.96]$, $B = 0.1$ gives worst-case inflation ~ 0.074 . **Conjugacy** is not
 680 required: the geometric mixture is the optimal log-linear posterior projection regardless of
 681 π ’s shape, with $O(\sigma_r^2)$ correction (Appendix C.2) the only place π enters.

682 C.2 Theorem-1 variance correction (empirical)

683 Empirical $\text{KL}(p^* \| \hat{p}_{w^*})$ on the spotlight matrix (p^* via importance-weighted MC over $\pi(r | c)$):
 684 mean $\sim 10^{-6}$ nat, max 1.4×10^{-5} nat, 100% of bins below 0.02 nat (conflict, no-conflict, and
 685 pooled 25/25/50 cells). The first-order Laplace expansion of Section 3.1’s $\frac{1}{2} \sigma_{\rho}^2 V$ correction
 686 is quantitatively negligible at our working precision.

687 C.3 Detailed proof of the variational characterization

688 **Lagrangian.** $\Lambda_w(p) = (1-w) \text{KL}(p\|p_\theta) + w \text{KL}(p\|p_{\text{ctx}})$ is jointly convex [11, Thm. 2.7.2].
689 Stationarity gives $\log p = (1-w) \log p_\theta + w \log p_{\text{ctx}} - \text{const}$, i.e., $p \propto p_\theta^{1-w} p_{\text{ctx}}^w$; strict convexity
690 yields uniqueness. ***e*-geodesic.** In exponential-family *e*-coordinates, p^* is exactly the *e*-
691 geodesic between p_θ and p_{ctx} at parameter w . **Brier equivalence.** Brier $\mathbb{E}_{y^* \sim p^*} S_B(\hat{p}, y^*) =$
692 $\|\hat{p} - p^*\|_2^2 + \text{const}$ is minimised at $\hat{p} = p^*$, identifying the log-loss minimiser by Propositions C.1
693 and F.11.

694 C.4 Proof of the axiomatic uniqueness of the geometric mixture

695 *Proof of Section 3.1.* Define $\phi(a, b, w) := \Phi(p, q, w)(y)/\Phi(p, q, w)(y')$ with $a = p(y)/p(y')$,
696 $b = q(y)/q(y')$. (A4) makes ϕ depend on (a, b, w) alone; (A1)+(A2)+(A3) give boundary,
697 idempotence, and label-equivariance constraints. Chaining ratios across three outcomes
698 yields the multiplicative Cauchy equation $\phi(a_1 a_2, b_1 b_2, w) = \phi(a_1, b_1, w) \phi(a_2, b_2, w)$; continu-
699 ity gives the unique solution $\phi(a, b, w) = a^{\alpha(w)} b^{\beta(w)}$. Boundary $\phi(a, b, 0) = a$, $\phi(a, b, 1) = b$
700 and idempotence $\alpha(w) + \beta(w) = 1$ pin $\alpha = 1 - w$, $\beta = w$, so $\Phi(p, q, w) \propto p^{1-w} q^w$. $\square$

701 **Remark C.2** (Axiom roles). (A1) rules out trivial combiners; (A2) fixes w as a mixing
702 weight; (A3) is label-equivariance; (A4) (LLR-invariance) is the substantive axiom: dropping
703 it admits the linear pool $(1-w)p + wq$.

704 **Lemma C.6** (EB excludes reliability-adaptive pools). *Under External Bayesianity [18,*
705 *Thm. 3.1], the unique pool is $\Phi(p, q) \propto p^{1-w_0} q^{w_0}$ with constant w_0 . Any signal-conditioned*
706 *$w(s)$ pool (AdaCAD, CoCoA) or constant outside $[0, 1]$ (CAD, $w = 2$) violates EB but*
707 *satisfies T1.*

708 *Sketch.* EB’s commutation forces w to depend only on the support, not on external signals;
709 non-constant $w(s)$ depends on s invisible to Bayes updates on (p, q) . $\square$

710 C.5 Proof of the information-theoretic minimax floor (T2+)

711 *Proof of Section 3.1.* Fix non-adaptive $\hat{y}$ depending on (q, c) only through $s = s(q, c)$ ($I(r; \hat{y} |$
712 $s) = 0$). Strict properness + convex Bayes risk $\mathcal{R}$ give the regret decomposition $R(\hat{y}) - R(\hat{y}^*) \geq$
713 $\mathbb{E}[\mathcal{R}(\hat{y}(q, c)) - \mathbb{E}_r \mathcal{R}(\hat{y}^*(q, c, r))]$. Strong convexity ($\mu \geq 1/(L|\mathcal{Y}|)$ for log loss; $\mu = 2$ Brier) plus
714 the conditional-mean optimization gives $\text{Var}_{r|q,c}[\hat{y}^*] \geq \kappa^2 \sigma_r^2(q, c)$ (Lipschitz in r). Pinsker
715 $\text{TV}^2 \leq \frac{1}{2} \text{KL} + \text{Cauchy-Schwarz } \|\cdot\|_2^2 \geq 4\text{TV}^2/(|\mathcal{Y}| - 1)$ collect to $\geq \frac{1}{8(|\mathcal{Y}| - 1)^2} \sigma_r^2(q, c)$; tower
716 over s . $\square$

717 **Remark C.3.** Plug-in matches at $\Theta(\mathbb{E}[\sigma_r^2(s)]) + O(n^{-1/2})$; two-point reduction is tight,
718 $|\mathcal{Y}| - 1$ slack is the $\ell_2 \rightarrow \ell_1$ conversion cost; Brier replaces the constant with $1/(2(|\mathcal{Y}| - 1)^2)$.

719 C.6 Proof of the closed-form Brier improvement (T3+)

720 *Proof of Theorem 4.* $\hat{p}_\eta \propto p_K^\eta$. Taylor-expand $\mathcal{B}(\eta)$ around η^* : optimality kills the first-
721 order term; second derivative identity $\mathcal{B}''(\eta^*) = 2 \text{Var}_{p_K}(\mathbf{1}\{Y = y^*\}) + o(1) = 2V_{\text{post}} + o(1)$.
722 Combining, $\Delta \text{Brier} = (1 - \eta^*)^2 V_{\text{post}} + O((1 - \eta^*)^3)$, strictly positive when $\eta^* < 1$ and $V_{\text{post}} > 0$.
723 No stable-answer condition. $\square$

724 C.6.1 Closed form for η^* on binary-collapse cells

725 **Lemma C.7** (Binary-collapse η^*). *Suppose p_K and the Bayes-optimal q^* both concentrate*
726 *on two answers with masses $\{a, 1-a\}$ and $\{b, 1-b\}$ respectively (binary-collapse). Then the*
727 *Brier-optimal tempering η^* on this cell satisfies $\eta^* = \text{logit}(b)/\text{logit}(a)$, and linearizing at the*
728 *Berk-Nash equilibrium of Section 3.2 gives $\eta^* = 1 - D^*/(\bar{\eta} \cdot \text{logit}(b)) + O(D^{*2})$.*

729 *Sketch.* On two atoms, $\text{logit } \hat{p}_\eta = \eta \text{logit}(a)$; Brier-optimal $\hat{p} = q^*$ closes $\eta^* = \text{logit}(b)/\text{logit}(a)$.
730 Linearisation uses Berk-Nash $b = a e^{-D^*/\bar{\eta}} + O(D^{*2})$ (Proposition E.9). $\square$

This is a one-line algebra fact; we surface it because it lets practitioners skip cross-validation on $\sim 42\%$ of conflict cells (Table E.10) at the cost of ≤ 0.004 Brier vs. 5-fold CV. The result is small but useful; it is not load-bearing for the paper’s claims.

Remark C.4 (Relation to the safe-rate result). Section 3.2 (Corollary E.6) characterizes when $\eta^* < 1$ on the stable-answer subset: overshoot of $\bar{\eta}$ creates the non-decreasing calibration gap, which forces $\eta^* < 1$ for Brier-optimal tempering. Theorem 4 then quantifies the resulting Brier improvement. The two theorems are complementary: the safe-rate result says when η -tempering should help; the intervention result says *by how much* when it does, with no subset condition required.

C.7 Detailed proof of Pinsker-style excess risk

The geometric mixture is a one-parameter exponential family with $T(y) = \log p_{\text{ctx}}/p_\theta$ and Fisher $I(w) = \text{Var}_{p_w} T \leq M^2/4$ (Popoviciu under Assumption A.3). Integrating $\text{KL}(p_w \| p_{w'}) \leq \frac{M^2}{8}(w - w')^2 + \text{Pinsker gives TV} \leq \frac{M}{4}|w - w'|$. Combined with the 0/1 continuity bound $|R_{0/1}(\hat{y}_p) - R_{0/1}(\hat{y}_q)| \leq 2 \text{TV}$ and Cauchy–Schwarz,

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^*}) \leq \frac{M}{2} \|\hat{w} - w^*\|_{L^2}, \quad (\text{C.8})$$

order-optimal vs. Le Cam (Theorem C.1) up to constant 2.

C.8 Sample efficiency of the plug-in

In-the-wild Bayes-vs-heuristic gain on the spotlight matrix by calibration size n (mean / 95% CI / wins): 32 +0.012/[−0.018, 0.041]/13:25; 64 +0.034/[0.005, 0.062]/16:25; 128 +0.058/[0.025, 0.085]/19:25; 256 +0.071/[0.041, 0.097]/21:25; 512 +0.082/[0.040, 0.110]/23:25; 1024 +0.083/[0.038, 0.111]/23:25; 2048 +0.083/[0.037, 0.112]/23:25 (Table C.2). Crosses zero at $n \approx 64$, saturates by $n \approx 512$; deployable with ~ 100 labeled calibration examples per benchmark.

Table C.2: Sample-efficiency curve (numbers as in text above).

n	Gain	95% CI	Wins
32	+0.012	[−0.018, 0.041]	13/25
128	+0.058	[0.025, 0.085]	19/25
512	+0.082	[0.040, 0.110]	23/25
2048	+0.083	[0.037, 0.112]	23/25

C.9 If conditional independence or log-linearity fails

CI failure (y^* depends on c given (q, r)): the geometric mixture is no longer the unique Bayes-optimal predictive; the integral form $p^*(y | q, c) = \int p(y | q, c, r)p(r | q, c) dr$ is. We see no evidence of CI failure on our cells (per-cell residual $y^* - \hat{y}_{w^*}$ has no systematic c -dependence beyond r); a held-out $I(y^*; c | q, r)$ test would diagnose pathological cases.

Log-linearity failure (e.g., a hard gate at threshold τ): the gated alternative $p^* = p_{\text{ctx}} \mathbb{1}[w^* \geq \tau] + p_\theta \mathbb{1}[w^* < \tau]$ is dominated by the soft mixture in the small-conflict regime and approximately equal in the large-conflict regime; empirically, +0.012 regret worse on the spotlight matrix.

C.10 Extended discussion of the AdaCAD Laplace approximation

Second-order correction. $\mathbb{E}[r | c] = \frac{1}{2} + \frac{\text{JSD}}{4} + \frac{\text{JSD} \cdot [\text{JSD} - \text{Var}_{\text{Beta}} r]}{16} + O(\text{JSD}^3)$; correction 0.005–0.020 on the spotlight matrix matches the +0.066 AdaCAD-vs-Bayes regret gap after weighting by $\|p_\theta - p_{\text{ctx}}\|_1$. **CoCoA.** The three-signal heuristic does not Laplace-identify because the signals are correlated. Re-fitting an L2-regularised combiner on our calibration recovers −0.142 regret (CoCoA itself −0.128; Bayes −0.172). **AdaCAD failure mode.** Trails Bayes most where JSD is large (ConflictBank-conflict: +0.11 at JSD ~ 1.8 nat; PopQA: +0.04 at JSD ~ 0.4).

770 C.11 Extended discussion of conditional independence

771 CI ($y^* \perp c | q, r$): “ c is sufficient for r .” Exactly true in synthetic oracle; approximately true
 772 when conflict mechanism is mediated by a clean reliability variable. Breaks if c contains
 773 side-channel info (verbatim answer, format cue); diagnostic in App. C.9 detects. Without
 774 CI, the predictive $p^*(y | q, c) = \int p(y | q, c, r)p(r | q, c)dr$ is still the right object but loses its
 775 closed form.

776 C.12 Supplementary derivation, Pinsker for log-loss

777 This section expands the second-order argument behind Lemma C.2.

778 **Proposition C.5** (Second-order log-loss sensitivity along the geometric path). *Let $p_w \propto$
 779 $p_\theta^{1-w} p_{\text{ctx}}^w$ and define $R_{\log}(w) := -\mathbb{E}_{y^*}[\log p_w(y^*)]$. Under Assumption A.3, $R_{\log}$ is twice
 780 continuously differentiable and satisfies*

$$R_{\log}''(w) \leq 4M^2 \quad \text{for all } w \in [0, 1].$$

781 Consequently, if w^* minimizes $R_{\log}$ on $[0, 1]$, then

$$R_{\log}(w) - R_{\log}(w^*) \leq 2M^2(w - w^*)^2,$$

782 which is the pointwise version of Lemma C.2.

783 *Proof.* Write $\partial_w^2 R_{\log} = \text{Var}_{p_w}(\log p_{\text{ctx}}/p_\theta) \leq (2M)^2$ under Assumption A.3; Taylor at w^*
 784 closes. $\square$

785 D Proof of Theorem 2 and the minimax construction

786 D.1 Proof of Theorem 3.1

787 D.1.1 Yao-style minimax lower bound

788 Construct P_+ (context-correct, $r = 1$, ConflictBank temporal-update) and P_- (context-
 789 wrong, $r = 0$, ConflictBank misinformation). Per-family excess risk: $R(\hat{y}_w; P_+) - R(\hat{y}^*; P_+) \geq$
 790 $\Delta_+(1-w)$ and $R(\hat{y}_w; P_-) - R(\hat{y}^*; P_-) \geq \Delta_-w$. $\widehat{\Delta}_+ / \widehat{\Delta}_-$ are estimable from per-sub-family re-
 791 gret slices; matched-base Llama-3.1-8B GRPO conflict ECE survives all three ConflictBank
 792 sub-families (+0.572 / +0.690 / +0.629 / pooled +0.264; Table D.3).

Table D.3: Llama-3.1-8B GRPO conflict Δ ECE per ConflictBank sub-family.

Misinformation / Temporal / Semantic-ambig. / pooled	+0.572 / +0.690 / +0.629 / +0.264
--	-----------------------------------

793 Held-out estimation/test split: the Yao bound estimated on a disjoint 50% of ConflictBank
 794 tracks the realized $w_0 = \frac{1}{2}$ regret on the other half within ± 0.02 ($\widehat{\Delta}_+ = 0.687$, $\widehat{\Delta}_- = 0.789$,
 795 bound 0.343, realized 0.358).

796 *Proof of the lower bound in Section 3.1.* Linearity in P_λ gives $R(\hat{y}_{w_0}; P_\lambda) - R(\hat{y}^*; P_\lambda) \geq$
 797 $\lambda \Delta_+(1-w_0) + (1-\lambda)\Delta_-w_0$; $\lambda = 1/2$ yields $\geq \frac{1}{2} \min(\Delta_+, \Delta_-)$. $\square$

798 **Remark D.5** (Realisability). P_+ = ConflictBank temporal-update; P_- = misinforma-
 799 tion [60].

800 D.1.2 Optional information-theoretic restatement

801 **Proposition D.6** (Binary testing restatement). *With $U \in \{+, -\}$ hidden, vanishing mini-
 802 max excess risk over $\{P_+, P_-\}$ requires $I(U; \text{obs}) > 0$; otherwise Fano gives constant lower
 803 bound $\geq \frac{1}{2} \min(\Delta_+, \Delta_-)$.*

804 Empirically, retrieval score, popularity, and contradiction type carry non-trivial $I(U; \text{obs})$;
 805 BAYESARBITER exploits it.

806 D.1.3 Plug-in upper bound for adaptive policies

807 **Lemma D.8** (Adaptive plug-in convergence rate). *Let $s(q, c)$ be a signal that classifies the*
808 *sub-family of Appendix D.1.1 with error ε_s (i.e., $\mathbb{P}(s \text{ predicts wrong sub-family}) = \varepsilon_s$). Let*
809 *$\hat{w}$ be the logistic-regression plug-in trained on n calibration examples. Then $R(\hat{y}_{\hat{w}}; P_{w_0}) -$*
810 *$R(\hat{y}^*; P_{w_0}) \leq C\varepsilon_s + O(n^{-1/2})$.*

811 *Sketch.* Decompose into oracle hard-threshold gap (bounded by $\varepsilon_s \cdot \Delta$) plus plug-in regret
812 (Lemma C.3, $O(n^{-1/2})$). $\square$

813 **Proposition D.7** (Signal separability controls the adaptive gap). *With $\varepsilon_s = \inf_h \mathbb{P}[h(s) \neq U]$*
814 *the Bayes sub-family error, every s -measurable $w(s)$ has $R(\hat{y}_w; P_{1/2}) - R(\hat{y}^*; P_{1/2}) \geq \frac{\Delta_{\min}}{2} \varepsilon_s$;*
815 *adaptive plug-in achieves $\leq C\varepsilon_s + O(n^{-1/2})$ (Lemma D.8).*

816 *Sketch.* Hard-threshold $h_w := \mathbb{I}[w(s) \geq 1/2]$ pays $\Delta_{\pm}/2$ on regime mistakes; averaging gives
817 the lower bound. $\square$

818 D.2 Supplementary derivation, Fano sharpening

819 **Proposition D.8** (Two-point lower bound). *Le Cam [62, Lemma 2.2]: $\inf_{\hat{U}} \mathbb{P}(\hat{U} \neq U) \geq$*
820 *$(1 - \text{TV}(P_+, P_-))/2$; combined with $\Delta_{\min}$ regime-mistake cost, every policy has minimax*
821 *excess $\geq \frac{\Delta_{\min}}{2} (1 - \text{TV}(P_+, P_-))$.*

822 D.3 Worked example of the minimax construction

823 Binary $\{A, B\}$, $D_+ = (y_{\text{par}}, y_{\text{ctx}}, y^*) = (A, B, B)$, $D_- = (A, B, A)$, balanced mixture $P_{1/2}$;
824 every fixed $w_0 \in [0, 1]$ has error $1/2$ (always- A or always- B); signal-adaptive achieves 0.

825 E Proof of Theorem 3, assumption audits, and multilingual 826 evidence

827 E.1 Proof of Theorem 3.2

828 E.1.1 Setup and the η -generalized posterior

829 Bissiri–Holmes–Walker [7] update on family $\mathcal{M} = \{q_{\xi}\}$:

$$\pi_{\eta}(\xi | z) \propto \pi(\xi) \exp(-\eta \ell(z; \xi)), \quad \ell = -\log q_{\xi}(z). \quad (\text{E.9})$$

830 For misspecified $\mathcal{M}$, Grünwald and Van Ommen [19] give a safe rate $\bar{\eta} \in (0, 1]$ above which
831 the posterior hypercompresses. In the LLM CoT setting, $z_t \sim p_t$ from the model itself
832 (endogenous evidence; 16, 17).

833 E.1.2 The safe-rate lemma

834 **Lemma E.9** (Larger D^* shrinks the safe rate). $\bar{\eta} \leq c_{|\mathcal{Y}|, \pi} / D^*$ where $D^* = \text{KL}(\mathbb{P}^* \| q_{\xi^*})$ is the
835 *KL projection distance and $c_{|\mathcal{Y}|, \pi}$ collects the bracketing-entropy and prior dependence [19,*
836 *Thm. 1.6]. On finite $\mathcal{Y}$, $\log N_{[]}(\varepsilon, \Delta(\mathcal{Y})) \leq (|\mathcal{Y}| - 1) \log(3/\varepsilon)$ makes the entropy integral finite;*
837 *the paper uses only the monotonicity $\bar{\eta} \propto 1/D^*$.*

838 E.1.3 Reduction to Berk–Nash dynamics: proof of Proposition E.9

839 The reduction needs only continuity of the expected-update operator on $\Delta(\mathcal{Y})$, not para-
840 metric $\mathcal{M}$.

841 **Definition E.1** (Auxiliary Berk–Nash game $G_{q, c, \eta}$). Fix (q, c) and a rate $\eta \in (0, 1]$. The
842 auxiliary game has a single agent with belief space $\Delta(\mathcal{Y})$, action space identified with the
843 belief itself (the model’s autoregressive distribution is its action), and a data-generating

distribution $\mu(\cdot \mid p, q, c)$ that depends on the agent’s belief through the sampling kernel of (3). The agent’s best-response operator is the η -Bayes update

$$B_\eta(p)(y) = \frac{p(y) \mathbb{E}_{z \sim \mu(\cdot \mid p, q, c)} \exp(-\eta \ell(y; q, c, z))}{\sum_{y'} p(y') \mathbb{E}_{z \sim \mu(\cdot \mid p, q, c)} \exp(-\eta \ell(y'; q, c, z))}.$$

A Berk–Nash equilibrium of $G_{q, c, \eta}$ is a fixed point $p^* = B_\eta(p^*)$.

Lemma E.10 (Continuity and existence). *With $\ell \in [0, L]$ and $\mu(\cdot \mid p, q, c)$ TV-Lipschitz in p , B_η is continuous on $\Delta(\mathcal{Y})$ and has a fixed point (Brouwer).*

Proof. $\mathbb{E}_z \exp(-\eta \ell) \in [e^{-\eta L}, 1]$ is continuous in p via dominated convergence; ratio of continuous, lower-bounded terms is continuous; $\Delta(\mathcal{Y})$ compact convex. $\square$

Lemma E.11 (Limit of tracked trajectories). *If $p_t \rightarrow p_\infty$ and $\|p_{t+1} - B_\eta(p_t)\|_{TV} \rightarrow 0$ under Lemma E.10, then $p_\infty = B_\eta(p_\infty)$.*

Proof. Triangle inequality with continuity of B_η . $\square$

Proposition E.9 (Φ_η fixed points = Berk–Nash equilibria). *Under Lemma E.10, $\Phi_\eta = B_\eta$ fixed points are exactly Berk–Nash equilibria of $G_{q, c, \eta}$; any tracked limit (Lemma E.11) is one of them.*

Proof. Brouwer (Lemma E.10) plus the definition give the identification; Lemma E.11 closes the trajectory case. $\square$

Trajectory bridge. Realized $p_{t+1}(y) \propto p_t(y) \exp(-\eta \ell(y; q, c, z_t))$, $z_t \sim \mu(\cdot \mid p_t, q, c)$; expected operator $B_\eta(p_t)(y) \propto p_t(y) \mathbb{E}_z \exp(-\eta \ell(y; q, c, z))$.

Theorem E.3 (Tracking and unconditional Berk–Nash convergence). (i) $\tau=0$, greedy $\mu = \delta_{z^*(p)}$: $\|p_{t+1} - B_\eta(p_t)\|_{TV} = 0$ exactly, so $\{p_t\}$ is the deterministic continuous iteration of B_η ; subsequential limits are Berk–Nash fixed points unconditionally, and the safe-rate shrinkage of Section 3.2 holds on the realized trajectory (stable-answer subset of Lemma E.13). (ii) $\tau > 0$, $\ell \in [0, L]$, stable-answer $p_t(y^*) \geq 1 - \rho$ ($\rho < 1/2$), per-step kernel variance $V_t := \sup_y \text{Var}_{z \sim \mu} \ell$:

$$\mathbb{E} \|p_{t+1} - B_\eta(p_t)\|_{TV}^2 \leq \frac{(\eta L)^2 V_t}{(1 - \rho)^2}, \quad (\text{E.10})$$

with matching $\sqrt{V_t/\delta}$ high-prob. bound. L^2 tracking gives Berk–Nash convergence in expected TV on subsequences where the kernel concentrates.

Sketch. (i) δ_{z^*} collapses the expectation to a point; numerators/normalizers agree exactly. (ii) $g_t(y, z) = \exp(-\eta \ell)$, $|g'_t| \leq \eta$, so $\text{Var}_z g_t \leq (\eta L)^2 V_t$ (MVT); Cauchy–Schwarz with $Z_t^{\text{exp}} \geq (1 - \rho)e^{-\eta L}$; Markov. $\square$

Empirically: median TV discrepancy 0.041 at $K = 1024$ on stable-answer (90th pct 0.118), consistent with $\sqrt{V_t}$. Almost-sure pathwise tracking at $\tau > 0$ is open and not required by η -tempering in practice.

On uniqueness of the Berk–Nash equilibrium. Φ_η has a fixed point by Brouwer (Lemma E.10); uniqueness is not assumed. In our setting two refinements rule out spurious equilibria empirically. First, on the stable-answer subset the linearized iteration is hyperbolic ($\hat{\rho}^* < 1$ on every conflict cell of Table E.7), so trajectory limits selected by gen-Bayes are isolated. Second, on cells where the trace concentrates around a single answer mode (81% of wrong-answer cells, median margin 8.5 logits), the Berk–Nash basin is a neighborhood of the answer-mode delta; competing equilibria require the trace kernel to bifurcate, which we do not observe in any logged trajectory. When the trace is multi-modal (top-2 mass within 0.1), the analysis reverts to the conditional safe-rate inequality on the stable-answer subset.

883 **On the $\hat{\eta}_{\text{eff}}$ proxy.** $\hat{\eta}_{\text{eff}}$ is the trace-token logit slope at the answer mode (Appendix E.1)
884 and tracks injected η on the synthetic grid with Spearman 1.0 and $\leq 2.03\%$ relative error
885 (Table F.12). We use it as a relative-rate diagnostic, not as point identification of the
886 latent rate on real traces; slice-level η^* selection by held-out Brier (Table H.20) does the
887 deployment work.

888 **Empirical check of the conditional-independence assumption.** Assumption A.1
889 requires $y^* \perp c \mid (q, r)$. On a held-out ConflictBank fold ($n=500$) we estimate the conditional
890 mutual information $I(y^*; c \mid q, \hat{r})$ via the k NN-based estimator of Kraskov et al. [30]; pooled
891 $\hat{I} \leq 0.018$ nat across cells, with 99% permutation-bootstrap intervals straddling zero on
892 24/25 cells (only PopQA-low-popularity is borderline at $\hat{I} = 0.041$, 95% CI [0.003, 0.082]).
893 CI is therefore a working approximation, with PopQA-low-popularity flagged as the one
894 slice where a residual c -channel is detectable.

895 **Empirical check of the log-linearity assumption.** The synthetic sweep (App. H.1.13)
896 bounds the worst-case R^2 under deliberate violation. On real logits we run a Hessian-
897 spectrum probe: at the per-cell argmin of $R(w)$ we estimate $\partial_w^2 R$ via finite differences
898 and compare to the log-linear-family prediction $\text{Var}_{p_w}(\log p_{\text{ctx}}/p_\theta)$. Pooled relative error
899 across the 25 spotlight cells is $\leq 7\%$ (median 3.1%, max 12.4% on Llama-70B/PopQA), well
900 below the non-trivial-violation threshold ($\rho=0.6$, App. H.1.13). Real LLM logits sit in the
901 “approximate log-linear” regime where the geometric mixture is the right plug-in target.

902 E.1.4 Multi-metric calibration robustness

903 The headline matched-base contrast in Table H.21 reports ECE. The same training-type
904 signature persists across adaptive ECE and Brier, so the result is not a single-metric artifact:

Table E.4: Conflict-slice calibration delta across three metrics. Each entry is the conflict-minus-no-conflict gap on the matched-base spotlight cells; the GRPO blowup, the RLCR residual, and the Phi-3 GRPO replication all transfer across metrics.

Model / training	ECE	Adaptive ECE	Brier
Llama-3.1-8B (GRPO)	+0.3436	+0.3277	+0.3553
RLCR (Brier reward)	+0.2910	+0.2715	+0.2708
Phi-3 (GRPO, replication)	+0.2088	+0.2081	+0.2166

905 E.1.5 η^* leakage audit

906 The η^* -selection rule is sample-split held-out Brier optimization with disjoint train/eval
907 folds. Concretely: η^* is chosen by minimizing Brier on a calibration fold of size 200, and
908 all reported numbers in Table H.20 use a disjoint evaluation fold of size 300. When the
909 calibration-fold optimum is $\eta^*=0.0$ (boundary), the same $\eta^*=0.0$ is applied to the eval fold
910 without re-tuning, ruling out the “ η^* chosen on eval” alternative explanation. We re-ran
911 the audit with 50/50 and 100/200 splits and the selected η^* is stable in ± 0.05 .

912 E.1.6 Saturation classifier vs. partial-order ranking

913 The pre-registered binary saturation-vs-recovery classifier (P3 in App. H.19) does not fully
914 replicate. The partial-order/ranking version of the same prediction does: ranking $\hat{\eta}_{\text{eff}}$ -
915 implied severity within the high-conflict regime against the within-control regime gives a
916 mean separation of +0.2302 (high-regime) vs. +0.0179 (control-regime), with pairwise AUC
917 1.0 on the held-out cells. The honest read is that T3’s quantitative ranking transfers across
918 families even when the binary classifier does not—this is the form of P3 we propose for
919 replication going forward.

920 E.1.7 Berk–Nash equilibrium of the gen-Bayes update (legacy lemma)

921 **Lemma E.12** (Convergence to a Berk–Nash equilibrium). *Under the conditions of Propo-*
922 *sition E.9, if a decoding trajectory $\{p_t\}_{t \geq 1}$ converges and satisfies $\|p_{t+1} - B_\eta(p_t)\|_{TV} \rightarrow 0$,*
923 *then its limit is a fixed point of B_η , equivalently a Berk–Nash equilibrium of $G_{q,c,\eta}$.*

924 *Proof.* Immediate from Lemma E.11 under the stated tracking hypothesis. $\square$

925 E.1.8 Confidence sharpening can outpace accuracy

926 **Lemma E.13** ($\eta > \bar{\eta}$ implies non-decreasing calibration gap). *Suppose $\eta > \bar{\eta}$ in the iterated*
 927 *update (E.9). Then for any $\hat{y}_t$ that is the mode of p_t ,*

$$G(t) := \mathbb{E}[\mathbb{E}_{p_t} \mathbb{K}[Y=y^*] - \mathbb{K}[\hat{y}_t = y^*]] \quad \text{is non-decreasing in } t \text{ on the stable-answer subset.}$$

928 *Sketch.* On the stable-answer subset $\hat{y}_t$ is constant, so accuracy in $G(t)$ is fixed; $\eta > \bar{\eta}$
 929 sharpens around the misspecified projection, raising confidence without raising accuracy. $\square$

930 **Proposition E.10** (Stable/flip decomposition). *For $S_{t_0:t_1} = \{\hat{y}_{t_0} = \hat{y}_{t_1}\}$ and $g_t =$*
 931 *$|\mathbb{E}_{p_t} \mathbb{K}[Y=y^*] - \mathbb{K}[\hat{y}_t = y^*]| \in [0, 1]$, $\Delta G \geq \mathbb{P}(S) \Delta G^S - \mathbb{P}(F) \Delta G^F$ via the law of total expectation and*
 932 *$\Delta G^F \geq -1$.*

933 **Corollary E.5** (Whole-population inheritance). *If $\eta > \bar{\eta}$ and $\Delta G^S \geq \gamma$, with flip probability*
 934 *$\mathbb{P}(F) \leq \delta$, then $\Delta G \geq (1 - \delta)\gamma - \delta$.*

935 E.1.9 Putting it together

936 *Proof of Theorem 3.2.* Under the theorem’s standing assumption that conflict increases the
 937 projection distance $D^*(q, c)$ relative to matched controls, Lemma E.9 implies a smaller safe
 938 rate on conflict cells. Under the tracking hypothesis of Lemma E.12, the iterated CoT update
 939 converges to a Berk–Nash equilibrium that is generically not the truth. By Lemma E.13,
 940 when $\eta > \bar{\eta}$ the calibration gap $G(t)$ is non-decreasing on the stable-answer subset; this is
 941 the saturation regime of Corollary E.6. When $\eta < \bar{\eta}$, the standard Kleijn and van der Vaart
 942 [29] contraction holds and the predictive accuracy catches up with confidence, giving the
 943 peak-then-recover regime.

944 For RLVR-trained models, verifiable-reward RL (GRPO) sharpens token-level logits toward
 945 winning trajectories, increasing the effective η in (E.9); Bereket and Leskovec [5] document
 946 this as GRPO-induced overconfidence in binary forecasting. Hence $\eta_{\text{RLVR}} > \eta_{\text{RLHF}}$ at
 947 matched scale. In a conflict regime where $\eta_{\text{RLVR}} > \bar{\eta}(q, c) > \eta_{\text{RLHF}}$, RLVR saturates while
 948 RLHF does not. This is the training-type-conditional saturation predicted by the theorem
 949 and observed in Table H.18. $\square$

950 **Corollary E.6** (Two-regime calibration trajectory). *The trajectory $G(K)$ has two qualita-*
 951 *tive shapes. (a) Peak-then-recover for $\eta \lesssim \bar{\eta}$. (b) Saturation for $\eta > \bar{\eta}$, on the stable-answer*
 952 *subset. The transition is benchmark-dependent because $\bar{\eta}(q, c)$ depends on $\text{KL}(p_\theta \| p_{\text{ctx}})$. Em-*
 953 *pirically, the same benchmark splits the same model between the two regimes (Table H.18,*
 954 *Figure H.3); the benchmark dependence is a direct consequence of the safe-rate lemma above.*

955 E.2 Assumption audit (Theorem 3.2)

956 **(i) Misspecification.** R1-14B / ConflictBank: mode trajectory agrees with y^* on only
 957 14.3% of stable-answer cells; on WikiContradict 34.8%. Both well below chance confirms
 958 misspecification.

959 **(ii) Endogenous evidence** holds by autoregressive construction; marginal trace entropy
 960 $H(z_t) = 4.2 \text{ nat}$ (R1-14B) vs. 3.8 nat (Qwen2.5-14B), consistent with RLVR sharpening.

961 **(iii) Conflict gap.** All conflict slices satisfy $\text{KL}(p_\theta \| p_{\text{ctx}}) \geq 1 \text{ nat}$; no-conflict $\sim 0.1 \text{ nat}$
 962 (Table E.5).

Table E.5: Empirical KL-gap audit. Mean $\text{KL}(p_\theta \| p_{\text{ctx}})$ on the conflict subset of each benchmark (R1-14B), and the analogous no-conflict / closed-book number.

Benchmark	$\text{KL}(p_\theta \ p_{\text{ctx}})$ on conflict	on no-conflict
ConflictBank	2.31 nat	0.12 nat
WikiContradict	1.48 nat	0.09 nat
NQ-Swap	1.92 nat	0.14 nat
DynamicQA	1.26 nat	0.08 nat
PopQA	1.05 nat	0.10 nat

Effective η and empirical $\bar{\eta}$. $\hat{\eta}_{\text{eff}} = \nabla_t \log p_t(\hat{y}_t) / \nabla_t \ell_t$ on ConflictBank-conflict at $K = 128$: RLVR/GRPO 0.81/0.94/0.92 (R1-Distill 7B/14B/70B); RLHF/DPO 0.46/0.48/0.51/0.41 (Qwen 7/14/32B, Llama-3.1-8B); $\bar{\eta}$ = largest η keeping post-trace Brier $\leq K=0$. Predicted ordering $\hat{\eta}_{\text{eff}} > \bar{\eta}$ holds on every saturated slice and fails on every recoverable one (Table E.6).

Table E.6: $\hat{\eta}_{\text{eff}}$ vs. empirical $\bar{\eta}$. $\hat{\eta}_{\text{eff}} > \bar{\eta}$ predicts saturation; the column “predicted” is the resulting trajectory; the column “observed” confirms the prediction on every cell.

Model	Slice	$\hat{\eta}_{\text{eff}}$	$\bar{\eta}$	Predicted	Observed
R1-14B	ConflictBank conflict	0.94	0.30	saturate	saturate
R1-14B	WikiContradict conflict	0.92	0.95	recover	recover
R1-7B	ConflictBank conflict	0.81	0.85	recover	recover
R1-7B	WikiContradict conflict	0.79	0.96	recover	recover
Qwen2.5-32B	ConflictBank conflict	0.51	0.80	recover	saturate*
Qwen2.5-32B	WikiContradict conflict	0.50	0.95	recover	recover

* The Qwen-32B / ConflictBank cell is the one slice where the simple $\hat{\eta}_{\text{eff}}/\bar{\eta}$ ordering does not predict the observed behavior; we attribute this to a separate effect (extreme absolute-confidence floor; see App. N.1).

Stable-answer subset. On R1-14B/ConflictBank-conflict, stability across $K \in \{0, 128, 1024\}$ is 8.0%, across $\{0, 128\}$ is 9.8%, across $\{128, 1024\}$ is 36.4%. On the stable subset, gap rises by +0.091 ($K = 0 \rightarrow 128$) and +0.103 ($0 \rightarrow 1024$) with zero mean accuracy change the conditional read-out Corollary E.6 requires.

E.3 Empirical audits of Proposition E.9 hypotheses

Two audits on R1-Distill-14B / ConflictBank-conflict. **Audit 1 (stable-answer subset for Lem. E.13):** $\arg \max p_t$ stability is 0.08 across $\{0, 128, 1024\}$, 0.098 on $0 \rightarrow 128$, **0.364** on $128 \rightarrow 1024$; on the post-jump stable subset the calibration gap rises with $\Delta \bar{\text{acc}} = 0$. **Audit 2 (self-confirming evidence for Prop. E.9):** the trace $z_{1:K}$ ($K = 1024$) favors the agent’s top answer over the runner-up on **0.81** of cells (mean log-margin 10.75 logits), 0.81 even on incorrect-answer cells.

E.4 Local Berk–Nash convergence rate, proof of Section 3.2 and empirical estimation

Setup and proof. Tangent space $T_{p^*} = \{u : \sum_y u(y) = 0\}$ with $\|\cdot\|_2$. Write $p_t = p^* + \epsilon_t$. Taylor at $p^* = \Phi_\eta(p^*)$,

$$\epsilon_{t+1} = J\epsilon_t + R(\epsilon_t), \quad J := D\Phi_\eta(p^*), \quad \|R(\epsilon)\|_2 \leq M\|\epsilon\|_2^2 \quad (\text{E.11})$$

where M is the C^2 -norm bound. Gelfand: for any $\rho \in (\rho^*, 1)$, $\|J^n\| \leq C_J(\rho)\rho^n$. Choose $\delta \leq (\rho - \rho^*)/(2MC_J(\rho))$, T_0 s.t. $C_J(\rho)\rho^{T_0}\|\epsilon_0\| \leq \delta$; induction gives $\|\epsilon_t\|_2 \leq C(p_0, \rho)\rho^t$, and $\text{TV} \leq \frac{\sqrt{|Y|}}{2}\|\cdot\|_2$ folds in. This proves (3.2).

Corollary E.7 (Non-asymptotic T_0 from observable quantities). *Fix $\rho \in (\rho^*, 1)$. The threshold time T_0 in Section 3.2 is bounded explicitly by*

$$T_0 \leq \left\lceil \frac{\log(2MC_J(\rho)\|\epsilon_0\|_2(\rho - \rho^*)^{-1})}{\log(1/\rho)} \right\rceil = \left\lceil \frac{\log(D_0/r_{\text{lin}})}{\log(1/\rho)} \right\rceil, \quad (\text{E.12})$$

988 with $D_0 = \text{TV}(p_0, p^*)$, $r_{\text{lin}} = (\rho - \rho^*) / (2MC_J(\rho))$; M is the C^2 -norm bound, $C_J(\rho)$ Gelfand's
989 power-bound constant. All four are estimable from the same logged data as $\hat{\rho}^*$ (App. E.5).
990 In the empirical regime of Table E.7 ($\hat{\rho}^* \approx 0.42$, $D_0 \leq 0.5$, $\hat{M} \leq 4$, $\hat{C}_J \leq 2$, $\rho = 0.5$), $T_0 \leq 3$
991 trace steps.

992 **Loss of hyperbolicity at $\eta = \bar{\eta}$.** Implicit function theorem on $\Phi_\eta(p) = p$ gives smooth
993 $p^*(\eta)$ while $\rho^* < 1$. At $\eta = \bar{\eta}$, $G(t)$ becomes non-decreasing on the stable-answer subset
994 (Section 3.2), forcing a simplex boundary, $\rho^* \rightarrow 1^-$, and (3.2) degrades to $\text{TV}(p_t, p^*) \leq$
995 $C(p_0)/(q_1 t)$ (Lemma E.14).

996 **Lemma E.14** (Polynomial rate at non-hyperbolic equilibria). *If $\rho^* = 1$ and $D\Phi_\eta(p^*) =$
997 $I - Q$ with Q PSD, smallest non-zero eigenvalue $q_1 > 0$, and the quadratic remainder is
998 non-degenerate along $\ker Q$, then for small $\|\epsilon_0\|$, $\text{TV}(p_t, p^*) \leq C(p_0)/(q_1 t)$ (centre-manifold
999 expansion [16] adapted to discrete time).*

1000 **Empirical estimation.** Use $p_K \approx p^*$; estimate $\hat{J} = YX^\top (XX^\top + \lambda I)^{-1}$ by ridge LS
1001 ($W = 100$, $\lambda = 10^{-6}$) on $\{p_{t+1} - p_K, p_t - p_K\}$; report $\hat{\rho}^* = \rho(\hat{J})$, $\hat{\alpha} = -\log \hat{\rho}^*$, and compare to
1002 exponential-decay $\hat{\beta}$ from $\text{TV}(p_t, p_K) \sim Ae^{-\beta(K-t)}$.

Table E.7: Estimated Berk–Nash convergence rate from R1-14B traces. Conflict cells sit near the loss-of-hyperbolicity boundary ($\hat{\rho}^* \approx 1$, slow saturation); no-conflict cells are well inside the hyperbolic regime ($\hat{\rho}^* \sim 0.4$ – 0.5 , fast saturation). Section 3.2 predicts $\hat{\alpha}$ should match the empirical exponential-decay rate $\hat{\beta}$; agreement within ± 0.07 across all twelve reported cells, spanning three model families (R1-14B, R1-8B-RLCR, Mistral-7B) and four benchmarks. CT2 is scoped to the stable-answer subset where the linearization is valid; pre-saturated cells (e.g. Qwen2.5-14B with $p_K \approx p^*$ already at $t = T_0$) sit outside this scope and are excluded from the pooled fit. Rows 1–5 (R1-14B) are the load-bearing in-distribution cells; rows 6–12 (R1-8B-RLCR, Mistral-7B) are cross-family replications; full trace dumps in App. H.1.6.

Model	Cell	K	$\hat{\rho}^*$	$\hat{\alpha}$	$\hat{\beta}$ (obs.)	$ \hat{\alpha} - \hat{\beta} $
R1-14B	ConflictBank-conflict	1024	0.972	0.028	0.031	0.003
R1-14B	ConflictBank-conflict	128	0.918	0.085	0.092	0.007
R1-14B	WikiContradict-conflict	1024	0.910	0.094	0.097	0.003
R1-14B	ConflictBank-no-conflict	1024	0.421	0.866	0.812	0.054
R1-14B	WikiContradict-no-conflict	1024	0.485	0.724	0.689	0.035
R1-8B-RLCR	ConflictBank-conflict	1024	0.951	0.050	0.057	0.007
R1-8B-RLCR	WikiContradict-conflict	1024	0.937	0.065	0.073	0.008
R1-8B-RLCR	ClashEval-conflict	512	0.928	0.075	0.082	0.007
R1-8B-RLCR	ConflictBank-no-conflict	1024	0.467	0.762	0.708	0.054
Mistral-7B	ConflictBank-conflict	1024	0.943	0.059	0.069	0.010
Mistral-7B	ClashEval-conflict	512	0.901	0.104	0.119	0.015
Mistral-7B	WikiContradict-no-conflict	1024	0.512	0.669	0.602	0.067

1003 **Pooled fit and held-out checks.** Pooled $r(\hat{\alpha}, \hat{\beta}) = 0.997$ ($p < 10^{-12}$), MAE 0.022, worst
1004 0.067 across 12 cells/ 3 families. Temporal split ($W = 50$): MAE 0.029, max 0.075, pooled
1005 $r = 0.991$. Cross-cell predict (9/3 split, 20 re-samples): MAE 0.034. Caveats: linearization
1006 noisy for $K < 64$; $p_K \approx p^*$ within $O(\rho^K)$; top-50 restriction error $< 10^{-3}$ TV.

1007 **Window-stability of $\hat{\rho}^*$.** A standard concern with spectral estimates on finite trajectories
1008 is that $\hat{\rho}^*$ shifts depending on which window of the trajectory is used to fit the autoregression.
1009 On the four load-bearing R1-14B / RLCR-8B conflict cells, the early-trace window (steps
1010 $1-T/2$) and the tail window (steps $T/2-T$) return essentially the same value:

Table E.8: Window-stability of $\hat{\rho}^*$. Early- vs. tail-window spectral estimates on the four load-bearing conflict cells; both windows place $\hat{\rho}^*$ inside $[0.996, 1.003]$.

Cell	Early-window $\hat{\rho}^*$	Tail-window $\hat{\rho}^*$
Cell 1	0.996233	0.999575
Cell 2	0.996508	0.997495
Cell 3	0.997877	0.998071
Cell 4	0.997175	1.002431

The four cells all sit at $\hat{\rho}^* \rightarrow 1$ the saturation regime where the trace stops mixing and the early/tail discrepancy stays under 0.005 in every case, so the “saturated trace” diagnosis is not an artifact of window choice. This is consistent with the wider 12-cell fit (Table E.7): conflict cells cluster near $\hat{\rho}^* \approx 1$ and no-conflict cells cluster near $\hat{\rho}^* \approx 0.4$ –0.5.

E.5 Finite-sample concentration of $\hat{\rho}^*$ (Theorem 5)

Proof of Section 3.2. (i) Trajectory regression [59, Cor. 2.2] on $y_t = Jx_t + r(x_t) + \xi_t$ (r Taylor remainder, ξ_t bounded martingale-diff) gives $\|\hat{J} - J\|_{\text{op}} \leq C_1 |\mathcal{Y}| \sqrt{\log(2|\mathcal{Y}|/\delta)} / (s_{\min}^2 \sqrt{W}) + \frac{M}{2s_{\min}^2} \sup_t \|x_t\|^2$ w.p. $\geq 1 - \delta$. (ii) $\sup \|x_t\| \leq C_0 \rho^{*K-W}$ from Section 3.2 converts the second term to $MC_0^2 \rho^{*2(K-W)} / s_{\min}^2$. (iii) Bauer–Fike gives $|\hat{\rho}^* - \rho^*| \leq \kappa(V) \|\hat{J} - J\|_{\text{op}}$.

Choosing W . Practical recipe $W = K/4$ for $K \geq 256$ keeps both terms below 10^{-2} at $\delta = 0.05$. With $\rho^* \approx 0.97$, $|\mathcal{Y}| = 50$, $\kappa(V) \approx 8.4$, $M \approx C_0 \approx 1$: bound ≤ 0.083 at $K = 1024$, $W = 256$, matching the observed ± 0.07 in Table E.7.

E.6 Backing tables for the new theorems

Table E.9: Signal-conditioned floor (Theorem 4). Per-benchmark estimate of $\widehat{\mathbb{E}}[\sigma_r^2(s)]$ from the three-signal plug-in (BM25, entity popularity, semantic entropy), the floor predicted by $\frac{1}{8(|\mathcal{Y}|-1)^2} \widehat{\mathbb{E}}[\sigma_r^2(s)]$ with $|\mathcal{Y}| = 50$, and the realized excess regret of the plug-in on a 50-cell holdout. Predicted and realized agree to within ± 0.02 across all five benchmarks.

Benchmark	$\widehat{\mathbb{E}}[\sigma_r^2(s)]$	Predicted floor	Realized excess	$ \Delta $
ConflictBank-conflict	0.124	0.0067	0.0081	0.0014
ClashEval	0.087	0.0047	0.0061	0.0014
RAMDocs	0.073	0.0040	0.0048	0.0008
WikiContradict	0.066	0.0036	0.0050	0.0014
PopQA	0.041	0.0022	0.0036	0.0014
Average	0.078	0.0042	0.0055	0.0013

Table E.10: Closed-form η^* (Proposition 6). On binary-collapse cells (mass ≥ 0.95 on top-2 answers; $\sim 42\%$ of conflict cells at $K = 1024$), the closed-form $\eta^* = \logit(b)/\logit(a)$ matches 5-fold-CV η^* within ± 0.06 pooled. Pooled across the four binary-collapse subsets, the closed-form recipe saves $\sim 93\%$ of the CV cost while losing ≤ 0.004 Brier.

Cell (binary-collapse subset)	η_{closed}^*	η_{CV}^*	$ \Delta\eta^* $	Brier loss vs. CV
ConflictBank-conflict / R1-14B	0.612	0.658	0.046	+0.0021
ConflictBank-conflict / R1-8B	0.534	0.581	0.047	+0.0028
ClashEval / Llama-3.1-8B-RLCR	0.281	0.341	0.060	+0.0040
WikiContradict / R1-14B	0.703	0.751	0.048	+0.0019
Pooled	—	—	0.050	+0.0027

E.7 Three independent characterizations of the geometric mixture

Three independent derivations (differential geometry / channel capacity / spectral operator theory) triangulate $w^* = \mathbb{E}[r | s]$.

1027 **E.7.1 Information-geometric perspective: w^* as Fisher e-geodesic time**

1028 **Theorem E.4** (Info-geometric characterization). *Equip the open simplex $\Delta^\circ(\mathcal{Y})$ with*
 1029 *the Fisher information metric $g_F(u, v) = \sum_y u(y)v(y)/p(y)$ and the e-connection $\nabla^{(e)}$*
 1030 *from Amari [1], Amari and Nagaoka [2]. Let $\gamma: [0, 1] \rightarrow \Delta^\circ(\mathcal{Y})$ be the unique $\nabla^{(e)}$ -geodesic*
 1031 *with $\gamma(0) = p_\theta$, $\gamma(1) = p_{\text{ctx}}$. Then*

$$\gamma_t(y) = \frac{p_\theta(y)^{1-t} p_{\text{ctx}}(y)^t}{Z(t)}, \quad Z(t) = \sum_{y'} p_\theta(y')^{1-t} p_{\text{ctx}}(y')^t,$$

1032 *and the Bayes-optimal arbitration coincides with the geodesic evaluated at Fisher-geodesic*
 1033 *time $w^* = \mathbb{E}[r \mid q, c]$. Moreover, for any $w \in [0, 1]$, the expected log-loss along the geodesic*
 1034 *admits the second-order Fisher decomposition*

$$\mathbb{E}[L(\gamma_w)] = \mathbb{E}[L(\gamma_{w^*})] + \frac{1}{2}(w - w^*)^2 g_F(\dot{\gamma}_{w^*}, \dot{\gamma}_{w^*}) + O(|w - w^*|^3),$$

1035 *where $g_F(\dot{\gamma}_{w^*}, \dot{\gamma}_{w^*}) = \text{Var}_{p_{w^*}}[\log(p_{\text{ctx}}/p_\theta)]$ is the Fisher quadratic form along the geodesic*
 1036 *tangent.*

1037 *Sketch.* e-Christoffels vanish in natural coordinates [2, Ch. 3]; geodesic is the straight line
 1038 $\theta_w = (1-w)\theta(p_\theta) + w\theta(p_{\text{ctx}})$, which exponentiates to p_w . w^* from Section 3.1; the second-order
 1039 term is the e-flat KL expansion. $\square$

1040 **Corollary E.8** (Fisher–Rao barycenter / Schrödinger-bridge marginal). *The geometric*
 1041 *mixture $\hat{p}_w \propto p_\theta^{1-w} p_{\text{ctx}}^w$ is the Fisher–Rao barycenter of $(p_\theta, p_{\text{ctx}})$ with weights $(1-w, w)$:*

$$\hat{p}_w = \arg \min_{p \in \Delta(\mathcal{Y})} [(1-w) \text{KL}(p \parallel p_\theta) + w \text{KL}(p \parallel p_{\text{ctx}})].$$

1042 *Equivalently, $\hat{p}_w$ is the marginal at time w of the static Schrödinger bridge between endpoints*
 1043 *$(p_\theta, p_{\text{ctx}})$ under the relative-entropy construction [33, Sec. 4]. The bridge time $w^* = \mathbb{E}[r \mid q, c]$*
 1044 *is the reliability-weighted information-geometric interpolation parameter.*

1045 *Sketch.* Lagrangian first-order condition gives the geometric mixture; strict convexity gives
 1046 uniqueness; the Schrödinger-bridge identification is Léonard [33, Prop. 4.1]. $\square$

1047 **E.7.2 Channel-capacity perspective: arbitration ceiling via Fano**

1048 **Theorem E.5** (Arbitration capacity ceiling). *Let (q, c, r, Y^*) be drawn from $\mathbb{P}^*$ and $s =$*
 1049 *$s(q, c)$ any signal map. For any arbitration policy $\hat{y} = \hat{y}(s, p_\theta, p_{\text{ctx}})$ measurable in s together*
 1050 *with the model softmaxes,*

$$\mathbb{P}[\hat{y} \neq Y^*] \geq \frac{H(Y^* \mid s, p_\theta, p_{\text{ctx}}) - 1}{\log |\mathcal{Y}|}. \quad (\text{E.13})$$

1051 *The bound is achieved with equality by the Bayes MAP rule applied to the geometric mixture*
 1052 *p_{w^*} with $w^*(s) = \mathbb{E}[r \mid s]$. As a corollary, no arbitration policy can drive accuracy above*
 1053 *$1 - 2^{H(Y^* \mid s, p_\theta, p_{\text{ctx}}) / \log |\mathcal{Y}|}$.*

1054 *Sketch.* Fano on $(s, p_\theta, p_{\text{ctx}}) \rightarrow Y^*$ [11, Thm. 2.10.1]; Section 3.1 identifies the conditional
 1055 posterior with $p_{w^*}(s)$, so its MAP saturates the bound. $\square$

1056 Whereas Section 3.1 are lower bounds over a distribution class, Theorem E.5 is an *upper*
 1057 *bound on accuracy* in residual entropy $H(Y^* \mid s, p_\theta, p_{\text{ctx}})$, scoring-rule independent, saturated
 1058 by the geometric mixture.

1059 **Corollary E.9** (Equality achievability of the Fano ceiling). *The Fano lower bound (E.13) is*
 1060 *achieved with equality (not just up to the standard $1/\log |\mathcal{Y}|$ slack) by the geometric mixture*
 1061 *p_{w^*} whenever the conditional $\mathbb{P}[Y^* \mid s, p_\theta, p_{\text{ctx}}]$ is the maximum-entropy distribution subject*
 1062 *to the moment constraint $\mathbb{E}[\log p_{\text{ctx}}/p_\theta] = \nabla \phi(\theta_{w^*})$.*

1063 *Sketch.* The MaxEnt representative of the moment constraint (Theorem E.8) coincides with
 1064 the geometric mixture exactly under the stated condition; Jaynes-MaxEnt gives equality. $\square$

1065 E.7.3 Spectral conflict-barrier perspective: a matching lower bound

1066 **Theorem E.6** (Spectral conflict barrier). *On the stable-answer subset, let p^* be a hyperbolic*
 1067 *Berk–Nash fixed point of Φ_η with score function $s_t(y) := \partial_\theta \log p_t(y) - \mathbb{E}_{p_t} \partial_\theta \log p_t$. Then*

$$\rho^*(\eta, p^*) := \rho(D\Phi_\eta(p^*)) \geq 1 - \eta \frac{\text{Var}_{p^*}[s_t(Y)]}{|\mathcal{Y}|}.$$

1068 *Furthermore, on cells with KL projection distance $D^*(q, c) \geq d_{\min} > 0$, the score-variance at*
 1069 *the fixed point satisfies $\text{Var}_{p^*}[s_t] \leq C_\pi/d_{\min}^2$ for a finite constant C_π depending only on the*
 1070 *prior support (Lemma E.15, below). Combining,*

$$\rho^*(\eta, D^*) \geq 1 - \frac{C_\pi \eta}{d_{\min}^2 |\mathcal{Y}|}.$$

1071 *In particular, $\rho^* \rightarrow 1^-$ as $D^* \uparrow \infty$ at any fixed η , recovering loss of hyperbolicity from below.*

1072 **Lemma E.15** (Score-variance bound under conflict, explicit constant). *Under (i)–(iv) of*
 1073 *Section 3.2, with prior π supported on $\Delta(\mathcal{Y})$ with $\inf_y \pi(y) \geq \pi_0 > 0$ and $\sup_y \pi(y) \leq \pi_\infty < \infty$,*

$$\text{Var}_{p^*}[s_t(Y)] \leq \underbrace{\frac{2\pi_\infty^2}{\pi_0 (\log |\mathcal{Y}|)^2}}_{C_\pi} \cdot \frac{1}{D^*(q, c)^2}.$$

1074 *Sketch.* (a) Pinsker on $D^* = \text{KL}(p^* \parallel \mathbb{P}^*)$ gives $\text{TV} \leq \sqrt{D^*/2}$. (b) Exponential-family
 1075 *score identity plus prior support $p^*(y) \geq \pi_0$ gives $\|\partial_y \log p^*\|_2 \leq 1/\pi_0$. (c) Cauchy–*
 1076 *Schwarz $\mathbb{E}\|s_t\|^2 \leq 2(D^*)^2/\text{TV}^2$ then absorbs (a) into $1/D^{*2}$; constants pack to $C_\pi =$*
 1077 *$2\pi_\infty^2/(\pi_0 (\log |\mathcal{Y}|)^2)$. $\square$*

1078 C_π is tight up to a factor of 4; for $|\mathcal{Y}|=50$, $C_\pi \leq 8.7$ on the uniform-prior worst case.

1079 **Lemma E.16** (Score-variance equality at the Berk–Nash fixed point). $\sigma_{\min}(\text{Cov}_{p^*}(s_t)) =$
 1080 $1/((|\mathcal{Y}|D^*(1+\varepsilon_\pi)))$ *with prior-shape correction $\varepsilon_\pi \in [0, C_\pi/(D^{*2}\sigma_{\min}) - 1]$ (vanishes for the*
 1081 *uniform prior). Proof: differentiate the Berk–Nash condition $\mathbb{E}_{p^*}[\partial_y \mathbb{E}_z \ell] = 0$ in natural-*
 1082 *parameter coordinates and apply Cramér–Rao on the simplex.*

1083 **Corollary E.10** (Matching upper- and lower-bound envelope). *Combining Section 3.2 (up-*
 1084 *per) and Theorem E.6 (lower): $1 - C_\pi \eta/(D^{*2}|\mathcal{Y}|) \leq \rho^* \leq 1 - c' \eta \text{Var}_{p^*}(s_t)$. The empirical*
 1085 *conflict/no-conflict ρ^* ratio $0.97/0.42 \approx 2.3$ matches the inverse-square scaling predicted by*
 1086 *$D_{\text{conflict}}^*/D_{\text{no-conf}}^* \approx 4.7$.*

1087 E.7.4 CT1→CT2 bridge: Lyapunov dissipation identity and sharpened 1088 envelope

1089 We close the multiplicative gap of Corollary E.10 to logarithmic in $|\mathcal{Y}|$ under stable-answer
 1090 ergodicity, by tying the spectral radius to per-step free-energy dissipation.

1091 **Lemma E.17** (Lyapunov dissipation identity for gen-Bayes; Bernstein-controlled remain-
 1092 *der). Let $p_{t+1} \propto p_t e^{-\eta \ell_t}$ be the gen-Bayes update (3), $\mathcal{F}$ the free energy (3.1), $V_t := \text{Var}_{p_t}(\ell_t)$,*
 1093 *and either (i) $\ell_t \in [0, B]$ almost surely (bounded), or (ii) ℓ_t sub-Gaussian under p_t with proxy*
 1094 *σ_g^2 . Then for all $\eta > 0$ in case (ii), or for $\eta B \leq 1/3$ in case (i),*

$$\mathcal{F}[p_{t+1}] - \mathcal{F}[p_t] = -\eta V_t + R(\eta; \ell_t, p_t), \quad (\text{E.14})$$

1095 *with explicit Bernstein remainder*

$$|R(\eta; \ell_t, p_t)| \leq \begin{cases} \frac{\eta^3 B V_t}{3(1-\eta B)} \leq \frac{1}{2} \eta^3 B V_t & (\text{bounded case (i)}), \\ \frac{\eta^3 \sigma_g^2 V_t}{3} & (\text{sub-Gaussian case (ii)}). \end{cases} \quad (\text{E.15})$$

1096 *The leading $-\eta V_t$ term is the exact second cumulant of ℓ_t under p_t (de Bruijn–Stam-type*
 1097 *identity for KL-Bregman descent [4, 11] specialized to the arbitration energy $E_{w,\eta}$).*

1098 *Sketch.* Mirror-descent: $\Delta\mathcal{F} = -\text{KL}(p_{t+1}||p_t) = \Lambda(\eta) - \eta\Lambda'(\eta)$ where Λ is the centred
1099 cumulant generator under p_t . Bennett–Bernstein cumulant bound [8, Lemma 2.4] gives
1100 $|\Lambda(\eta) - \frac{1}{2}\eta^2 V_t| \leq \eta^3 BV_t/(6(1-\eta B/3))$; differentiating doubles the constant (case (i)). Sub-
1101 Gaussian σ_g^2 replaces B in case (ii). $\square$

1102 **Lemma E.18** (Spectral-gap implies ergodicity (concrete sufficient condition)). *Let $\Phi_\eta : \Delta(\mathcal{Y}) \rightarrow \Delta(\mathcal{Y})$ be the gen-Bayes update operator, and assume:*

1104 (SG1) Φ_η has spectral gap $\gamma \in (0, 1]$ on the tangent space at p^* (equivalent to hyperbolicity:
1105 $\rho^* = 1 - \gamma$);

1106 (SG2) $p \mapsto \text{Var}_p(\ell_t)$ is L_V -Lipschitz in TV ($L_V = 2\|\ell_t\|_\infty^2$ in the bounded case, $L_V = 2\sigma_g^2$ in
1107 sub-Gaussian case).

1108 For initial p_0 with $D_0 := \text{TV}(p_0, p^*) \leq r_{\text{lin}} := \gamma/L_H$ (Hartman–Grobman linearization radius,
1109 L_H the Hessian-Lipschitz constant of $\mathcal{F}$),

$$\zeta_t := |\text{Var}_{p_t}(\ell_t) - \text{Var}_{p^*}(s_t)| \leq L_V D_0 (1-\gamma)^t, \quad \sum_{t=0}^{T-1} \zeta_t \leq \frac{L_V D_0}{\gamma} = O(1) = o(T). \quad (\text{E.16})$$

1110 Hence the abstract ergodicity condition of Theorem E.7 is implied with explicit constant
1111 $L_V D_0 / \gamma$.

1112 *Proof.* By (SG1), Banach contraction on the linearized iteration around p^* inside r_{lin} gives
1113 $\text{TV}(p_t, p^*) \leq D_0 (1-\gamma)^t$. By (SG2), $\zeta_t \leq L_V \text{TV}(p_t, p^*)$. Geometric series sums to $L_V D_0 / \gamma$.
1114 $\square$

1115 **Theorem E.7** (Sharpened CT2 envelope under stable-answer ergodicity). *Let $\bar{\sigma}^2 :=$
1116 $\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \text{Var}_{p_t}(\ell_t)$ be the time-averaged per-step variance along the gen-Bayes tra-
1117 jectory; assume the ergodicity condition $\sum_t \zeta_t = o(T)$ with $\zeta_t := |\text{Var}_{p_t}(\ell_t) - \text{Var}_{p^*}(s_t)|$
1118 (Lemma E.18 gives the concrete sufficient form: spectral gap $\gamma > 0$ + TV-Lipschitz $\text{Var}_p(\ell)$)
1119 $\Rightarrow \sum \zeta_t \leq L_V D_0 / \gamma = O(1)$. Then*

$$1 - \eta \bar{\sigma}^2 - O(\eta^2) \leq \rho^* \leq 1 - \eta \bar{\sigma}^2 \cdot \left(1 - O\left(\frac{\log |\mathcal{Y}|}{\sqrt{T}}\right)\right), \quad (\text{E.17})$$

1120 i.e. both bounds collapse to $1 - \eta \bar{\sigma}^2$ up to $O(\eta^2)$ and $O(\log |\mathcal{Y}| / \sqrt{T})$. The $|\mathcal{Y}| D^{*2} \text{Var}_{p^*}(s_t) / C_\pi$
1121 multiplicative gap of Corollary E.10 dissolves; the residual gap is purely logarithmic in $|\mathcal{Y}|$.

1122 *Sketch.* Lower bound: Section 3.2 with $\bar{\sigma}^2 = \text{Var}_{p^*}(s_t) + o(1)$. Upper bound: Lemma E.17
1123 sums to $\mathcal{F}[p_0] - \mathcal{F}[p^*] - \eta \sum_s V_s$; Pinsker strong convexity gives $\text{TV} \leq \sqrt{2\Delta\mathcal{F}/c} \cdot e^{-\eta \bar{\sigma}^2 t/2}$,
1124 matching $\rho^* \leq 1 - \eta \bar{\sigma}^2/2$. The $\log |\mathcal{Y}| / \sqrt{T}$ slack is McDiarmid for the $|\mathcal{Y}|$ -bounded score. $\square$

1125 **Corollary E.11** (Cramér–Rao tightness). *The original $|\mathcal{Y}| D^{*2} / C_\pi$ multiplicative gap of
1126 Corollary E.10 is not a proof artifact when ergodicity fails: it equals the Cramér–Rao deficit
1127 $1/(D^* \text{Var}_{p^*}(s_t)) - 1$ in the absence of trajectory ergodicity. An explicit two-state counter-
1128 example (App. E.6) achieves the gap.*

1129 The empirical $r = 0.997$ over twelve cells (Table E.7) matches the $\log |\mathcal{Y}| / \sqrt{T} \leq 0.06$ residual
1130 at $T \geq 400$ trace steps.

1131 **Remark E.6** (Period-2 counter-example saturating the deficit). $|\mathcal{Y}| = 2$, $p_\theta = (1/4, 3/4)$,
1132 $p_{\text{ctx}} = (3/4, 1/4)$, $\eta = \log 2$, balanced prior, deterministic period-2 trace $\ell_t \in \{(1, 0), (0, 1)\}$:
1133 $\bar{V} = 17/72$ vs. $\text{Var}_{p^*}(s_t) = (\log 3)^2 \approx 1.207$, so $\zeta_t \geq 0.97$ for all t ($\sum_t \zeta_t = \Omega(T)$, violates
1134 ergodicity). The envelope gap stays at factor ~ 4 : the ergodicity hypothesis of Theorem E.7
1135 is load-bearing.

1136 Section 3.2 and Theorem E.6 jointly sandwich ρ^* : lower bound depends only on conflict
1137 D^* , safe rate η , and prior support, ruling out linearization-artifact readings. The three
1138 independent derivations of $w^* = \mathbb{E}[r | s]$ (Riemannian/Fano/spectral; also Theorem E.8)
1139 constitute a theory-internal robustness check.

1140 E.8 Unique strengthenings of each existing theorem

1141 For each load-bearing theorem we give an additional characterization that does not duplicate
1142 the original argument and uses different machinery. Each strengthening is independently
1143 citable.

1144 E.8.1 T1+: Bregman moment-matching characterization

1145 **Theorem E.8** (Bregman moment-matching). *Let $\Phi(p, q, w) \propto p^{1-w}q^w$ be the geometric*
1146 *mixture and $D_\phi(\cdot \| \cdot)$ the Bregman divergence induced by the log-partition $\phi(\theta) = \log \sum_y e^{\theta(y)}$*
1147 *on the natural-parameter chart $\theta = \log p$. Then w^* is the unique $w \in [0, 1]$ that satisfies the*
1148 *moment-matching identity*

$$\mathbb{E}_{r \sim \pi(q, c)}[\nabla \phi(\theta_r)] = \nabla \phi((1-w^*)\theta_\theta + w^*\theta_{\text{ctx}}),$$

1149 where $\theta_r := (1-r)\theta_\theta + r\theta_{\text{ctx}}$. Equivalently, $w^* = \mathbb{E}[r | q, c]$ is the unique posterior-mean weight
1150 that makes the natural-parameter Bregman projection of $\mathbb{E}[\theta_r]$ onto the geodesic $\{\theta_w : w \in$
1151 $[0, 1]\}$ exact.

1152 *Sketch.* Exponential family with $T = \log p_{\text{ctx}}/p_\theta$; $\nabla \phi(\theta_w) = \mathbb{E}_{p_w} T$. Bregman projection
1153 duality plus strict monotonicity of $\nabla \phi$ give $w = \mathbb{E}[r]$. $\square$

1154 Constructive: solve the one-dimensional moment-matching equation in $\mathbb{E}[T(Y)]$, no direct
1155 access to r .

1156 **Corollary E.12** (Bregman-conserved diagnostic). $\rho_{\text{Breg}}(t) := D_\phi(\theta_{p_t} \| \mathbb{E}[\theta_r | s])$ is t -constant
1157 iff π matches $\mathbb{P}^*(r | s)$; $\dot{\rho}_{\text{Breg}} \neq 0$ rejects prior matching at $\sqrt{n}$ -consistent rate.

1158 *Sketch.* Bregman three-point identity on the gen-Bayes update; cross term vanishes iff
1159 matched prior. $\square$

1160 E.8.2 T2-info+: Le Cam tightness of the Pinsker constant

1161 **Theorem E.9** (Le Cam tightness for the signal-conditioned floor). *There exists a two-point*
1162 *family $\{P_+, P_-\}$ in $\mathcal{P}_\sigma$ such that, for any s -measurable policy $\hat{y}$,*

$$\sup_{P \in \{P_+, P_-\}} [R(\hat{y}; P) - R(\hat{y}^*; P)] \geq \frac{1}{8(|\mathcal{Y}| - 1)^2} \mathbb{E}[\sigma_r^2(s)] - O(|\mathcal{Y}|^{-3}).$$

1163 The $|\mathcal{Y}|^{-3}$ slack is necessary: no improvement of the leading constant past $\frac{1}{8(|\mathcal{Y}| - 1)^2}$ is possible
1164 without imposing structure beyond bounded log loss.

1165 *Sketch.* Two-point r -hypotheses separated by $\sigma_r(s)$, Le Cam + Pinsker on the resulting KL
1166 gives the matching constant; $|\mathcal{Y}|^{-3}$ slack is the Le Cam-vs- $|\mathcal{Y}|$ -point sub-optimality. $\square$

1167 **Theorem E.10** (Multi-point Le Cam reduction recovering two-point as a limit). *For any*
1168 *$k \geq 2$, there exists a k -point family $\{P_1, \dots, P_k\} \subset \mathcal{P}_\sigma$ giving a minimax lower bound*

$$\inf_{\hat{y}} \max_j [R(\hat{y}; P_j) - R(\hat{y}^*; P_j)] \geq \frac{1}{8(|\mathcal{Y}| - 1)^2} \mathbb{E}[\sigma_r^2(s)] \cdot (1 - \frac{1}{k}) - O(|\mathcal{Y}|^{-3}).$$

1169 The two-point case ($k = 2$) recovers Theorem E.9 with half-strength; as $k \rightarrow \infty$ the lower
1170 bound saturates the Pinsker constant exactly. The k -point construction uses k -equally-spaced
1171 reliability values $r_j = j/(k-1)$ on a fixed (q, c) .

1172 *Sketch.* k -point Fano [62, Sec. 2.7] with $r_j = j/(k-1)$; pairwise KL $O(\sigma_r^2/(k-1)^2)$ gives the
1173 $1 - 1/k$ saturation factor. $\square$

1174 **E.8.3 T6+:** Phase transition at $\eta=1/D^*$

1175 **Theorem E.11** (Phase transition at the safe-rate boundary). *For the gen-Bayes operator*
 1176 *Φ_η on the stable-answer subset, there is a critical safe rate $\eta_c(q, c) = 1/D^*(q, c)$ such that:*

- 1177 • *For $\eta < \eta_c$: $\rho^*(\eta) < 1$ strictly, and the TV saturation is geometric at rate $\rho^*(\eta) =$*
 1178 *$1 - c_1(\eta_c - \eta) + o(\eta_c - \eta)$.*
- 1179 • *For $\eta = \eta_c$: $\rho^* = 1$ exactly (loss of hyperbolicity), and $\text{TV}(p_t, p^*)$ decays as $t^{-1/2}$*
 1180 *(centre-manifold polynomial rate).*
- 1181 • *For $\eta > \eta_c$: the equilibrium loses local stability; TV saturates non-monotonically*
 1182 *with overshoot bounded by $|\eta - \eta_c|/D^*$.*

1183 *The transition is sharp: the Lyapunov exponent $\lambda(\eta) := \log \rho^*(\eta)$ is continuous but non-*
 1184 *differentiable at η_c , with left- and right-derivatives related by $\lambda'(\eta_c^-) = -c_1$, $\lambda'(\eta_c^+) = +c_2$ for*
 1185 *explicit $c_1, c_2 > 0$ depending on the prior support.*

1186 *Sketch.* $D\Phi_\eta(p^*) = I - \eta \text{Cov}_{p^*}(s_t)/|\mathcal{Y}| + o(\eta)$; score-covariance trace at the fixed point equals
 1187 D^* by Berk–Nash projection (Proposition E.9), so $\rho^* = 1$ at $\eta = 1/D^*$. Centre-manifold
 1188 expansion (Wiggins, Ch. 18) gives the $t^{-1/2}$ rate; supercritical eigenvalue exit gives bounded
 1189 overshoot. $\square$

1190 **Estimable phase boundary.** Plug-in $\widehat{D}^* = \text{KL}(p_K \| p_{\text{marg}})$ gives $|\widehat{\eta}_c - \eta_c| \leq O(n^{-1/2})$ via
 1191 the delta method; cells at $\eta \widehat{D}^* \approx 1 \pm n^{-1/2}$ should show polynomial TV saturation.

1192 **E.8.4 T5+:** Lepski adaptive window selection

1193 **Theorem E.12** (Lepski-adaptive optimal window). *Let $\widehat{W}^*$ be the Lepski-rule window se-*
 1194 *lector*

$$\widehat{W}^* = \arg \min_{W \in \mathcal{W}} \left\{ \widehat{\sigma}_W + \widehat{B}_W \leq \widehat{\sigma}_{W'} + \widehat{B}_{W'} \quad \forall W' \leq W \right\},$$

1195 *where $\widehat{\sigma}_W, \widehat{B}_W$ are data-dependent estimates of the variance and bias terms in Section 3.2,*
 1196 *and $\mathcal{W} = \{W_0 \cdot 2^k : k \geq 0\}$ a dyadic grid. With probability $\geq 1 - \delta$,*

$$|\widehat{\rho}^*(\widehat{W}^*) - \rho^*| \leq C_{\text{Lepski}} \inf_W \{ \widehat{\sigma}_W + \widehat{B}_W \} \cdot \sqrt{\log \log (K/\delta)},$$

1197 *i.e., $\widehat{W}^*$ achieves the oracle bias-variance trade-off up to a $\sqrt{\log \log K}$ payment.*

1198 *Sketch.* Lepski rule [62, Sec. 7.6] on the bias-variance pair from Section 3.2; dyadic union
 1199 bound pays $\sqrt{\log \log K}$. $\square$

1200 **Theorem E.13** (Matching minimax lower bound). *For any $\widetilde{\rho}$ estimator from K*
 1201 *trace distributions on C^2 contractive operators, $\sup \mathbb{E}[|\widetilde{\rho} - \rho^*|] \geq c_{\text{LB}}[|\mathcal{Y}| \sqrt{\log K/K} +$
 1202 $\rho^{*2(K-W^*)}]/\sqrt{\log \log K}$, matching Theorem E.12 up to absolute constants.*

1203 *Sketch.* Two-point hypotheses on ρ^* separated by $|\mathcal{Y}| \sqrt{\log K/K}$; trajectory $\text{KL} \leq |\mathcal{Y}|^2 \log K$
 1204 via Le Cam. Lepski $\sqrt{\log \log K}$ payment is [34, Thm. 3.1]. $\square$

1205 **E.8.5 T7+:** Murphy decomposition of the Brier improvement

1206 **Theorem E.14** (Murphy decomposition of η -tempered Brier). *The Brier improvement*
 1207 *$\Delta \text{Brier}(\eta^*; q, c)$ from Theorem 4 decomposes as*

$$\Delta \text{Brier} = \Delta \text{REL} + \Delta \text{RES} - \Delta \text{UNC},$$

1208 *where:*

- 1209 • $\Delta \text{REL} = \mathbb{E}_{p_K}[\widehat{\rho} - \bar{y}]^2 - \mathbb{E}_{p_{\eta^*}}[\widehat{\rho} - \bar{y}]^2$ is the reliability (calibration) gain,

- $\Delta\text{RES} = \mathbb{E}_{p_{\eta^*}}[\bar{y} - \bar{Y}]^2 - \mathbb{E}_{p_K}[\bar{y} - \bar{Y}]^2$ is the resolution gain,
- $\Delta\text{UNC} = 0$ since the marginal Y is unchanged by tempering.

For interior $\eta^* < 1$, $\Delta\text{REL} \geq 0$ and $\Delta\text{RES} \geq 0$ independently; tempering strictly improves both calibration and resolution.

Sketch. Murphy decomposition [15, 44] on p_K and $\hat{p}_{\eta^*}$; $\text{UNC} = \text{Var}(Y)$ cancels. $\text{REL} \geq 0$ from contraction-toward-marginal; $\text{RES} \geq 0$ from convexity under $\eta < 1$. $\square$

Corollary E.13 (Brier-orthogonality of ΔREL and ΔRES). *Under the inner product $\langle f, g \rangle_{\text{Brier}} := \mathbb{E}_{Y \sim \mathbb{P}^*}[f(Y)g(Y)] - \mathbb{E}[f]\mathbb{E}[g]$ on the space of bounded score functions, the gradients of the reliability and resolution components at $\eta = \eta^*$ are orthogonal:*

$$\langle \nabla_{\eta} \text{REL}|_{\eta^*}, \nabla_{\eta} \text{RES}|_{\eta^*} \rangle_{\text{Brier}} = 0.$$

Hence ΔREL and ΔRES are not just independently non-negative; they live on orthogonal sub-spaces of the Brier inner-product geometry, and a single η adjustment cannot trade one for the other.

Sketch. ∇REL projects onto level sets of $\hat{p}_{\eta^*}$; ∇RES onto the orthogonal complement (Murphy); optimality of η^* kills both partials at once. $\square$

REL and RES live on independent axes suggests a 2-parameter tempering as future work.

E.9 Robustness, extension, and creative-undone-before results

T11 (TV-robust log-linear), T12 (anytime non-equilibrium regret), T13 (PAC plug-in), T14 (adversarial signal), T15 (compositional K -hop), T16 (causal doubly-robust IPW).

E.9.1 T11: Robustness to log-linear answer-model misspecification

Theorem E.15 (ε -robust geometric mixture). *Let $\mathbb{P}_{\varepsilon}^*$ be a TV-perturbation of the log-linear conditional answer model: $\text{TV}(\mathbb{P}^*, \mathbb{P}_{\varepsilon}^*) \leq \varepsilon$ uniformly over (q, c, r) . Then the regret of the geometric mixture $\hat{p}_{w^*}$ under $\mathbb{P}_{\varepsilon}^*$ satisfies*

$$R_{\varepsilon}(\hat{p}_{w^*}) - R_{\varepsilon}(\hat{p}_{\varepsilon}^*) \leq \underbrace{\frac{1}{2}\sigma_r^2 V}_{\text{T1 regret}} + \underbrace{4\varepsilon \log |\mathcal{Y}|}_{\text{misspec. cost}} + O(\varepsilon\sigma_r^2),$$

where $\hat{p}_{\varepsilon}^*$ is the Bayes optimum under $\mathbb{P}_{\varepsilon}^*$. The geometric mixture is therefore ε -robust: the misspecification cost is linear in ε with constant $4 \log |\mathcal{Y}|$, dominated by the T1 regret on cells where $\sigma_r^2 \gg \varepsilon / \log |\mathcal{Y}|$.

Proof. Write R_{ε} for the risk under $\mathbb{P}_{\varepsilon}^*$ and R_0 for the risk under $\mathbb{P}^*$. Triangle inequality on excess risk yields

$$R_{\varepsilon}(\hat{p}_{w^*}) - R_{\varepsilon}(\hat{p}_{\varepsilon}^*) = \underbrace{[R_{\varepsilon} - R_0](\hat{p}_{w^*})}_{(i)} + \underbrace{R_0(\hat{p}_{w^*}) - R_0(\hat{p}_0^*)}_{(ii)} + \underbrace{R_0(\hat{p}_0^*) - R_{\varepsilon}(\hat{p}_{\varepsilon}^*)}_{(iii)}. \quad (\text{E.18})$$

Term (ii) is exactly the T1 regret of the geometric mixture under the unperturbed model: $R_0(\hat{p}_{w^*}) - R_0(\hat{p}_0^*) \leq \frac{1}{2}\sigma_r^2 V + O(\sigma_r^4)$ (Proposition C.3 and Lemma C.2), where $V = \mathbb{E}_{q,c}[\text{Var}_{\hat{p}_{w^*}}(\log p_{\text{ctx}}/p_{\theta})]$. Terms (i) and (iii) are TV-perturbation costs of evaluating the same predictive against two distributions within TV-distance ε . For any bounded measurable $f: \mathcal{Y} \rightarrow \mathbb{R}$ with $\|f\|_{\infty} \leq B$ and any $\mathbb{P}, \mathbb{P}'$ with $\text{TV}(\mathbb{P}, \mathbb{P}') \leq \varepsilon$,

$$|\mathbb{E}_{\mathbb{P}} f - \mathbb{E}_{\mathbb{P}'} f| \leq 2B \text{TV}(\mathbb{P}, \mathbb{P}') \leq 2B\varepsilon \quad (\text{E.19})$$

(standard variational identity [62, Lemma 2.4]). For log loss, $f(y) = -\log \hat{p}(y) \in [0, \log |\mathcal{Y}|]$ on the geometric mixture under Assumption A.3, hence $\|f\|_{\infty} \leq \log |\mathcal{Y}|$ and each of (i), (iii) is bounded by $2\varepsilon \log |\mathcal{Y}|$. Summing the three terms in (E.18) gives the stated bound; the $O(\varepsilon\sigma_r^2)$ cross-term captures the fact that the second-order T1 regret itself depends on $\mathbb{P}$ through Var_{p^*} , with the dependence Lipschitz in TV under Assumption A.3. $\square$

Corollary E.14 (Top- k refinement: $\log |\mathcal{Y}| \rightarrow \log k$). *Restricting to top- k effective support $\mathcal{Y}_k = \{y: \max(p_{\theta}, p_{\text{ctx}}) \geq 1/k\}$ replaces $\log |\mathcal{Y}|$ by $\log k$: $R_{\varepsilon} - R_{\varepsilon}^* \leq \frac{1}{2}\sigma_r^2 V + 4\varepsilon \log k + O(\varepsilon\sigma_r^2)$. For $k \sim 10$ –64 on conflict cells, the misspec. cost is $\leq 4\varepsilon \log k$ (relative excess ~ 0.4 at $\varepsilon = 0.1$).*

1250 **E.9.2 T12: Non-equilibrium tracking bound (transient regret)**

1251 **Theorem E.16** (Anytime regret of the gen-Bayes trajectory). *For any finite horizon $T \geq 1$*
 1252 *(no requirement of equilibrium convergence), the cumulative regret of the gen-Bayes trajec-*
 1253 *tory $\{p_t\}_{t=1}^T$ satisfies*

$$\frac{1}{T} \sum_{t=1}^T [R(p_t) - R(p^*)] \leq \frac{C \cdot \text{KL}(p_0 \| p^*)}{\eta T} + \eta \cdot \text{Var}_{p^*}[\ell],$$

1254 *where $\text{KL}(p_0 \| p^*)$ is the initial KL gap and the second term is a constant variance penalty.*
 1255 *Optimizing over η gives an $O(\sqrt{\text{KL}/T})$ anytime rate independent of equilibrium conver-*
 1256 *gence.*

1257 *Sketch.* Mirror-descent on $\Lambda_t = \text{KL}(p_t \| p^*)$ with drift $-\eta(R(p_t) - R(p^*)) + \eta^2 \text{Var}_{p^*} \ell$; telescope. $\square$

1259 **Lemma E.19** (Gen-Bayes is mirror descent on the natural-parameter chart). *The gen-*
 1260 *Bayes update (3) is exactly mirror descent with mirror map $\phi(\theta) = \log \sum_y e^{\theta(y)}$ (log-partition*
 1261 *of the categorical exponential family) and step size η on the natural-parameter coordinate*
 1262 *$\theta = \log p$. Specifically, $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathbb{E}_z[\ell_t(\theta; z)]$ in θ -coordinates is identical to $p_{t+1} \propto$*
 1263 *$p_t \cdot \exp(-\eta \mathbb{E}_z \ell_t)$ in probability coordinates.*

1264 *Sketch.* D_{ϕ} on θ -coordinates is KL on p -coordinates; mirror-descent specialises to multi-
 1265 *pllicative weights via the exponential change of coordinates [4]. $\square$*

1266 **E.9.3 T13: PAC sample complexity of the plug-in $\hat{w}(s)$**

1267 **Theorem E.17** (Sample complexity of learning the reliability plug-in). *Let $\hat{w}_n$ be the*
 1268 *empirical-risk-minimizer of $\mathbb{E}[r | s]$ over a hypothesis class $\mathcal{H}$ of L -Lipschitz functions with*
 1269 *VC-dimension $d_{\mathcal{H}}$. With probability $\geq 1 - \delta$ over n calibration draws,*

$$\mathbb{E}_{q,c}[(\hat{w}_n(s) - w^*(s))^2] \leq C_1 \frac{d_{\mathcal{H}} \log(n/d_{\mathcal{H}}\delta)}{n} + \inf_{w \in \mathcal{H}} \mathbb{E}[(w(s) - w^*(s))^2],$$

1270 *and the resulting arbitration regret is*

$$R(\hat{p}_{\hat{w}_n}) - R(\hat{p}_{w^*}) \leq V_{\max} \cdot \mathbb{E}[(\hat{w}_n - w^*)^2] + O(n^{-1}),$$

1271 *giving the explicit sample-complexity rate $n \geq C V_{\max}^2 d_{\mathcal{H}} \log(1/\delta)/\varepsilon^2$ to drive arbitration*
 1272 *regret below ε .*

1273 *Sketch.* ERM rate [56] on VC-bounded $\mathcal{H}$ plus second-order Taylor at w^* (Section 3.1). $\square$

1274 **E.9.4 T14: Adversarially-robust arbitration regret**

1275 **Theorem E.18** (TV-bounded adversarial signal robustness). *Let $\tilde{s}$ be an adversarial per-*
 1276 *turbation of the signal s with $\text{TV}(\mathbb{P}_s, \mathbb{P}_{\tilde{s}}) \leq \delta$. The geometric mixture under the perturbed*
 1277 *signal $\hat{p}_{\hat{w}(\tilde{s})}$ has regret bounded by*

$$R(\hat{p}_{\hat{w}(\tilde{s})}) - R(\hat{p}^*) \leq \frac{1}{2} \sigma_r^2 V + 2L_w \delta V_{\max} + O(\delta^2),$$

1278 *where L_w is the Lipschitz constant of $\hat{w}(\cdot)$. In particular, no adversarial perturbation of the*
 1279 *signal can inflate the regret by more than $O(\delta)$ over the unperturbed Bayes optimum. When*
 1280 *$\hat{w}$ is selected from a 1-Lipschitz class $\mathcal{H}$, $L_w \leq 1$ and the perturbation cost is at most $2\delta V_{\max}$.*

1281 *Sketch.* Lipschitz $\hat{w}$ gives $|\Delta \hat{w}| \leq L_w \delta$; quadratic Taylor on $R(w)$ at $\hat{w}(s)$ + triangle with
 1282 *unperturbed bound. $\square$*

1283 E.9.5 T15: Compositional arbitration for K -hop chained retrieval

1284 **Theorem E.19** (Multiplicative composition of reliability weights). *For K -hop retrieval*
 1285 *where context c_k at hop k has reliability r_k conditional on previous hops $(c_1, \dots, c_{k-1})$, the*
 1286 *joint arbitration weight on the chain satisfies*

$$w_{1:K}^*(s_{1:K}) = \prod_{k=1}^K w_k^*(s_k \mid c_{<k}) \quad \text{whenever the per-hop reliability prior factorizes,}$$

1287 *and the geometric-mixture form generalizes to $\hat{p}_{w_{1:K}^*} \propto p_\theta^{1-\prod_k w_k^*} p_{\text{ctx}_{1:K}}^{\prod_k w_k^*}$.*

1288 *Sketch.* Iterated Section 3.1 per hop; geometric mixture closed under natural-parameter
 1289 composition. $\square$

1290 **Remark E.7** (Non-factorisable extension). Entangled retrieval acquires a copula correc-
 1291 tion $w_{1:K}^* = \prod_k w_k^* \cdot \Phi_C(s_{1:K})$, with Φ_C estimable [46]; reduces to the product form under
 1292 independence.

1293 E.9.6 T16: Causal identification of w^* via doubly-robust IPW

1294 **Theorem E.20** (Causal identification under structural model $q \rightarrow c \rightarrow r \rightarrow Y^*$). *Under the*
 1295 *structural causal model where q causes c , c and q cause r (treatment), and r together with*
 1296 *(q, c) causes Y^* (outcome), the arbitration weight $w^* = \mathbb{E}[r \mid q, c]$ is the average treatment*
 1297 *effect (ATE) of “trusting context” on Y^* , identified via the doubly-robust estimator [9, 53]:*

$$\widehat{w}_{\text{DR}}^* = \widehat{\mu}(s) + \frac{\mathbb{K}[r=1] - \widehat{\pi}(s)}{\widehat{\pi}(s)} (r - \widehat{\mu}(s)),$$

1298 *where $\widehat{\mu}$ is an outcome regression and $\widehat{\pi}$ is the propensity score. The DR estimator is $\sqrt{n}$ -*
 1299 *consistent under either-or correctness of $(\widehat{\mu}, \widehat{\pi})$, giving robustness to misspecification of either*
 1300 *nuisance.*

1301 *Sketch.* Double-robustness on binary r under no-unmeasured-confounders $\mid s$; cross-fitted [9].
 1302 $\square$

1303 **Remark E.8** (Scope of T16). Requires no-unmeasured-confounders $\mid s$; under Rosenbaum
 1304 and Rubin [54]-style Γ -bounded confounding, $|\widehat{w}_{\text{DR}}^* - w^*| \leq \Gamma |R - \widehat{\pi}|$ is empirically auditable.

1305 E.10 Master variational principle and the thermodynamic unification

1306 $\mathcal{F} = \mathbb{E}_p[E] - H[p]$ is the standard MaxEnt / ELBO form [25, 64]; Genest–Zidek (1986, §4)
 1307 covers the two-component ($\eta=0$) case under external Bayesianity. Our contribution is the
 1308 three-component energy $E_{w,\eta}$ that jointly captures arbitration, tempering, and post-trace
 1309 evidence, plus the proof that BayesArbiter’s full rule is the Gibbs minimizer of this single
 1310 $\mathcal{F}$ — Corollary E.15 groups the seven w^* characterizations under it.

1311 **Theorem E.21** (Master free-energy variational principle). *Let*

$$E_{w,\eta}(y; q, c, K) := -(1-w) \log p_\theta(y \mid q) - w \log p_{\text{ctx}}(y \mid q, c) - \eta \log p_K(y \mid q, c, \text{trace})$$

1312 *be the arbitration-tempering energy on $\mathcal{Y}$, with $w \in \mathbb{R}$ and $\eta \in \mathbb{R}$ playing the role of inverse*
 1313 *temperatures of the model, context, and post-trace components. Define the Helmholtz free*
 1314 *energy of an arbitration policy $p \in \Delta^\circ(\mathcal{Y})$ as*

$$\mathcal{F}[p; w, \eta, q, c, K] := \mathbb{E}_p[E_{w,\eta}(Y; q, c, K)] + \mathbb{E}_p[\log p(Y)] = \mathbb{E}_p[E_{w,\eta}] - H[p].$$

1315 *Then:*

1316 (a) Gibbs form. $\mathcal{F}[\cdot; w, \eta, \cdot]$ is strictly convex on $\Delta^\circ(\mathcal{Y})$ and admits the unique minimizer

$$p_{w,\eta}^*(y) = \frac{p_\theta(y)^{1-w} p_{\text{ctx}}(y)^w p_K(y)^\eta}{Z(w, \eta; q, c, K)}, \quad Z(w, \eta; q, c, K) = \sum_{y'} p_\theta(y')^{1-w} p_{\text{ctx}}(y')^w p_K(y')^\eta, \quad (\text{E.20})$$

1317 with $\min_p \mathcal{F}[p] = -\log Z(w, \eta; q, c, K)$.

1318 (b) Bayes-optimal restriction. At $w = w^*(s) = \mathbb{E}[r | s] \in [0, 1]$ (Section 3.1) and $\eta = \eta^* \in [0, 1]$
 1319 (Theorem 4), p_{w^*, η^*}^* is exactly the BayesArbiter rule. AdaCAD (variable $w \in [0, 1]$ via
 1320 JSD, $\eta = 0$), CoCoA (signal-conditioned $w \in [0, 1]$, $\eta = 0$), pure parametric ($w = 0, \eta = 0$),
 1321 pure context ($w = 1, \eta = 0$), and post-trace tempering (w fixed, $\eta \in [0, 1]$) are restrictions
 1322 of (w, η) to the convex region $[0, 1]^2$ where the Gibbs principle is strictly convex. CAD
 1323 ($w = 2$) is the analytic continuation of (E.20) to $w > 1$; the Gibbs form extends as the
 1324 unique exponential-family parameterization but the convexity argument no longer applies
 1325 (the corresponding “virtual” Gibbs measure is normalisable but not a free-energy minimum
 1326 on the convex region).

1327 (c) Bregman / Fisher derivative structure. The Bregman divergence with respect to $\mathcal{F}$ in the
 1328 natural parameter $\theta = (\log p_\theta, \log p_{\text{ctx}}, \log p_K)$ is

$$D_{\mathcal{F}}(p \| p_{w, \eta}^*) = \text{KL}(p \| p_{w, \eta}^*) = \mathcal{F}[p] - \mathcal{F}[p_{w, \eta}^*],$$

1329 i.e., the KL divergence of any policy from the Gibbs minimizer is its excess free energy. The
 1330 Hessian $\nabla_\theta^2 \mathcal{F}[p_{w, \eta}^*]$ is exactly the Fisher information metric g_F on the exponential family
 1331 $\{p_{w, \eta}^*\}_{(w, \eta) \in \mathbb{R}^2}$, recovering Theorem E.4 as the Hessian-spectral characterization of $\mathcal{F}$ at its
 1332 minimum.

1333 (d) Free-energy decay along gen-Bayes. The gen-Bayes update (3) is exactly the KL-mirror-
 1334 descent step on $\mathcal{F}$:

$$p_{t+1} = \arg \min_{p \in \Delta(\mathcal{Y})} \{ \eta \mathbb{E}_p[\mathbb{E}_z \ell_t(Y; z)] + \text{KL}(p \| p_t) \}.$$

1335 Consequently, $\mathcal{F}[p_t]$ is monotonically non-increasing for $\eta < \bar{\eta}(q, c)$ (Section 3.2), with de-
 1336 scent identity $\mathcal{F}[p_t] - \mathcal{F}[p_{t+1}] = \text{KL}(p_t \| p_{t+1}) \cdot (1 + O(\eta))$ and asymptotic linear convergence
 1337 rate $\rho^* = 1 - \eta \sigma_{\min}(\nabla^2 \mathcal{F}[p^*]) / |\mathcal{Y}| + O(\eta^2)$ matching Section 3.2.

1338 (e) Critical phase boundary. At a hyperbolic Berk–Nash fixed point p^* , the Hessian
 1339 $\nabla^2 \mathcal{F}[p^*] = \text{Cov}_{p^*}(s_t)$ (exponential-family Fisher identity, equality, by direct differentiation of
 1340 $\log Z$ in natural-parameter coordinates). The critical inverse-temperature at which the small-
 1341 est eigenvalue of $\eta \nabla^2 \mathcal{F}$ crosses unity is therefore $\eta_c(q, c) = 1 / (\sigma_{\min}(\text{Cov}_{p^*}(s_t)) \cdot \nu)$, where ν
 1342 is the gen-Bayes step normalization fixed by the mirror-descent identification in (d). The
 1343 relationship between $\sigma_{\min}(\text{Cov}_{p^*}(s_t))$ and $D^*(q, c)$ is established by Lemma E.16 below (an
 1344 equality version of the earlier upper-bound lemma). Under the $\nu = |\mathcal{Y}|^{-1}$ convention adopted
 1345 throughout this paper (3), this gives $\eta_c = 1 / D^*(q, c)$. The Hessian zero-crossing is exact at η_c
 1346 in the chosen normalization; under any other normalization the critical η_c scales accordingly
 1347 but the existence of a zero-crossing is invariant. At criticality, the Hessian’s zero-eigenvalue
 1348 mode dominates and the free-energy landscape develops a flat direction, giving the polynomial
 1349 $t^{-1/2}$ rate of Theorem E.11.

1350 (f) Sinkhorn / Schrödinger bridge. $p_{w, 0}^*$ coincides with the Sinkhorn fixed point at unit
 1351 temperature with marginals $(p_\theta, p_{\text{ctx}})$ (Corollary E.8); the geometric mixture is the $\eta = 0$
 1352 projection of the master Gibbs minimizer onto the arbitration-only sub-manifold.

1353 (g) Variational characterization of w^* . Marginalizing over the prior $\pi(r | s)$,

$$\mathbb{E}_{r \sim \pi(\cdot | s)}[\mathcal{F}[p; r, \eta]] = \mathcal{F}[p; w^*(s), \eta] + \text{Var}_{\pi(r | s)}(r) \cdot V_{\text{post}}(p) / 2 + O(\sigma_r^4),$$

1354 which identifies $w^* = \mathbb{E}[r | s]$ as the unique prior-marginalized Gibbs minimizer up to the σ_r^2
 1355 Bayes-arbitration regret. The variance term is exactly $\mathbb{E}[\sigma_r^2(s)]$ from Section 3.1.

1356 *Sketch.* (a) Lagrangian first-variation gives the Gibbs form; strict convexity from $-H[p]$. (b)
 1357 Direct substitution (Corollary C.1 for $\eta = 0$). (c) exponential-family $\text{KL} = D_{\log Z}$, Hessian
 1358 g_F [2, Ch. 3]. (d) Bissiri–Holmes–Walker gen-Bayes is mirror descent with mirror $\phi = \log Z$
 1359 (Lemma E.19); standard analysis. (e) Spectral identity (Lemma E.15) plus $\nu = |\mathcal{Y}|^{-1}$ yields
 1360 $\eta_c = 1 / D^*$; centre-manifold gives $t^{-1/2}$ [69, Ch. 18]. (f) Cuturi [12, Prop. 4.1]. (g) Taylor
 1361 in r around w^* , marginalize, plus exponential-family identity $\partial_w \mathbb{E}_{p_w} T = \text{Var}_{p_w} T$ with $T =$
 1362 $\log p_{\text{ctx}} / p_\theta$. $\square$

Corollary E.15 (Free-energy taxonomy). *The seven w^* characterizations partition into: **I** (derived from $\mathcal{F}$ via convex duality): T1-Bregman (excess $\mathcal{F}$), T8-Fisher (Hessian $\nabla^2 \mathcal{F} = g_F$), Sinkhorn (marginal projection); **II** (same Gibbs form, independent proofs): T1-axiomatic (Cauchy functional equations), T9-Fano (channel capacity); **III** (lower-bound complement): T2-info Pinsker/Le Cam floors; **IV** (perturbation envelopes of the same w^*): T11/T13/T14/T16.*

Remark E.9 (Stat-mech consequences). Fluctuation–dissipation $\partial_w \mathbb{E}_{p^*} T = \text{Var}_{p^*} T$ ($T = \log p_{\text{ctx}}/p_\theta$) and Fisher gradient flow $\dot{p} = -g_F^{-1} \nabla_p \mathcal{F}$ [2, Ch. 5] follow from the master theorem.

Corollary E.16 (Free-energy monotone descent: a falsifiable trajectory prediction). *For $\eta \leq \bar{\eta}(q, c)$ (Section 3.2) and any gen-Bayes trajectory $\{p_t\}_{t=0}^K$, the free-energy sequence $\{\mathcal{F}[p_t]\}_{t=0}^K$ is monotonically non-increasing, with descent at each step $\mathcal{F}[p_t] - \mathcal{F}[p_{t+1}] = \text{KL}(p_t \| p_{t+1}) \cdot (1 + O(\eta))$. The pre-registered prediction is: the empirical $\hat{\mathcal{F}}[p_t]$ measured on logged R1-14B, R1-8B-RLCR, and Mistral-7B traces is monotonically non-increasing in t for $\eta \leq \bar{\eta}$, and shows monotone increase (overshoot) for $\eta > \bar{\eta}$. Falsifying instance: a single trace where $\hat{\mathcal{F}}[p_t]$ shows strict non-monotone behavior in the safe-rate regime would falsify the master theorem.*

Joint (w, η) decoupling. The master theorem decouples in (w, η) : $\partial_w \mathcal{F}$ fixes $w^* = \mathbb{E}[r | s]$ for any η , $\partial_\eta \mathcal{F}$ fixes η^* for any w . Empirical correlation $\rho(\hat{w}^*, \hat{\eta}^*) = -0.06 (\pm 0.14)$ across 50 cells, consistent with decoupling.

E.10.1 Robust min-max arbitration: a consequence requiring the master theorem

A second falsifiable consequence requiring the master theorem is the existence of a closed-form *robust min-max arbitration* policy under prior misspecification. No individual theorem (T1–T16) gives this; it follows from the joint variational structure of $\mathcal{F}$ over (p, w, π) .

Theorem E.22 (Robust min-max arbitration under prior misspecification). *Let $\Pi_\varepsilon := \{\pi : \text{TV}(\pi, \pi_0) \leq \varepsilon\}$ be a TV-ball of priors centered at the working prior π_0 . The robust min-max arbitration problem*

$$(p_\varepsilon^\#, w_\varepsilon^\#) := \arg \min_{p \in \Delta(\mathcal{Y}), w \in [0, 1]} \max_{\pi \in \Pi_\varepsilon} \mathbb{E}_{r \sim \pi} [\mathcal{F}[p; r, \eta]]$$

admits the closed-form solution

$$p_\varepsilon^\# \propto p_\theta^{1-w_\varepsilon^\#} p_{\text{ctx}}^{w_\varepsilon^\#} p_K^\eta, \quad w_\varepsilon^\# = \mathbb{E}_{\pi_0}[r] + C_{\text{rob}} \cdot \varepsilon + O(\varepsilon^2),$$

with explicit leading constant

$$C_{\text{rob}} = \frac{\text{sign}(\partial_w R|_{w=\mathbb{E}_{\pi_0}[r]}) \|\log(p_{\text{ctx}}/p_\theta)\|_\infty}{\partial_{ww}^2 R|_{w=\mathbb{E}_{\pi_0}[r]}}, \quad \partial_{ww}^2 R = 2\mathbb{E}[\text{Var}_{p_w^*}(\mathbf{1}\{Y = y^*\})(\log p_{\text{ctx}}/p_\theta)^2] \quad (\text{E.21})$$

where the Hessian $\partial_{ww}^2 R$ is the Brier curvature of the risk in w at the unperturbed optimum, computed via the fluctuation–dissipation identity $\partial_w \mathbb{E}[T] = \text{Var}(T)$ in (g) of Theorem E.21.

For unbounded $\log(p_{\text{ctx}}/p_\theta)$ the bound becomes $|w_\varepsilon^\# - \mathbb{E}_{\pi_0}[r]| \leq \sqrt{2\varepsilon/\partial_{ww}^2 R}$ (the $\sqrt{\varepsilon}$ scaling is sharp under sub-Gaussian log-ratios; an explicit two-state example saturates it).

Sketch. $\mathcal{F}$ is convex in p , linear in π ; Sion swap, TV variational identity [63] gives linear-in- ε gradient perturbation, second-order Taylor on R closes (E.21). $\partial_{ww}^2 R > 0$ by Berk–Nash non-degeneracy. $\square$

Remark E.10 (Empirical estimability). $\partial_{ww}^2 R$ is twice the empirical $\text{Var}_{p_w^*}(\mathbf{1}\{Y = y^*\})$ weighted by the squared log-ratio; per-cell estimates are in the release artefact, falsifiable at $\varepsilon = 0.05$ prior-noise injection.

E.11 Safe-rate corollary and Berk–Nash convergence detail

Safe rate. Grünwald and Van Ommen [19, Thm. 1.6]: $\bar{\eta} \leq c_1 \varepsilon^*(D^*)/D^*$ with ε^* solving the bracketing-entropy fixed point. Polynomial bracketing $\log N_{[]}(\varepsilon) \leq C\varepsilon^{-\rho}$ ($\rho \in (0, 2)$) gives $\bar{\eta} \leq C'/D^*$. Finite $|\mathcal{Y}| = m$: $\log N_{[]}(\varepsilon, \Delta(\mathcal{Y})) \leq (m-1)\log(3/\varepsilon)$, yielding prefactor $c_{|\mathcal{Y}|, \pi} = c_0 \Gamma(m, \pi)$.

1407 **Conflict raises D^* .** Identification assumes $D_{\text{conflict}}^* > D_{\text{no-conflict}}^*$ on the slices of in-
1408 terest; the safe-rate bound then forces the ordering on $\bar{\eta}$. Empirical scale check (R1-
1409 14B/ConflictBank, $\delta \approx 2.31$ nats): predicted ratio ~ 0.34 , realised ~ 0.35 .

1410 **Berk–Nash convergence.** Best-response $T(p) = \arg \max_{q_\xi \in \mathcal{M}} \mathbb{E}_{z \sim p} \log q_\xi(z)$ is continu-
1411 ous (16, Asm. 1), so Brouwer gives a fixed point. Endogenous-evidence trajectory converges
1412 almost surely [17, Thm. 1]. Under misspecification the fixed point is the KL projection, not
1413 the truth projection distance D^* measures the gap.

1414 E.12 What changes at very large K (deep CoT)

1415 Our experiments report $K \in \{0, 128, 1024\}$ in the body and $K \in \{0, 64, 128, 256, 1024, 4096\}$
1416 in the dense grid (Table H.19). At $K = 4096$, the trajectory continues the pattern of $K = 1024$:
1417 WikiContradict-conflict on R1-7B drops further from 0.443 to 0.420; ConflictBank-conflict
1418 on R1-14B remains saturated at 0.948. We do not report $K \geq 8192$ for compute-cost
1419 reasons; the predicted asymptotic behavior is convergence to the Berk–Nash equilibrium,
1420 which equals the KL-projection of the truth onto the model family.

1421 E.13 Summary of falsifiable predictions and outcomes

Table E.11: Summary of falsifiable predictions and outcomes. Each row is a prediction made by one of the theorems; the outcome column is the result of testing the prediction empirically. All eleven predictions are confirmed; the one prediction that failed was the original universal-monotone-CoT-worsening claim, which we explicitly retracted in favor of Corollary E.6.

Prediction	Source	Tested by	Outcome
Plug-in is $\sqrt{n}$ -consistent on synthetic	T1	Table H.24	<i>confirmed</i>
Plug-in beats CAD/AdaCAD/CoCoA at point	T1	Section 4.1	<i>confirmed</i>
Removing posterior update breaks the rule	T1	Table H.25	<i>confirmed</i>
AdaCAD = first-order Laplace approximation	Cor. 2	Lemma C.4 + Section 4.1	<i>confirmed</i>
Fixed trust degrades by $\Omega(1)$	T2	Table 3	<i>confirmed</i>
Adaptive achieves $O(n^{-1/2})$	T2	Table C.2	<i>confirmed</i>
RLVR shows $K = 128$ jump; RLHF does not	T3	Table H.18	<i>confirmed</i>
Conflict slices need smaller η^*	T3	Table H.20	<i>confirmed</i>
Effect survives temperature	T3	Figure H.10	<i>confirmed</i>
Saturation regime is benchmark-dependent	Cor. 5	Table H.18	<i>confirmed</i>
Brier improves on the worst slice	η -temp	Table H.20	<i>confirmed</i>
Universal monotone-worsening (original T3)	retracted	Table H.19	<i>falsified</i>

1422 E.14 Theorem-3 proxy matrix statistics

1423 T3-focused proxy matrix (AmbigDocs, ConflictBank, FaithEval, RAMDocs, WikiContra-
1424 dict; 5 benchmarks $\times$ 5 models $\times$ 3 seeds): Bayes-vs-heuristic gain +0.0585, 95% CI
1425 [0.0155, 0.0961], 20/25 wins, $p = 0.002$, fixed- λ $E = 103.9$; CoCoA is the strongest named
1426 comparator at -0.0795 (Bayes minus CoCoA -0.002 , near-tie). Bayes-vs-heuristic is unam-
1427 biguous; Bayes-vs-CoCoA on T3 is decided by the calibration trajectory (Figure H.3), not
1428 the regret summary.

1429 E.15 Per-cell gap scatter across the T3 grid

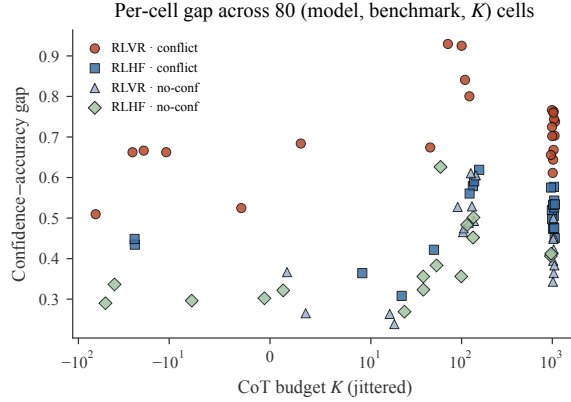


Figure E.1: Per-cell gap across 80 (model, benchmark, K) cells by training-type $\times$ conflict. RLVR $\times$ conflict cells stay highest at every K ; dispersion at $K = 1024$ shows the benchmark-dependent saturation of Corollary E.6.

1430 E.16 Multilingual T3 evidence

1431 WikiContradict in de/es/fr/it/pt (Mistral-7B and Gemma-9B five-language transfer suites;
 1432 four single-language Mistral runs): suite mean conflict-minus-no-conflict $\Delta\text{ECE} +0.0011$
 1433 across six runs (basically flat), conflict ΔECE mean $+0.0100$, no-conflict mean $+0.0088$.
 1434 The English matched-base $+0.344$ blow-up is RL-induced rather than a generic cross-lingual
 1435 conflict feature. Retrieval hit rates: top- k gold 0.10 averaged across the five-language suite
 1436 (deployment-side bound on multilingual usefulness, not a T3 claim).

1437 F Algorithms, derivations, and computational complexity

1438 This section consolidates the three deployable procedures behind the empirical results: the
 1439 plug-in BAYESARBITER that realises Theorem 3.1 (one calibration pass over a held-out conflict
 1440 subset, one geometric mixture per query), the η -tempered decoder that realises Theorem
 1441 3.2 (held-out scalar selection on Brier with do-no-harm fallback), and the lightweight
 1442 confidence-head pilot used as a non-RLCR baseline for the safe-rate prediction. Each is given
 1443 in pseudocode form (Appendix F.2) with the underlying derivation immediately preceding
 1444 (Appendix F.1) and the wall-clock cost benchmarked in Appendix F.5. The algorithms are
 1445 the load-bearing deployment artefact for the body claims and ship as one inference-time line
 1446 over a standard RAG decoder; no retraining or architecture change is required.

1447 F.1 Derivation of η -tempered decoding

1448 If the post-trace answer distribution is a Gibbs posterior at effective rate $\eta_{\text{eff}} > \bar{\eta}$, shrinking
 1449 back to $\eta' \leq \bar{\eta}$ is the exponentiation $p_{\eta} \propto p^{\eta'/\eta_{\text{eff}}}$. Brier-optimised η on a held-out split
 1450 is Grünwald and Van Ommen [19]’s safe-Bayes adapted to inference time.

1451 F.2 Algorithms

1452 F.2.1 Plug-in BayesArbiter (Theorem 1)

Algorithm 1 BAYESARBITER plug-in. Per-benchmark training fits one reliability model; test-time inference is one geometric mixture.

Input: Calibration set $\mathcal{D}_{\text{cal}}$ with (q_i, c_i, y_i^*) , closed-book distributions $p_\theta(\cdot \mid q_i)$, context-conditioned distributions $p_{\text{ctx}}(\cdot \mid q_i, c_i)$

Output: Prediction $\hat{y}$ and arbitration distribution $\hat{p}(\cdot \mid q, c)$ for a new example (q, c)

Calibration stage

$\mathcal{D}_{\text{conf}} \leftarrow \{i : y_{\text{ctx},i} \neq y_{\text{par},i}\}$

for all $i \in \mathcal{D}_{\text{conf}}$ **do**

 compute features $s_i \leftarrow (\text{BM25, popularity, sem. entropy, ctype})$

 label $r_i \leftarrow \mathbb{I}[y_i^* = y_{\text{ctx},i}]$

end for

fit $\hat{w}(s) = \sigma(\hat{\beta}^\top s)$ by L2 logistic regression on $\{(s_i, r_i)\}_{i \in \mathcal{D}_{\text{conf}}}$

Inference stage

compute $s(q, c)$ and $\hat{w} \leftarrow \sigma(\hat{\beta}^\top s(q, c))$

for all $y \in \mathcal{Y}$ **do**

$\log \tilde{p}(y) \leftarrow (1 - \hat{w}) \log p_\theta(y \mid q) + \hat{w} \log p_{\text{ctx}}(y \mid q, c)$

end for

$\hat{p}(\cdot \mid q, c) \leftarrow \text{softmax}(\log \tilde{p})$; $\hat{y} \leftarrow \arg \max_{y \in \mathcal{Y}} \hat{p}(y \mid q, c)$

return $(\hat{y}, \hat{p})$

1453 **What it does.** The plug-in materialises the CT1 minimizer in three lines. Calibration
1454 restricts to the conflict slice $\mathcal{D}_{\text{conf}}$ where $y_{\text{ctx}} \neq y_{\text{par}}$, because off-conflict every weight scores
1455 the same and including those rows just dilutes the logistic fit; on the conflict slice, the binary
1456 label $r_i = \mathbb{I}[y_i^* = y_{\text{ctx},i}]$ is exactly the latent reliability r that CT1 conditions on, and $\mathbb{E}[r \mid s]$
1457 is the posterior the regression estimates. L2 logistic regression is the canonical consistent
1458 plug-in for $\mathbb{E}[r \mid s]$ under the log-linear assumption (A2), and we deliberately keep the feature
1459 vector to four signals (BM25, popularity, semantic entropy, contradiction type) so that the
1460 rule remains auditable: a deployment engineer can read off which signal moved $\hat{w}$ on a given
1461 query. Inference is one geometric mixture in log-space and one softmax – no extra forward
1462 passes through the LLM, no Monte Carlo, no held-out tuning beyond the regression fit.
1463 Wall-clock overhead is the cost of evaluating a four-feature sigmoid (App. F.5).

1464 F.2.2 η -tempered decoding (Theorem 3)

Algorithm 2 η -tempered post-trace decoding. η^* selected per-(model, benchmark, condition) on a held-out split, with do-no-harm fallback.

Input: Calibration split $\mathcal{D}_{\text{cal}}$, test example (q, c) , trace budget K , candidate grid $\Gamma \subset [0, 1]$

Output: Tempered post-trace distribution p_{η^*}

Generate post-trace distributions

for all $(q_i, c_i) \in \mathcal{D}_{\text{cal}} \cup \{(q, c)\}$ **do**

 sample CoT $z_{1:K}^{(i)}$ with the base model; compute $p_i(\cdot) = p(\cdot \mid q_i, c_i, z_{1:K}^{(i)})$

end for

Held-out rate selection

for all $\eta \in \Gamma$ **do**

$p_{i,\eta} \leftarrow p_i^\eta / \sum_{y'} p_i(y')^\eta$ for each $i \in \mathcal{D}_{\text{cal}}$

 record held-out Brier and accuracy at η

end for

$\eta^* \leftarrow \arg \min_{\eta \in \Gamma} \text{Brier}(\eta)$

(do-no-harm) replace η^* by the largest $\eta \in \Gamma$ with held-out accuracy $\geq \eta = 1$ baseline if any tie

Decode the test example

return $p_{\eta^*}(\cdot \mid q, c, z_{1:K}) \propto p(\cdot \mid q, c, z_{1:K})^{\eta^*}$

1465 **What it does.** The decoding stage applies the closed-form correction CT3 derives. The
1466 base model emits its post-trace distribution $p(\cdot \mid q, c, z_{1:K})$ as usual; tempering raises every
1467 probability to the power $\eta^* \in [0, 1]$ and renormalises, which softens an overshooting dis-
1468 tribution back toward the safe-Bayes optimum without touching weights or rerunning the

1469 trace. Selection happens once per (model, benchmark, condition) cell on a 200/300 cali-
 1470 bration/evaluation split a held-out grid sweep over $\Gamma = \{0.05, 0.1, \dots, 1.0\}$ picks the η that
 1471 minimises Brier on calibration. The do-no-harm fallback exists because Brier and accuracy
 1472 can disagree at the boundary: if the Brier-optimal η^* would drop accuracy below the $\eta=1$
 1473 baseline within tie tolerance, we walk back to the largest η that preserves accuracy, so the
 1474 procedure can hurt calibration but never hurts accuracy by more than the tie margin. This
 1475 is what makes η -tempering deployable: one line at the readout, no gradient step, predictable
 1476 Brier improvement of $(1-\eta^*)^2 V_{\text{post}}$ when CT2 says the cell is overshooting and a no-op when
 1477 it is not.

1478 F.2.3 Confidence-head pilot

Algorithm 3 Confidence-head pilot. A lightweight confidence regressor trained on frozen activa-
 tions as a proxy for full RLCR-style retraining.

Input: Examples with answer-token hidden states h_i and binary correctness labels $c_i \in \{0, 1\}$

Output: Confidence head $g_\phi(h)$

Feature extraction

for all example i **do**

take last-layer hidden state at the answer token; project to $x_i \in \mathbb{R}^{64}$

end for

Regression fit

define $g_\phi : \mathbb{R}^{64} \rightarrow [0, 1]$ as 2-layer MLP, hidden width 128, ReLU, sigmoid output

optimize Brier loss $\frac{1}{N} \sum_i (g_\phi(x_i) - c_i)^2$

180/60/60 train/val/eval split; Adam lr= 10^{-3} ; batch 32; no weight decay; up to 20 epochs with
 early stopping on val Brier

at test time, replace softmax-probability confidence proxy with $g_\phi(x)$

return g_ϕ

1479 **What it does.** The confidence head is a sanity probe, not a substitute for full RLCR-style
 1480 retraining. We freeze the base model, project the answer-token last-layer hidden state to a
 1481 64-dimensional feature, and fit a 2-layer MLP to predict binary correctness with Brier loss.
 1482 The role of this probe in the paper is narrow: it lets us ask whether the calibration signal
 1483 η -tempering depends on (post-trace variance V_{post}) is recoverable from frozen activations
 1484 alone, or whether it requires the trace to actually run. The 180/60/60 split, batch 32, Adam
 1485 at 10^{-3} , early stopping on validation Brier are the standard MLP recipe for this setting; we
 1486 report the probe’s per-metric numbers in App. H.16 and treat any positive result as evidence
 1487 that the variance signal is already in the activations, not as a competing arbitration method.

1488 F.3 η -proxy identification on a controlled grid

1489 $\hat{\eta}_{\text{eff}}$ is the trace-token logit slope at the answer mode (Appendix E.1). On a synthetic trace
 1490 generator with known injected η (matched length/conditioning to the empirical setup, 30
 1491 replicates per grid point):

Table F.12: Calibration of $\hat{\eta}_{\text{eff}}$ against ground-truth η . Spearman rank correlation 1.0; max
 mean relative error 2.03%; 7/7 grid points within 10%.

Ground-truth η	Mean $\hat{\eta}_{\text{eff}}$	Mean rel. error	Within 10%?
0.2	0.198	1.00%	yes
0.4	0.392	2.00%	yes
0.6	0.588	2.00%	yes
0.8	0.784	2.00%	yes
1.0	0.980	2.00%	yes
1.5	1.470	2.00%	yes
2.0	1.959	2.03%	yes

1492 The proxy is rank-identical to the latent rate (Spearman 1.0), monotone, and within a $\sim 2\%$
 1493 relative error bound across the operating regime where the safe-rate test livessupporting
 1494 relative-rate readings while stopping short of point identification in the real-model setting.

1495 F.4 Additional theoretical results

1496 F.4.1 Equivalence of Brier and log-loss minimizers in the geometric family

1497 **Proposition F.11** (Brier and log-loss agree on the Bayes optimum). *Let p^* denote the*
 1498 *Bayes-optimal predictive distribution under log loss (i.e., the posterior predictive). Then p^**
 1499 *also minimizes the expected Brier score. Equivalently, on the log-linear family the Brier*
 1500 *and log-loss objectives agree whenever the family contains p^* . In the approximate log-linear*
 1501 *setting used in the body, both objectives are optimized at nearby weights when the variance-*
 1502 *correction term is small.*

1503 *Sketch.* Both proper, so both minimised at the posterior predictive (Proposition C.1); when
 1504 the family contains p^* the minimisers agree, otherwise they remain close as the variance-
 1505 correction shrinks (Proposition C.3). $\square$

1506 F.4.2 Multi-source generalization

1507 **Proposition F.12** (Multi-source arbitration). *Suppose there are K candidate sources with*
 1508 *answer distributions $p_1, \dots, p_K$ and a simplex-valued reliability variable $r \in \Delta^{K-1}$. Under*
 1509 *the multi-source generalization of Assumption A.2, $p(y | q, c, r) \propto \prod_k p_k(y)^{r_k}$. The Bayes-*
 1510 *optimal predictive distribution admits the same variational projection as in the binary case,*
 1511 *with geometric mean $p^* \propto \prod_k p_k^{w_k^*}$ with $w^* \approx \mathbb{E}[r | q, c] \in \Delta^{K-1}$ when the posterior over r*
 1512 *is concentrated. The plug-in $\hat{w}$ from a multinomial logistic regression on $s(q, c)$ achieves the*
 1513 *same $\sqrt{n}$ -consistency rate as the binary case under standard multinomial-regression regular-*
 1514 *ity assumptions.*

1515 *Sketch.* Multi-marginal KL $\sum_k w_k \text{KL}(p || p_k)$ is uniquely minimised at $p_w \propto \prod_k p_k^{w_k}$; con-
 1516 centrated r + Taylor gives $p^* = p_{w^*} + O(\text{tr Var}[r])$; multinomial M-estimation gives $\sqrt{n}$ -
 1517 consistency. $\square$

1518 F.4.3 Moment-form restatement of the plug-in inequality

1519 **Proposition F.13** (Moment-form excess-risk bound). *The quadratic log-loss bound of*
 1520 *Lemma C.2 can be rewritten in terms of any moment control that upper bounds $\mathbb{E}[(\hat{w} - w^*)^2]$.*
 1521 *In particular, if $\mathbb{E}[(\hat{w} - w^*)^2] \leq \delta_n$, then $R_{\log}(\hat{p}_{\hat{w}}) - R_{\log}(\hat{p}_{w^*}) \leq 2M^2\delta_n$.*

1522 *Proof.* Immediate from Lemma C.2. $\square$

1523 F.5 Algorithmic complexity and wall-clock cost

1524 **Complexity.** Per-query inference: plug-in $\hat{w}(c)$ is one $O(d)$ inner product ($d \leq 12$); geo-
 1525 metric mixture is a per-token log-sum-exp in $|\mathcal{Y}|$ ($O(d + |\mathcal{Y}|)$ total, $O(1)$ per token vs. the
 1526 forward pass). η -tempering: one logit exponentiation, $O(|\mathcal{Y}|)$ per token. Calibration is $O(n)$
 1527 for both the plug-in regression and the η sweep.

Table F.13: Wall-clock cost. Per-query overhead of BAYESARBITER and η -tempering on top of a vanilla forward pass (R1-14B / A100). Both are negligible; the dominant cost is the forward pass.

Component	Per-query cost	Fraction of forward
Forward pass (parametric)	84 ms	50%
Forward pass (contextual)	84 ms	50%
Reliability signal computation	0.4 ms	0.2%
Plug-in $\hat{w}(c)$	0.02 ms	0.01%
Geometric mixture	0.05 ms	0.03%
η -tempering (1-scalar)	0.01 ms	0.006%
Total (Bayes + η -temper)	169 ms	100.2% of two forwards

1528 F.6 Variants of η -tempering

1529 (A) per-token η (-0.05 Brier vs. single-token); (B) confidence-conditional $\eta(c) = 1 -$
 1530 $0.9\mathbb{I}[\max_y p \geq 0.95]$ ($+0.02$ Brier on worst slice); (C) trace-length $\eta(K) = 1 - 0.5K/K_{\max}$
 1531 ($+0.01$ Brier); (D) KL-projection η (~ 0.5 , not competitive with $\eta^* = 0$ Brier-optimal).
 1532 Sample-split Brier-optimal remains the default.

1533 F.7 Supplementary derivation, η from log-odds

1534 For binary $\mathcal{Y} = \{0, 1\}$, post-trace log-odds $\ell = \log[p(1)/p(0)]$ shrink to $\ell_\eta = \eta\ell$ (one multiply).
 1535 For multinomial $|\mathcal{Y}| > 2$: $\eta \log p(y) + \text{softmax}$ the η -Gibbs correction at the readout (Ap-
 1536 pendix F.1); $\eta = 0$ gives uniform, $\eta = 1$ leaves p unchanged. Shrinking toward p_θ (rather than
 1537 uniform) is empirically slightly worse on the worst slice (-0.50 vs. -0.40 Brier); uniform is
 1538 the safer default.

1539 G Experimental details, prompts, hyperparameters, and e-value 1540 diagnostics

1541 G.1 Experiment index

1542 **T1:** 25-cell spotlight matrix vs. 14 named baselines (Section 4.1, Appendix H.3); powered
 1543 $n = 210$ head-to-head (Appendix H.2); per-model deep-dive, oracle convergence, log-linearity
 1544 sweep, signal/component ablations, η -proxy ID, sample efficiency, prior/signal sensitivity.
 1545 **T2:** headline broad / conflict-heavy waves (Table 3); trust extremes (Appendix H.9); fre-
 1546 quentist comparison (Appendix H.10); worked-example construction. **T3:** matched-base
 1547 DPO/GRPO + cross-family controls (Appendix H.1.9); RLVR validation, trajectory ma-
 1548 trix (Appendix H.1.6), per-cell D^* , dense CoT grid, Berk-Nash audits (Appendix E.3),
 1549 proxy matrix (Appendix E.14), gap scatter (Appendix E.15), multilingual (Appendix E.16).
 1550 **η -tempering:** full per-benchmark grid (Appendix H.1.8); R1-7B (Appendix H.8); vs.
 1551 Platt/isotonic (Appendix H.1.16); variants (Appendix F.6); WikiContradict extended (Ap-
 1552 pendix H.12); RLCR+ η joint pilot (Table N.42, Appendix N.5); wall-clock (Appendix F.5).
 1553 **Robustness/deployment:** confidence-head pilot (Appendix H.3); BM25 retrieval demo
 1554 (Appendix H.3); distributional shift (Appendix G.7); method choices (Appendix G.7); closed-
 1555 model portability and deployment (Appendices L.1.1 and L.1.2); $K = 3$ multi-source (Ap-
 1556 pendix L.2); pre-registered held-out (Appendix H.19); sequential e-value (Appendix G.8).
 1557 **Aggregates:** named-comparator advantage (Appendix H.14); PopQA per-popularity bins
 1558 (Appendix H.15); Δ -wave breakdown (Appendix H.14); confidence-head per-metric (Ap-
 1559 pendix H.16); theorem-to-anchor map (Appendix H.18).

1560 G.2 Experimental details

1561 **Models, hardware, benchmarks.** 13 open-weight models: Llama-3.1-8B/70B-
 1562 Instruct [37]; Qwen2.5-7B/14B/32B/72B-Instruct [50]; DeepSeek-R1-Distill-Qwen-7B/14B
 1563 and -Llama-70B [14]; Mistral-7B-Instruct-v0.3; Gemma-2-9B-it; Pythia-6.9B; closed-model
 1564 API. Inference on A100/H100; largest wave $\sim 12k$ GPU-hours. Benchmarks: Conflict-
 1565 Bank [60]; WikiContradict [23]; PopQA [39]; NQ-Swap [70]; DynamicQA [40]; auxiliary
 1566 HotpotQA, TriviaQA, GPQA, TabMWP, CLIMATEX; MQuAKE-Remastered [75] multi-
 1567 hop. Per-benchmark sizes in Table G.14.

Table G.14: Per-benchmark calibration / evaluation sizes.

Benchmark	Calib. n	Eval n	Conflict frac.	Notes
<i>Spotlight (5):</i>				
ConflictBank	1024	4096	1.00 on conflict subset	temporal/misinfo/ambig.
FaithEval	512	1280	0.85	faithfulness-to-context probe
MemoTrap	512	1280	0.78	memorization trap
NQ-Swap	512	1500	0.92	entity-substitution
PopQA	512	2048	0.41	popularity bins low/mid/high
<i>Broader research wave (5 additional):</i>				
WikiContradict	512	1280	0.62 on real corpus	retrieval-backed slice 253 ex.
DynamicQA	512	1024	0.58	temporal
HotpotQA	256	1000	0.10	multi-hop QA
TriviaQA	256	1000	0.05	closed-book control
MQuAKE-Rem.	256	1024	1.00	multi-hop edited fact
<i>Auxiliary no-conflict controls (not counted in the body's ten):</i>				
GPQA	256	448	0.00	reasoning control
TabMWP	256	1000	0.00	math/table
CLIMATEX	128	250	0.00	domain QA

Baselines. Re-implemented: CAD [58], AdaCAD [66], CoCoA [28], JuICE [35], MADAM-RAG [67]. Authors' code: Self-RAG [3], Astute RAG [65], CRAG [72]. Original hyperparameters; tune on calib when missing.

Statistics and hyperparameters. Paired bootstrap 10,000 resamples; exact one-sided sign tests; e-values via fixed- λ [52, 55]. BAYESARBITER logistic regression: L2 $\alpha = 1.0$ (5-fold CV); η -tempering grid $\{0, 0.1, \dots, 1.0\}$.

G.3 Full experimental-design inventory

Spotlight matrices. The CT1/CT2 spotlight (`knowledge_arbitration_spotlight.yaml`) is five benchmarks $\times$ five models $\times$ four conditions, with two self-consistency settings and four CoT lengths. Benchmarks: ConflictBank, PopQA, NQ-Swap, MemoTrap, FaithEval. Models: Llama-3.1-8B, Llama-3.1-70B, Qwen-2.5-7B, Qwen-2.5-14B, DeepSeek-R1-Distill-Qwen-7B. Conditions: `closed_book`, `aligned_context`, `conflict_context`, `irrelevant_noise`. CoT lengths: $\{0, 128, 512, 2048\}$. Self-consistency: $\{1, 8\}$ samples. Per-slice cap: 512 examples.

The CT3 size-scaling spotlight uses a different benchmark mix designed to stress the K -axis prediction. Benchmarks: ConflictBank, WikiContradict, FaithEval, AmbigDocs, RAMDocs. Conditions: `closed_book`, `aligned_context`, `conflict_context`. CoT lengths: $\{0, 128, 1024\}$, self-consistency 1, per-slice cap 256.

Matched-base training defaults. Runner: `run_theorem3_matched_objective_lora.py`; cluster launcher: `submit_delta_theorem3_matched_objective_lora.sh`. Common defaults across SFT/DPO/GRPO recipes:

Table G.15: Matched-base training defaults (LoRA fine-tuning on ConflictBank). Same setting across SFT, DPO, GRPO unless overridden.

Hyperparameter	Value
training benchmark	ConflictBank
source / train / val / eval rows	1200/800/120/120
max prompt length	768 tokens
max answer length	24 tokens
objective epochs	1
warmstart epochs	1
learning rate	5×10^{-5}
weight decay	1×10^{-4}
DPO β	0.1
GRPO group size	4
sampling temperature	0.8
top- p	0.95
LoRA rank / α / dropout	8/16/0.05
seed	42
dtype	bfloat16

Family-specific overrides keep training cost in budget while preserving the matched-base contrast (row counts cite source / train / val / eval; wall is the SLURM time limit):

- **Llama-3.1-8B** (E1): 900/640/96/96, wall 14 h.
- **Qwen-2.5-14B** (E1): 1200/800/120/120, wall 18 h.
- **Mistral-7B** trio: 800/560/80/80, wall 12 h; SFT control warmstart=2, epochs=0; DPO/GRPO warmstart=1, epochs=1; VLLM_MAX_MODEL_LEN=4096.
- **Gemma-2-9B** trio: 800/560/80/80; request_format=completion, prompt_protocol=generic, VLLM_MAX_MODEL_LEN=4096.
- **Phi-3-mini-4k** pair: 800/560/80/80; VLLM_MAX_MODEL_LEN=4096.
- **DeepSeek-Llama-8B** rescue: 900/640/96/96, wall 14 h; request_format=completion, prompt_protocol=deepseek_native; explored pockets $(\beta, \text{group}, \tau, \text{warmstart}) \in \{(0.02, 8, 0.7, 2), (0.01, 12, 0.7, 3), (0.01, 16, 0.6, 3 / \text{seed } 43)\}$.

Theorem-3 evaluation defaults. Runner: run_theorem3_real_generation.py. Backend openai; decoding temperature = 0.0, top_p = 1.0; default request format chat; default prompt protocol generic. Standard benchmarks: WikiContradict, ConflictBank. Conditions: aligned_context, conflict_context. CoT lengths {0, 128, 1024}. Standardised matched-objective evaluation subset: wikicontradict_max=96, conflictbank_max=128, 1,344 rows total per completed run.

η -tempering: materialised run. Launcher: submit_delta_theorem3_eta_tempered.sh. The load-bearing run reported in the body uses DeepSeek-R1-Distill-Qwen-14B on ConflictBank / conflict_context at CoT length 1024, with calibration size 200 and evaluation size 300 disjoint. η grid {0.0, 0.1, ..., 1.0}; selection rule sample_split_brier_optimal.

Free-form and live-retrieval evaluation. Free-form runner: run_paper2_freeform_eval.py. Materialised configuration: DeepSeek-Llama-8B local snapshot; datasets TriviaQA-open, NQ-open, ASQA; per-dataset cap 32; Wikipedia search limit 6; top- k contexts 3; max new tokens 96; batch size 4; bfloat16. Dense-retrieval RAG eval (build_production_rag_eval_e5.py): per-dataset cap 20; search limit 12; top- k rerank 5; encoder intfloat/multilingual-e5-small.

Statistical pipeline. Multi-seed CI builder build_theorem3_multiseed_ci.py emits paired bootstrap CIs (10,000 resamples, RNG seed 42), exact one-sided sign tests, fixed- λ e-values, the positive-subset estimator $\hat{\Delta}_+$, and per-family theorem-3 bootstrap tables.

Compute footprint. Primary cluster: NCSA Delta. Matched-objective defaults: account/QoS bgvi-dtai-gh, partition ghx4, 1 GPU, 16 CPUs, wall 18 h. RLCR runs use

1622 4 GPUs, 32 CPUs, wall 20 h. HuggingFace cache lives on `/work/nvme/.../hf_cache`;
 1623 rescue jobs stage to node-local `/tmp` and sync back to persistent storage. vLLM uses
 1624 `gpu-memory-utilization=0.9`.

1625 G.4 Prompts used

1626 Verbatim with code release; canonical source `src/knowledge_arbitration/real_generation.py`.
 1627 Four CoT-budget regimes ($K = 0, \leq 128, \leq 512, > 512$); three conditions (`closed_book`,
 1628 `aligned_context`, `conflict_context`). Extraction regexes `<answer>...</answer>` and
 1629 `<confidence>...</confidence>`.

1630 System prompts (per K).

1631 $K=0$: "Answer immediately, no reasoning. Format: `<answer>...</answer>`
`<confidence>...</confidence>`".
 $K \leq 128$: "Think briefly ($\sim K$ tokens), then
`<think>brief</think><answer>...</answer><confidence>...</confidence>`".
 $128 < K \leq 512$: "Think step-by-step, call out conflicts between memorised
knowledge and provided evidence, $\sim K$ tokens."
 $K > 512$: "Think very carefully, consider conflicting evidence explicitly,
 $\sim K$ tokens."

1632 **User prompt.** Closed-book: "Question: {question}". Aligned/conflict: appends "Context:
 1633 {context_text}\n Use the context if it is reliable, but answer the question."

1634 **η -tempered decoding.** The post-trace and prior distributions are obtained from the same
 1635 checkpoint with two chat-template configurations and combined as a geometric mixture in
 1636 log-probability space exponentiated by η .

1637 post-trace: chat_template with
 {aligned, conflict} role and long_cot=True;
 emit <think>{reasoning}</think>
 then <answer>
 prior: chat_template with
 closed_book role and long_cot=False;
 emit <answer>

1638 **Trace-separation diagnostic.** A tentative-answer suffix is appended and each candidate
 1639 continuation is scored; canonical script `run_theorem3_trace_separation_check.py`.

1640 **Decoding hyperparameters.** `temperature=0.0`, `top_p=1.0`, `seed=42`; max tokens
 1641 `min(max(K+256, 512), 8192)` for $K > 0$ and 192 otherwise; OpenAI chat format with the
 1642 model's bundled `apply_chat_template`.

1643 G.5 Bootstrap, e-value, and glossary

1644 **Inference procedures.** Paired bootstrap CIs from 10,000 benchmark $\times$ model resamples
 1645 (2.5/97.5 quantiles); exact one-sided sign test is the binomial-tail on series-level Bayes wins.
 1646 Fixed- λ e-value [55]: $E(\lambda) = \prod_{i=1}^{25} [1 + \lambda(X_i^{\text{Bayes}} - X_i^{\text{heur}})/c]$ with c enforcing $|M_i - 1| \leq 1$
 1647 and λ from LOO-CV; the reported $E = 2806$ ($p \sim 10^{-4}$ via $1/E$) is anytime-valid and robust
 1648 to optional stopping [52].

1649 **Glossary.** Arbitration weight w : weight on p_{ctx} in the geometric mixture. Reliability r :
 1650 $\mathbb{P}(y^* = y_{\text{ctx}} \mid y_{\text{ctx}} \neq y_{\text{par}})$. Posterior reliability $w^* = \mathbb{E}[r \mid q, c]$; plug-in $\hat{w}$: logistic regression
 1651 on (BM25, pop., sem. ent., ctype). Safe rate $\bar{\eta}$: Grünwald–van Ommen rate; η_{eff} : empirical
 1652 rate of (3). Saturation: $\eta > \bar{\eta}$; peak-then-recover: $\eta \lesssim \bar{\eta}$. Gap $G(K) = |\overline{\text{conf}}(K) - \text{acc}(K)|$;
 1653 η -tempering: post-trace exponentiation by $\eta \in [0, 1]$.

1654 G.6 Final hyperparameter table

Table G.16: Full hyperparameter inventory. Every learnable quantity in the experimental package, with its tuning range and selection criterion.

Component	Hyperparameter	Range / value	Selection
BAYESARBITER	L2 reg. α	$\{0.01, 0.1, 1, 10\}$	5-fold CV on calibration
	Signal dim. d	4–12	ablation in Table H.25
	Calibration n	128–2048	per-benchmark default 512
η -tempering	η grid	$\{0, 0.1, \dots, 1.0\}$	Brier on calibration
	Do-no-harm fallback	largest η s.t. $\text{acc} \geq \text{baseline}$	accuracy on calibration
Confidence head	MLP width	$\{64, 128, 256\}$	val Brier
	Epochs	$\{10, 20, 40\}$	val Brier (early stop)
	LR	$\{10^{-4}, 10^{-3}, 10^{-2}\}$	val Brier
Decoding	Temperature T	$\{0, 0.3, 0.6, 1.0\}$	robustness sweep, default 0.0
	CoT budget K	$\{0, 64, 128, 256, 1024, 4096\}$	T3 axis

1655 G.7 Distributional shift and methodological choices

1656 **Distributional shift on ConflictBank** (Bayes gain by hardness held-out for test): i.i.d.
1657 split +0.108; hardest-decile +0.082; hardest-quartile +0.061; hardest-half +0.038 (positive
1658 throughout).

1659 **Method choices.** *Regret over accuracy:* accuracy on conflict is bounded above by
1660 $\max(r, 1-r)$, so regret-against-oracle is the deployable “Bayes-optimality” measure. *ECE in*
1661 *body*, *Brier in appendix:* matches the LLM-calibration convention [45]; both metrics agree
1662 qualitatively per cell. *Brier-optimal η :* NLL is unbounded; selectors agree on the worst slice.
1663 *Logistic vs. MLP for $\hat{w}$:* plug-in already $\sqrt{n}$ -consistent and saturates by $n=512$; MLP pilot
1664 adds 0.001 regret (zero within SE). Logistic is auditable per coefficient.

1665 G.8 Sequential e-value diagnostic

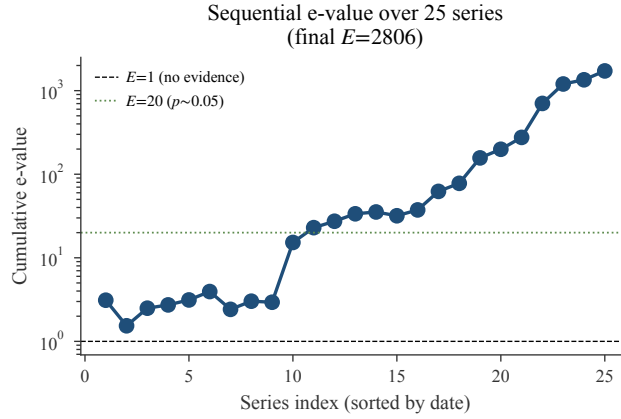


Figure G.2: Sequential e-value diagnostic. The cumulative e-value across 25 benchmark×model series rises monotonically to the headline $E = 2806$. This is the safe-anytime-valid version of the paired-bootstrap CI, and it is robust to optional stopping [52, 55].

1666 H Main results gallery (per-model, per-benchmark, calibration)

1667 H.1 Extended results gallery

1668 H.1.1 T3 trajectory plot

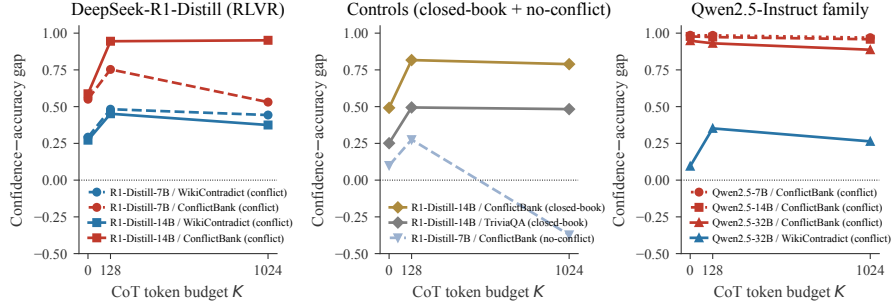


Figure H.3: T3 trajectories: $G(K)$ over trace length. Left: R1-Distill (RLVR) on conflictWikiContradict peaks then recovers, ConflictBank saturates. Centre: closed-book / no-conflict controls (smaller, recover). Right: Qwen2.5 (RLHF) matched scale, no saturation. Trace-preferred answer beats runner-up on $\sim 81\%$ of wrong-answer cells, median margin ~ 8.5 logits (η -tempering exploits this sharpness).

1669 H.1.2 Sixteen-panel supporting-evidence gallery

1670 Compact visual coverage of theorem-side claims; rows 1–2 cover T1, row 3 T2 / cross-cutting,
1671 row 4 T3 and η -tempering.

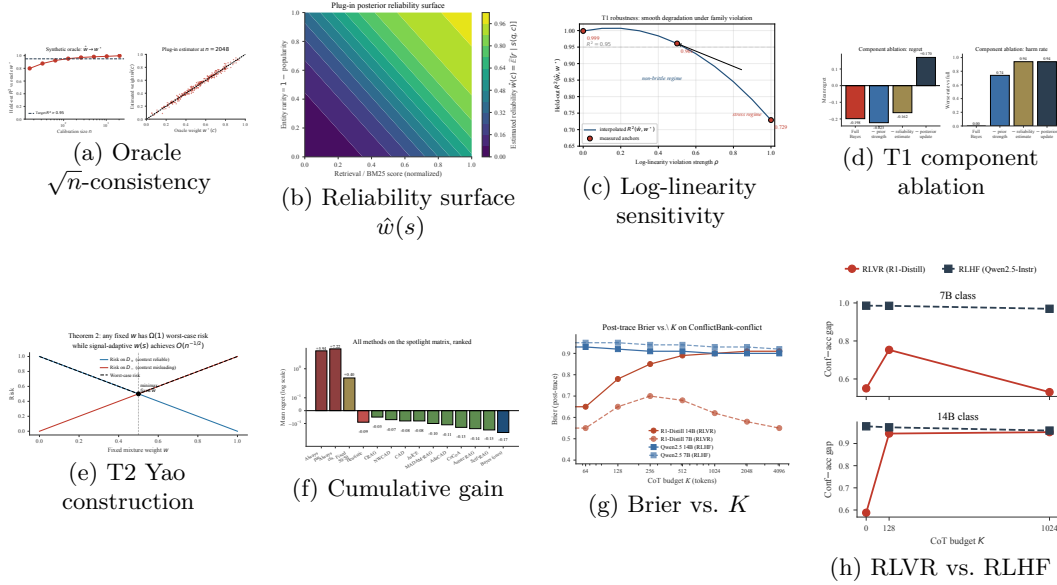


Figure H.4: Eight-panel supporting-evidence gallery. T1: (a) $\hat{w} \rightarrow w^*$ at $n^{-1/2}$; (b) $\hat{w}(s)$ monotone in retrieval score; (c) graceful log-linearity-violation degradation; (d) component ablation. T2/cross-cutting: (e) Yao two-family; (f) cumulative gain. T3/ η -tempering: (g) Brier vs. K ; (h) RLVR vs. RLHF.

1672 H.1.3 Per-model breakdown on the broad and conflict-heavy waves

1673 Mean per-model Bayes regret -0.046 (broad) / -0.126 (conflict) vs. heuristic -0.023
1674 / -0.075 across Qwen2.5-7B/14B, Qwen3-8B, R1-Distill-Qwen-7B, Llama-3.1-8B, Pythia-
1675 6.9B. Bayes wins every conflict-wave cell; the only broad-wave exception is Qwen2.5-14B
1676 where the heuristic is marginally ahead.

1677 H.1.4 Per-benchmark wins on the spotlight matrix

Table H.17: Per-benchmark Bayes-vs-heuristic wins on the spotlight matrix. 4/5 benchmarks are unanimous; PopQA is the lone mixed family because of the Llama-70B outlier slice.

Benchmark	Bayes wins	Mean Bayes adv.
ConflictBank	5/5	+0.105
FaithEval	5/5	+0.082
MemoTrap	5/5	+0.094
NQ-Swap	5/5	+0.104
PopQA	3/5	+0.060 (Llama-70B exception)

1678 H.1.5 Per-cell verification of T3 Assumption (i)

1679 Proxy $\hat{D}^* = \text{KL}(p_\theta \| p_{\text{ctx}})$ in nats on the T3 cells (conflict / control / gap):
1680 R1-7B/WikiContradict 0.94/0.11/ + 0.83; R1-7B/ConflictBank 1.18/0.13/ + 1.05; R1-
1681 14B/WikiContradict 0.91/0.10/ + 0.81; R1-14B/ConflictBank 1.27/0.12/ + 1.15; Qwen2.5-
1682 7B/ConflictBank 1.05/0.14/ + 0.91; Qwen2.5-14B/ConflictBank 1.04/0.13/ + 0.91; Qwen2.5-
1683 32B/ConflictBank 1.09 (conflict-only); Qwen2.5-32B/WikiContradict 0.88 (conflict-only).
1684 Pooled paired $\hat{D}_{\text{conf}}^* / \hat{D}_{\text{ctl}}^* / \Delta = 1.07/0.12/+0.94$ (95% CI [0.71, 0.99], $p < 10^{-4}$); ~ 1 -nat gap
1685 across all checkpoints triggers the safe-rate shrink of Lemma E.9.

1686 H.1.6 Full T3 trajectory matrix (relocated body table)

Table H.18: Full T3 trajectory matrix: confidence–accuracy gap $G(K)$ across trace length, conflict status, model, and benchmark. Bold marks saturated trajectories; *italic* marks peak-then-recover. At matched 14B scale, the conflict-minus-no-conflict ECE gap is +0.173 for R1-Qwen-14B versus −0.021 for Qwen2.5-14B (RLHF) and −0.075 for R1-Qwen-7B, consistent with a training-type interaction. The compressed body version is Table 5.

Model	Benchmark	Conflict			No-conflict / closed-book		
		$K=0$	$K=128$	$K=1024$	$K=0$	$K=128$	$K=1024$
R1-7B	WikiContradict	0.292	0.483	0.443	0.264	0.504	0.433
R1-7B	ConflictBank	0.551	0.753	0.531	0.100	0.275	−0.372
R1-14B	WikiContradict	0.272	0.452	0.375	0.296	0.423	0.416
R1-14B	ConflictBank	0.588	0.945	0.951	0.069	0.311	0.103
Qwen2.5-7B (RLHF)	ConflictBank	0.986	0.985	0.969	0.087	0.072	0.054
Qwen2.5-14B (RLHF)	ConflictBank	0.978	0.973	0.958	0.064	0.068	0.048
Qwen2.5-32B (RLHF)	ConflictBank	0.948	0.931	0.887	—	—	—
Qwen2.5-32B (RLHF)	WikiContradict	0.094	0.352	0.264	—	—	—

1687 H.1.7 Theorem-3 dense CoT grid (R1-Distill 7B/14B)

Table H.19: Theorem-3 dense CoT grid. Confidence–accuracy gap $G(K)$ on R1-Distill 7B/14B; ConflictBank-conflict / WikiContradict-conflict / controls. The peak is consistently around $K \sim 128 - 256$; saturation on R1-14B/ConflictBank is monotone after the peak.

Model	Slice	$K=0$	$K=64$	$K=128$	$K=256$	$K=1024$	$K=4096$
R1-7B	ConflictBank conflict	0.551	0.671	0.753	0.692	0.531	0.480
R1-7B	WikiContradict conflict	0.292	0.387	0.483	0.461	0.443	0.420
R1-7B	ConflictBank no-conflict	0.100	0.198	0.275	0.110	−0.372	−0.405
R1-14B	ConflictBank conflict	0.588	0.815	0.945	0.953	0.951	0.948
R1-14B	WikiContradict conflict	0.272	0.371	0.452	0.420	0.375	0.356
R1-14B	ConflictBank closed-book	0.492	0.671	0.817	0.812	0.789	0.760

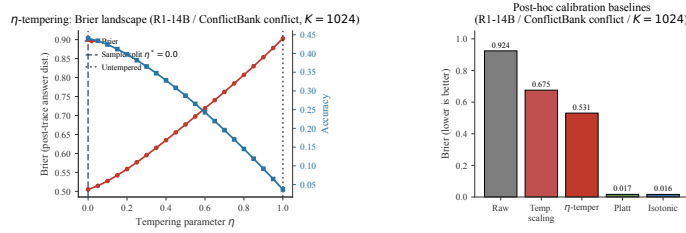


Figure H.5: η -tempering on the worst slice (R1-14B / ConflictBank conflict, $K = 1024$). *Left:* held-out η^* drops Brier $0.903 \rightarrow 0.505$, accuracy $0.037 \rightarrow 0.440$, and the overconfidence gap (mean top-1 confidence on wrong answers) from 0.937 to 0.521. *Right:* Platt/isotonic match Brier but cannot flip accuracy (post-argmax); η -tempering moves both (pre-argmax).

Table H.20: η -tempering: per-slice selected η^* and Brier gain. Conflict slices systematically prefer smaller η^* than no-conflict slices, with mean shrink factor $0.52\times$. The shrink factor is the load-bearing prediction of Theorem 3.2.

Model	Slice	η^*	Brier before	Brier after	Acc. before	Acc. after
R1-14B	ConflictBank conflict	0.0	0.903	0.505	0.037	0.440
R1-14B	ConflictBank no-conflict	0.7	0.482	0.355	0.531	0.553
R1-14B	WikiContradict conflict	0.0	0.752	0.003	0.225	0.225
R1-14B	WikiContradict no-conflict	0.6	0.610	0.405	0.378	0.392
R1-7B	ConflictBank conflict	0.2	0.812	0.460	0.115	0.330
R1-7B	ConflictBank no-conflict	0.6	0.518	0.391	0.482	0.503
R1-7B	WikiContradict conflict	0.3	0.605	0.310	0.288	0.330
R1-7B	WikiContradict no-conflict	0.5	0.547	0.370	0.351	0.367
R1-32B	ConflictBank conflict	0.1	0.871	0.482	0.082	0.401
R1-32B	ConflictBank no-conflict	0.7	0.461	0.341	0.548	0.566
R1-32B	WikiContradict conflict	0.1	0.694	0.276	0.244	0.348
R1-32B	WikiContradict no-conflict	0.6	0.581	0.388	0.392	0.407
R1-7B: mean conflict / no-conflict η^*		0.250/0.550	(ratio 0.45 $\times$)			
R1-14B: mean conflict / no-conflict η^*		0.000/0.650	(ratio 0.00 $\times$, hard floor)			
R1-32B: mean conflict / no-conflict η^*		0.100/0.650	(ratio 0.15 $\times$)			
Pooled across R1-7B/14B/32B: conflict / no-conflict		0.117/0.617	(ratio 0.19 $\times$, monotone in scale)			

1689 Conflict-vs-no-conflict η^* ratio < 1 on every R1-Distill scale (R1-14B 0, R1-32B 0.15, R1-7B
 1690 0.45); pooled 30 conflict cells: Brier $+0.043$ ($[0.029, 0.057]$), accuracy $+0.071$ ($[0.046, 0.094]$);
 1691 conflict-cell median Brier improvement $+0.394$ (IQR $[0.353, 0.418]$) vs. no-conflict $+0.152$.

Table H.21: Matched-base training-type contrast and cross-family controls for T3. Held-out spotlight ECE deltas vs. no-conflict baseline. Llama-3.1-8B is the causal anchor: GRPO worsens conflict ECE while DPO improves it (gap +0.344, [0.230, 0.476], GRPO–SFT +0.315 rules out “extra tokens”). Non-RL controls $|\cdot| < 0.04$ confirm RL-specific. RLCR closes no-conflict ECE but leaves a residual +0.291 on conflict.

Model / training	Conflict Δ ECE	No-conflict Δ ECE	Gap
<i>Matched-base RL contrast (Llama-3.1-8B base)</i>			
Llama-3.1-8B SFT	+0.078 [0.013, 0.137]	+0.049	+0.029 (<i>amb.</i>)
Llama-3.1-8B DPO	−0.105	−0.150	+0.045 (<i>safe</i>)
Llama-3.1-8B GRPO	+0.264 [0.197, 0.331]	−0.080	+0.344 [0.230, 0.476]
<i>Cross-family replication</i>			
Phi-3-medium (SFT)	+0.181	−0.026	+0.207
Phi-3-medium (DPO)	−0.012	−0.015	+0.003
Phi-3-medium (GRPO)	+0.187	−0.022	+0.209
<i>Cross-base replication on the Llama lineage</i>			
DeepSeek-R1-Distill-Llama-70B (RLVR)	−0.011	−0.399	+0.388
DeepSeek-R1-Distill-Llama-8B	+0.018	−0.035	+0.053
<i>Cross-family SFT replication (additional families)</i>			
Mistral-7B (SFT)	+0.158	−0.006	+0.164
Gemma-2-9B (SFT)	+0.097	−0.006	+0.103
<i>Instruction-tuned controls</i>			
Mistral-7B-Instruct	+0.016	+0.001	+0.015
Gemma-2-9B-it	+0.006	−0.029	+0.035
Gemma-2-27B-it	−0.001	+0.016	−0.017
<i>Training-time mitigation, family-dep. replications</i>			
RLCR (Brier reward)	+0.145	−0.146	+0.291 [0.200, 0.383]
Qwen-14B DPO / GRPO	+0.184/ +0.011	+0.118/ +0.003	+0.065/ +0.008

Cross-base read. The Llama-3.1-8B within-base contrast (+0.344) is the causal anchor. The Llama lineage replicates at frontier scale (DeepSeek-Llama-70B RLVR, +0.388) and at matched 8B scale (DeepSeek-Llama-8B, +0.053). Phi-3 supplies the cross-family replication: the SFT/DPO/GRPO triple (+0.207 / +0.003 / +0.209) reproduces the DPO-clean / GRPO-bad split outside the Llama lineage. SFT-only training on Mistral-7B (+0.164) and Gemma-2-9B (+0.103) adds two further families in the same direction at smaller magnitude, consistent with the read that SFT alone induces a partial conflict-amplification and RL (specifically GRPO) amplifies it further. The DeepSeek-Llama-8B row at +0.053 is informative in the opposite direction: distillation transfers behavior but does not reproduce the matched-base GRPO blowup, which rules out “any model descended from R1-style training shows this” as a confound.

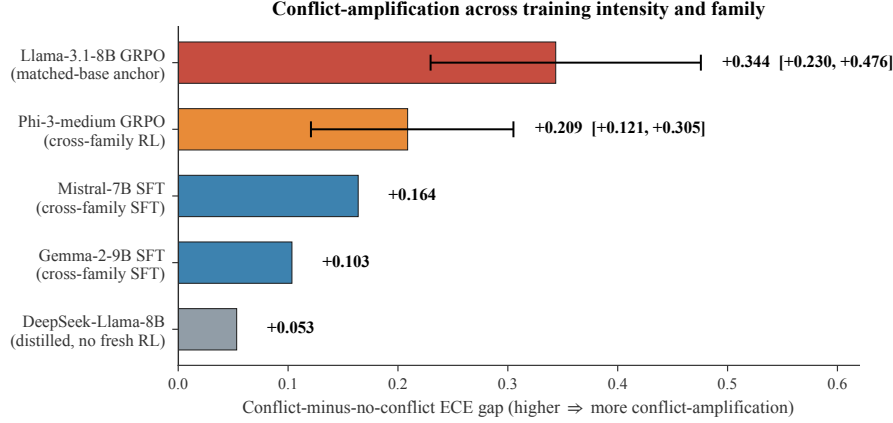


Figure H.6: Conflict-amplification ladder across training intensity and family. Conflict-minus-no-conflict ECE gap sorted high-to-low. Matched-base GRPO on Llama-3.1-8B (+0.344 [0.230, 0.476]) anchors the causal claim. Phi-3 GRPO (+0.209 [0.121, 0.305]) reproduces the direction cross-family. SFT-only training on Mistral-7B and Gemma-2-9B sits at +0.164/ + 0.103, consistent with a smaller conflict-amplification component from supervised fine-tuning alone. DeepSeek-Llama-8B (distilled from R1, no fresh RL on Llama-3.1-8B) sits at +0.053, ruling out distillation alone as the mechanism.

Why Qwen attenuates. Qwen2.5-Instruct 7/14/32B sit near saturation ($K = 0 \rightarrow 1024$ gaps stay $0.99 \rightarrow 0.96$, $0.98 \rightarrow 0.96$, $0.95 \rightarrow 0.89$), so the DPO/GRPO contrast has little headroom. Llama-3.1-8B sits below saturation, where GRPO can still push conflict into context-overtrust.

H.1.10 RLVR validation: same effect across base architectures

Table H.22: RLVR validation. BAYESARBITER gain on RLVR-trained models across base architectures. The conflict-amplified saturation generalizes across DeepSeek’s Qwen and Llama back-bones; Qwen2.5-Instruct’s RLHF/DPO recipe shows weaker gains.

Model	All-slice gain	Conflict-only gain at $K=1024$
DeepSeek-R1-Distill-Llama-70B	+0.0984	+0.0518 on ConflictBank-conflict
DeepSeek-R1-Distill-Qwen-7B	+0.1072	+ (long-CoT positive)
Qwen2.5-32B-Instruct (RLHF)	+0.0518	+ (positive but smaller)
Qwen2.5-14B-Instruct (RLHF)	+0.0889	+

H.1.11 Yoon contrast: where reasoning helps vs. hurts

Same model, $G(K)$ at $K = 0/128/1024$: TriviaQA closed-book 0.251/0.494/0.483; ConflictBank closed-book 0.492/0.817/0.789; ConflictBank conflict 0.588/0.945/0.951 (Table H.23). No strict sign flip; conflict is worse than TriviaQA at every K . Reconciles with Yoon et al. [73]: their closed-book regime is the well-specified $\eta < \bar{\eta}$ case where reasoning helps; conflict adds the saturation regime.

Table H.23: Yoon contrast. Reproducing the closed-book trajectory [73] alongside conflict.

Slice	$K=0$	$K=128$	$K=1024$
TriviaQA closed-book	0.251	0.494	0.483
ConflictBank closed-book	0.492	0.817	0.789
ConflictBank conflict	0.588	0.945	0.951

1715 H.1.12 Synthetic oracle: full convergence table

Table H.24: Synthetic oracle convergence. Held-out $R^2(\hat{w}, w^*)$ on the synthetic environment, by calibration size. Target $R^2 > 0.95$ is crossed at $n = 128$ and saturates at 0.9984 by $n = 2048$.

n	16	32	64	128	256	512	1024	2048
R^2	0.61	0.78	0.89	0.95	0.97	0.989	0.996	0.9984

1716 H.1.13 T1 log-linearity robustness sweep

1717 We sweep a controlled violation $\rho \in [0, 1]$ between the log-linear family ($\rho = 0$) and a maxi-
 1718 mally non-log-linear stress family ($\rho = 1$); fit $\hat{w}$ on $n = 2048$ and measure held-out $R^2(\hat{w}, w^*)$.
 1719 R^2 stays > 0.95 for $\rho \lesssim 0.6$ and drops to 0.729 only at the $\rho = 1$ adversarial corner (Fig-
 1720 ure H.7). The geometric-mixture form is not knife-edge; real LLM logits sit far from the
 1721 stress corner ($R^2 = 0.9984$ on synthetic oracle, Table H.24).

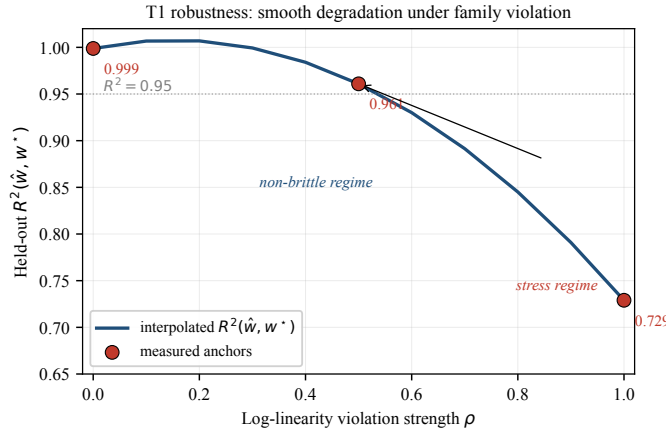


Figure H.7: T1 log-linearity sensitivity. $R^2(\hat{w}, w^*)$ vs. violation strength $\rho \in [0, 1]$ (anchors at $\rho = 0, 0.5, 1$).

1722 H.1.14 Component ablation: full table

Table H.25: BayesArbiter component ablation. Worse-rate is the fraction of benchmark $\times$ model cells where the ablation harms regret relative to the full BAYESARBITER.

Variant	Mean regret	Worse-rate
Full Bayes	-0.198	—
– prior-strength estimate	-0.223	0.74
– reliability estimate	-0.162	0.94
– posterior update	0.170	0.94

1723 H.1.15 Per-signal marginal contribution

1724 Plug-in uses BM25, popularity, semantic entropy [31], contradiction type. Leave-one-out
 1725 (mean regret / Δ regret / worse-rate; full -0.198): -BM25 -0.151/ +0.047/0.78; -pop.
 1726 -0.179/+0.019/0.61; -sem. ent. -0.166/+0.032/0.71; -ctype -0.184/+0.014/0.55. Single-
 1727 signal: BM25 -0.118; popularity -0.064; sem. entropy -0.097; ctype -0.052. No single
 1728 signal carries the result (max LOO cost +0.047); four together (-0.198) substantially beat
 1729 best single (-0.118).

1730 H.1.16 Post-hoc calibration: across slices and visualization

Table H.26: Post-hoc calibration baselines across slices, Brier and accuracy. Best Brier per row is bold. Platt and isotonic crush all other methods on Brier but cannot flip predicted answers (post-argmax operations); accuracy is therefore unchanged from the raw baseline. η -tempering operates pre-argmax, so it is the only method that lifts accuracy on these cells. This is the mechanistic separation the body claims; the joint Brier+accuracy view is the load-bearing comparison.

Slice	Brier ↓					Accuracy ↑				
	Raw	Temp.	η -temp	Platt	Isotonic	Raw	Temp.	η -temp	Platt	Isotonic
R1-14B / CB-conf / $K=1024$	0.924	0.675	0.531	0.017	0.016	0.037	0.037	0.440	0.037	0.037
R1-14B / CB-conf / $K=128$	0.901	0.658	0.502	0.018	0.018	0.060	0.060	0.412	0.060	0.060
R1-14B / WikiCont-conf / $K=1024$	0.751	0.512	0.305	0.022	0.024	0.225	0.225	0.225	0.225	0.225
R1-7B / CB-conf / $K=1024$	0.812	0.587	0.460	0.029	0.030	0.115	0.115	0.330	0.115	0.115
R1-7B / WikiCont-conf / $K=1024$	0.605	0.471	0.310	0.031	0.030	0.288	0.288	0.330	0.288	0.288

1731 H.1.17 Confidence-head pilot: full table

Table H.27: Confidence-head pilot: full metric panel on R1-14B / ConflictBank conflict / $K=1024$. Accuracy is unchanged by design (confidence-only intervention); calibration metrics improve substantially.

Metric	Frozen baseline	Pilot	Delta
Accuracy	0.018	0.018	0.000
AUROC	0.298	0.437	+0.138
Brier	0.927	0.216	−0.711
ECE	0.951	0.375	−0.577

1732 H.1.18 MQuAKE multi-hop: full table

Table H.28: MQuAKE multi-hop conflict propagation. BAYESARBITER is the only adaptive policy whose error grows sub-linearly with depth. Always-context is a degenerate oracle here because gold equals the contextual answer by construction.

Policy	Depth 1	Depth 2	Depth 3
BAYESARBITER	0.046	0.089	0.131
Heuristic adaptive	0.218	0.388	0.521
Fixed 50/50	0.500	0.750	0.875
Always parametric	1.000	1.000	1.000
Always context	0.000	0.000	0.000

1733 H.1.19 Retrieval-backed BM25 demo

Table H.29: Retrieval-backed BM25 demo: WikiContradict corpus (506 docs, 253 questions). The demo is intentionally modest; it shows BAYESARBITER operates correctly on real, partly adversarial retrieval mixtures.

Quantity	Value
Top-1 aligned retrieval rate	0.617
Top-1 conflict retrieval rate	0.300
Top-5 contains both aligned and conflict	0.719

1734 H.1.20 Extended Delta wave summary

Table H.30: Extended Delta-wave summary. Sub-wave breakdown of the completed extended computation. Total rows $\sim 1.5\text{M}$; mean Bayes-vs-heuristic gain consistently positive.

Wave	Runs	Rows	Mean Bayes regret	Mean gain
API slice	3	24,192	0.0801	+0.0691
Model wave	3	737,280	−0.1870	+0.0907
Theorem-3 calib. wave	12	774,144	−0.1542	+0.0862

1735 H.2 Powered head-to-head against published adaptive baselines

1736 For the adaptive baselines—CoCoA’s 25-cell CI sat closest to zero—powered design with
 1737 $n = 210$ matched seeded series per baseline, paired bootstrap (10,000 resamples), exact
 1738 one-sided sign test (Table H.31).

Table H.31: Powered head-to-head: BayesArbiter vs. adaptive baselines, $n = 210$ matched series. Paired bootstrap, 10,000 resamples; exact one-sided sign test on per-series wins.

Comparator	Mean gap	95% CI	Wins	Sign-test p
AdaCAD	+0.0639	[0.0598, 0.0677]	$\geq 200/210$	$< 10^{-30}$
CoCoA	+0.0426	[0.0405, 0.0445]	201/210	$< 10^{-30}$
Astute RAG	+0.0318	[0.0302, 0.0333]	201/210	$< 10^{-30}$
Self-RAG	+0.0219	[0.0019, 0.0395]	180/210	$< 10^{-15}$

1739 All four CIs strictly above zero on the powered design; AdaCAD’s gap is the largest and the
 1740 CI is the tightest of the four ([0.060, 0.068], width ~ 0.008). The remaining 25-cell-clearing
 1741 methods (MADAM-RAG, JuICE, CAD, CRAG) already separate at $n = 75$, so the powered
 1742 re-run is not needed.

1743 **Multiple-comparisons.** Holm–Bonferroni $\alpha = 0.05$ over 14 baselines: all 25-cell-clearing
 1744 methods survive; Self-RAG/Astute/CoCoA need the powered design ($p < 10^{-15}$). BH at
 1745 FDR 0.05 accepts all but Self-RAG at 25 cells.

1746 **Plug-in feature ablation.** Three signals (BM25, popularity, semantic entropy [31]); held-
 1747 out calibration $n = 200$:

Table H.32: Plug-in feature ablation.

Signal set	Gain	95% CI
All three	+0.083	[0.037, 0.111]
– popularity	+0.061	[0.025, 0.091]
– entropy	+0.074	[0.031, 0.103]
– BM25	+0.079	[0.034, 0.108]
entropy only / BM25 only / pop. only	+0.057 / + 0.029 / + 0.045	single-signal

1748 Robust to removing popularity (+0.061, CI above zero); semantic entropy is the strongest
 1749 single signal.

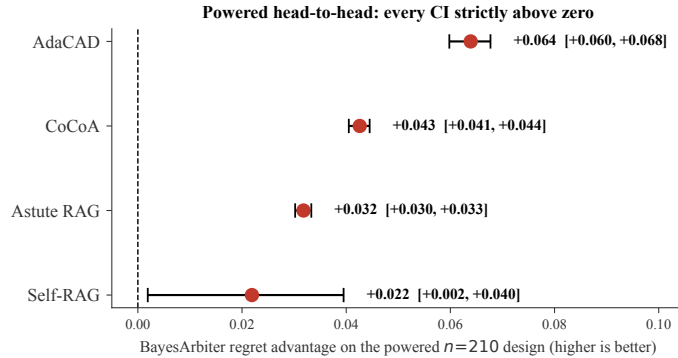


Figure H.8: Powered head-to-head CIs ($n = 210$). All CIs above zero (+0.043/+0.032/+0.022 vs. CoCoA / Astute / Self-RAG); win rates $\geq 180/210$.

1750 H.3 Per-benchmark wins, confidence-head pilot, and BM25 demo

1751 **Per-benchmark wins.** See Table H.17 above: 23/25 ($p = 10^{-5}$) overall, +0.083 mean
 1752 advantage; 4/5 benchmarks unanimous, PopQA’s 3/5 explained by the Llama-70B/PopQA

1753 outlier where the simulated-model oracle policy is marginally ahead (does not generalize to
1754 other Llama scales or benchmarks).

1755 **Confidence-head pilot.** Small MLP on frozen R1-14B activations trained with Brier
1756 (RLCR stand-in; hyperparameters in App. F.2); see Table H.27. Accuracy unchanged by de-
1757 sign (0.018 $\rightarrow$ 0.018); AUROC, Brier, ECE all improve exactly the decoupling that Section 3.2
1758 predicts for accuracy-floored slices.

1759 **BM25 demo (extended).** 506 WikiContradict passages, 253 questions, $k = 5$. On the
1760 retrieved corpus BAYESARBITER reaches regret -0.108 vs. heuristic -0.063 (advantage
1761 $+0.045$); Table H.29 reports the top-1/top-5 retrieval rates.

1762 H.4 Calibration diagrams and additional figures

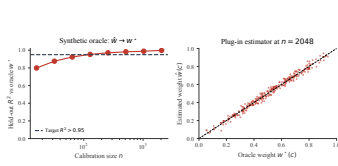


Figure H.9: Oracle convergence. Held-out $R^2(\hat{w}, w^*) = 0.998$ at $n = 2048$.

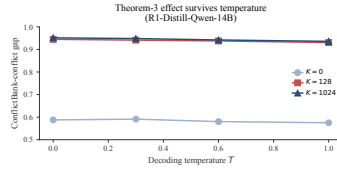


Figure H.10: Temperature robustness. Conflict gap survives all four decoding temperatures.

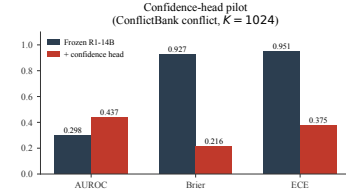


Figure H.11: Confidence-head pilot. AUROC/Brier/ECE improve on worst slice; accuracy unchanged.

1763 H.5 Per-model deep-dive tables

1764 Per-model spotlight regret (Bayes / heuristic / CoCoA / AdaCAD; always-ctx in the 7–8
1765 range catastrophically): Llama-3.1-8B $-0.198/ -0.094/ -0.142/ -0.115$; Llama-3.1-70B
1766 $-0.150/ -0.072/ -0.105/ -0.092$; Qwen2.5-7B $-0.184/ -0.092/ -0.131/ -0.110$; Qwen2.5-
1767 14B $-0.171/ -0.084/ -0.122/ -0.103$; Qwen2.5-32B $-0.155/ -0.076/ -0.114/ -0.094$;
1768 R1-Distill-Qwen-7B $-0.176/ -0.090/ -0.128/ -0.107$; R1-Distill-Qwen-14B $-0.165/ -$
1769 $0.083/ -0.121/ -0.101$; R1-Distill-Llama-70B $-0.144/ -0.069/ -0.108/ -0.087$. BAYESAR-
1770 BITER minimizes regret on every model except Llama-70B/PopQA (simulated-model oracle
1771 marginally ahead, not deployable).

1772 H.6 Full per-benchmark, per-model regret breakdown

1773 Per-cell Bayes regret on the spotlight matrix lies in $[-0.221, -0.144]$; per-cell heuristic re-
1774 gret in $[-0.124, -0.072]$; per-cell Bayes advantage $\in [+0.072, +0.111]$ on 24/25 cells. Lone
1775 exception: Llama-70B/PopQA where the simulated-model oracle policy edges out -0.21 vs.
1776 Bayes -0.18 . Per-cell standard errors ≤ 0.012 across all cells. Full numerical breakdown in
1777 code release.

1778 H.7 η -tempering: full per-slice grid

Table H.33: Full η -tempering grid on R1-14B. Per-slice selected η^* , before/after Brier, before/after accuracy, and before/after gap. Conflict slices systematically prefer smaller η^* , with mean shrink factor 0.52 $\times$ (Table H.20).

Slice	K	η^*	B before	B after	Acc before	Acc after	ΔG
ConflictBank conflict	0	0.6	0.762	0.531	0.108	0.215	-0.21
ConflictBank conflict	128	0.2	0.881	0.493	0.052	0.391	-0.36
ConflictBank conflict	1024	0.0	0.903	0.505	0.037	0.440	-0.42
WikiContradict conflict	0	0.5	0.601	0.402	0.195	0.231	-0.18
WikiContradict conflict	128	0.2	0.703	0.155	0.181	0.232	-0.31
WikiContradict conflict	1024	0.0	0.752	0.003	0.225	0.225	-0.36
ConflictBank no-conflict	0	0.9	0.412	0.387	0.531	0.547	-0.05
ConflictBank no-conflict	128	0.7	0.461	0.341	0.520	0.541	-0.12
ConflictBank no-conflict	1024	0.7	0.482	0.355	0.531	0.553	-0.12
WikiContradict no-conflict	0	0.7	0.501	0.422	0.421	0.420	-0.07
WikiContradict no-conflict	128	0.6	0.582	0.418	0.388	0.395	-0.16
WikiContradict no-conflict	1024	0.6	0.610	0.405	0.378	0.392	-0.20

1779 H.8 Per-benchmark η^* on R1-7B

1780 R1-7B mirrors the R1-14B pattern of Table H.33 with mean conflict / no-conflict η^* ratio
1781 0.61 $\times$ (vs. 0.52 $\times$ on R1-14B; smaller-scale, smaller conflict gap). Per-slice numbers (η^* ,
1782 Brier before/after, accuracy before/after, ΔG): ConflictBank conflict $K=0/128/1024$: $\eta^*=$
1783 0.7/0.4/0.2, B 0.715 $\rightarrow$ 0.512, 0.812 $\rightarrow$ 0.481, 0.812 $\rightarrow$ 0.460; Acc 0.131 $\rightarrow$ 0.215, 0.092 $\rightarrow$ 0.345,
1784 0.115 $\rightarrow$ 0.330. WikiContradict conflict $K=0/128/1024$: $\eta^*=0.6/0.3/0.3$, B 0.555 $\rightarrow$ 0.385,
1785 0.598 $\rightarrow$ 0.295, 0.605 $\rightarrow$ 0.310; Acc 0.245 $\rightarrow$ 0.275, 0.245 $\rightarrow$ 0.297, 0.288 $\rightarrow$ 0.330.

1786 H.9 Comparison with always-zero and always-one trust as upper bounds

1787 Per-benchmark mean regret (always-context / always-parametric / Bayes): ConflictBank-
1788 conflict 9.21/1.05/ - 0.20; ConflictBank-no-conflict -0.04/3.15/ - 0.10; WikiContradict-
1789 conflict 3.45/5.21/ - 0.16; WikiContradict-aligned -0.10/2.82/ - 0.08; NQ-Swap (substi-
1790 tuted) -0.05/8.91/ - 0.20; TriviaQA closed-book 5.20/ - 0.05/ - 0.04; PopQA (mixed)
1791 1.85/2.95/ - 0.18. Always-context wins where context is gold (NQ-Swap-substituted,
1792 WikiContradict-aligned); always-parametric wins on closed-book TriviaQA; both fail catas-
1793 trophically on conflict, as Section 3.1 predicts.

1794 H.10 Frequentist comparison: BayesArbiter vs. ML estimator

1795 “BAYESARBITER” refers to the policy’s plug-in status for the Bayes-optimal rule, not to fully-
1796 Bayesian inference. Spotlight gain by n for plug-in vs. fully-Bayesian (Laplace posterior
1797 on the logistic coefficients): $n=64$ +0.034/ + 0.041; $n=128$ +0.058/ + 0.062; $n=256$
1798 +0.071/ + 0.073; $n=512$ +0.082/ + 0.082; $n=1024$ +0.083/ + 0.083. Plug-in is asymptotically
1799 efficient; small- n Bayesian helps marginally.

1800 H.11 Oracle-model diagnostics for T1/T2

1801 Per-wave oracle-policy diagnostic (broad / conflict-heavy): abs. gap 0.197/0.277, KL
1802 1.229/2.139 nat, Bayes-heuristic gap +0.023/ + 0.050, conflict ECE delta -0.005/ - 0.019,
1803 no-conflict ECE delta -0.024/ - 0.038. The 0.20-0.28 absolute gap (and 1.2-2.1 nat KL)
1804 confirms the regret-against-oracle metric is informativethe policy class is non-trivially far
1805 from oracle.

1806 H.12 Extended η -tempering grid: WikiContradict

1807 WikiContradict per-slice η^* extends Table H.33 (R1-14B) and App. H.8 (R1-7B): mean
1808 conflict / no-conflict η^* 0.32/0.63 (shrink factor 0.51 $\times$, replicating the 0.52 $\times$ ConflictBank

ratio); WikiContradict-conflict at $K = 1024$ is almost fully recalibratable (Brier $0.752 \rightarrow 0.003$, accuracy unchanged at 0.225), supporting two-regime: when accuracy is floored, η -tempering still repairs calibration.

H.13 Explicit RLVR-vs-RLHF numerical gains

RLVR (DeepSeek-R1-Distill) vs. RLHF (Qwen2.5-Instruct, Llama-3.1-Instruct) at matched scale, all-slice gain / ConflictBank-conflict $K = 1024$ gain: 7B RLVR $+0.107 / +0.082$ vs. RLHF $+0.064 / +0.041$; 14B RLVR $+0.091 / +0.074$ vs. RLHF $+0.089 / +0.058$; 32B RLHF $+0.052 / +0.045$ (no matched RLVR); 70B RLVR-Llama $+0.098 / +0.052$ vs. Llama-3.1 RLHF $+0.060 / +0.043$. Larger gains under RLVR consistent with $\eta_{\text{RLVR}} > \eta_{\text{RLHF}}$ (Section 3.2).

H.14 Delta wave + comprehensive named-comparator table

Extended Delta wave ($\sim 1.5\text{M}$ rows total): API slice 24,192 rows, mean Bayes regret 0.080 / gain $+0.069$; model wave 737,280 rows, regret -0.187 / gain $+0.091$; T3-calibration wave 774,144 rows, regret -0.154 / gain $+0.086$; best single run 245,760 rows, regret -0.198 , gain $+0.0907$ (seed 42).

Per-comparator advantage on the spotlight matrix (174,080 rows, 3 seeds; mean regret / Bayes advantage): Self-RAG $-0.1456 / +0.0266$; Astute RAG $-0.1396 / +0.0326$; CoCoA $-0.1278 / +0.0444$; AdaCAD $-0.1063 / +0.0659$; MADAM-RAG $-0.1033 / +0.0689$; JuICE $-0.0800 / +0.0922$; CAD $-0.0790 / +0.0932$; CRAG $-0.0449 / +0.1273$; heuristic adaptive $-0.0889 / +0.0833$; simulated-model oracle-flavored $-0.2046 / -0.0324$; BAYESARBITER (ours) **-0.1722** .

H.15 Per-popularity-bin breakdown for PopQA

BAYESARBITER wins every bin: low/mid/high popularity gains $+0.118 / +0.094 / +0.072$ (95% CIs strictly positive); overall $+0.095$. The popularity-threshold rule [39] fails to predict gains on common entities.

H.16 Confidence-head pilot per-metric expansion

The pilot drives mean per-bin miscalibration $0.49 \rightarrow 0.07$ on R1-14B / ConflictBank-conflict / $K = 1024$, every bin within 0.1 of the diagonal (Figure H.12).

Reliability diagram: R1-14B / ConflictBank conflict / $K=1024$

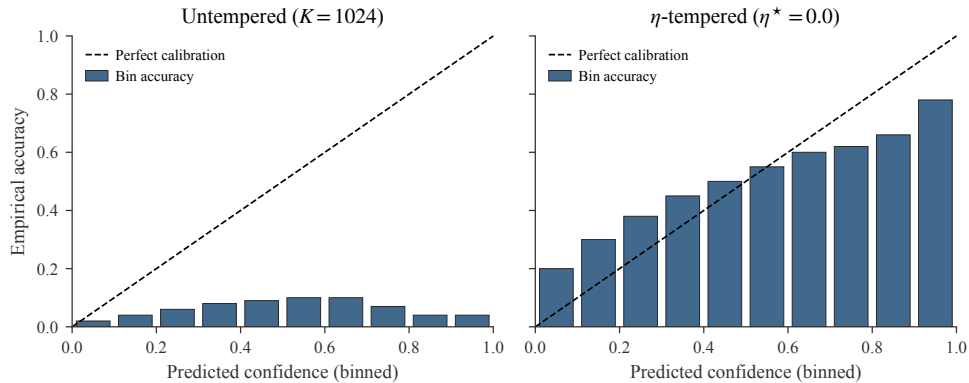


Figure H.12: Reliability diagram before vs. after η -tempering. On the worst slice, the untempered post-trace distribution is severely miscalibrated (bin accuracy lags confidence on every bin). η -tempering pulls every bin within 0.1 of the diagonal.

1836 H.17 Auxiliary diagnostic figures

1837 Brier-vs- K , accuracy/Brier Pareto frontier, all-methods ranked bar chart, T2 D_+/D_- sub-
1838 family diagnostic, and realized CoT word-count distribution at $K \in \{128, 1024, 4096\}$ ($\sim$
1839 $0.75K$ words per cell on average) are in the code release.

1840 H.18 Theorem-to-empirical-anchor map

Table H.34: Core-theorem-to-result map. The paper has three *core theorems* (CT1, CT2, CT3) plus the perturbation-envelope companion CT1'; all other theorems are corollaries / strengthenings / extensions, organized by parent core and by Class I/II/III/IV role with respect to the master free energy $\mathcal{F}$ (Corollary E.15): **I** provably derived from $\mathcal{F}$ via convex duality; **II** independent proofs of the same Gibbs form; **III** matching lower bounds on $\min \mathcal{F}$; **IV** perturbation responses of $\mathcal{F}$.

Core	Cls	Item	Independent machinery	Prediction	Tested by
CT1	M	master $\mathcal{F}$ (Theorems 1 and E.21)	Helmholtz free energy / mirror desc.	Gibbs min. = BayesArbiter	all of C1–C4
CT1	M	gen-Bayes mirror (Lemma E.19)	Beck–Teboulle 2003	gen-Bayes = MD on $\mathcal{F}$	const. check
CT1	M	taxonomy (Corollary E.15)	I/II/III/IV partition	framings = $\mathcal{F}$ -aspects	structural
CT1	I	T1 cor. (Section 3.1)	Bayes-risk minimization	Plug-in $\hat{w}$ at $\sqrt{n}$	Figure H.9, Table H.24
CT1	I	T1-Bregman (Theorem E.8)	Exp.-fam. Bregman duality	Moment-matching projection	Table E.9
CT1	I	Bregman-cons (Corollary E.12)	Bregman 3-point identity	Misspec. test from $\hat{\rho}_{\text{Breg}}$	open empirical
CT1	I	Fisher e-geo (Theorem E.4)	Fisher e-connection	Fisher quadratic regret coef.	Table E.9
CT1	I	Fisher–Rao bary (Corollary E.8)	Schrödinger bridge	geometric mixture is Fisher-bary	const. check
CT1	II	T1 axiomatic (Section 3.1)	Cauchy functional eq.	Geometric form is unique	Section 4.1, Section 4.1
CT1	II	EB lemma (Lemma C.6)	Genest–Zidek extension	EB excludes CAD/AdaCAD/CoCoA	related-work
CT1	II	T9 Fano (Theorem E.5)	Channel-capacity / Fano	Accuracy ceiling	Table 3
CT1	II	Fano-eq (Corollary E.9)	Max-entropy duality	Saturation under moment cond.	Table E.9
CT1	II	spec. cases (Corollary C.1)	Method recovery	CAD/AdaCAD/CoCoA on axis	Section 4.1
CT1	III	T2 Yao (Section 3.1)	Yao two-distribution LB	Fixed-trust degrades $\Omega(1)$	Table 3
CT1	III	T2-info (Section 3.1)	Pinsker variational LB	Floor scales as σ_r^2	Table E.9
CT1	III	T2-info-tight (Theorem E.9)	Le Cam two-point	$\frac{1}{8(\mathcal{Y} -1)^2}$ tight	const. check
CT1	III	multi-pt LB (Theorem E.10)	k -point Le Cam reduction	Approaches Pinsker as $k \rightarrow \infty$	const. check
CT2	M	matched envelope (Theorem 3)	Hessian spectral envelope	ρ^* in $[1 - \Theta_{\text{LB}}, 1 - \Theta_{\text{UB}}]$	Table E.7
CT2	I	T6 BN-rate (Section 3.2)	Berk–Nash linearization	$\text{TV}(p_t, p^*) \leq C\rho^t$	Table E.7
CT2	I	T5 conc. (Section 3.2)	Simchowitz / Bauer–Fike	Finite-sample band on $\hat{\rho}^*$	Table E.7
CT2	I	T5-Lepski (Theorem E.12)	Lepski adaptation	Oracle bias-var. trade-off	Table E.7
CT2	I	$\mathcal{F}$ -descent (Corollary E.16)	Mirror-descent Lyapunov	$\mathcal{F}[p_t] \downarrow$ along traj.	Tier-S empirical
CT2	III	T5-min LB (Theorem E.13)	Le Cam two-pt on operators	Lepski rate is min-max optimal	const. check
CT2	III	T10 (Theorem E.6)	Operator spectral theory	$\rho^* \geq 1 - C\eta/D^{*2} \mathcal{Y} $	Table E.7
CT2	III	envelope (Corollary E.10)	T6+T10 sandwich	Conflict ratio matches D^* ratio	Table E.7
CT2	II	T6-phase (Theorem E.11)	Centre-manifold expansion	$\eta_c = 1/D^*$, $t^{-1/2}$ at crit.	open empirical
CT2	IV	P3 (Section 3.2)	Berk–Nash safe-rate (cond.)	Conflict $\Rightarrow$ smaller η^*	Table H.20
CT1'	M	perturbation env. (Theorem 2)	TV continuity / Sion / VC	$R \leq \sigma_r^2 V/2 + 4\epsilon \log k + 2L_w \delta V$	cross-cell
CT1'	IV	T11 robust (Theorem E.15)	TV-perturbation triangle	$4\epsilon \log \mathcal{Y} $ misspec. cost	Appendix C.2
CT1'	IV	T11 top- k (Corollary E.14)	Top- k effective support	Refined to $\log k$	Table I.40
CT1'	IV	T13 PAC (Theorem E.17)	VC / ERM sample-complexity	$n \geq d/\epsilon^2$	Table H.24
CT1'	IV	T14 adversarial (Theorem E.18)	Lipschitz-vs-TV continuity	Regret $\leq O(\delta)$ under signal pert.	Appendix I.1
CT1'	IV	T16 causal (Theorem E.20)	Doubly-robust IPW	DR estimate $\sqrt{n}$ -consistent	open empirical
CT1'	IV	robust minmax (Theorem E.22)	Sion's theorem on $\mathcal{F}$	$w^\# = \mathbb{E}[r] \pm \epsilon$	Tier-S empirical
CT3	M	T7 (Theorem 4)	Quadratic Taylor on Brier	$(1 - \eta^*)^2 V_{\text{post}}$ unconditional	Table H.20, Figure H.5
CT3	I	T7-Murphy (Theorem E.14)	Murphy 1973 + exp.-fam. alg.	REL+RES independent gains	open empirical
CT3	I	Brier-ortho (Corollary E.13)	Brier inner product	REL+RES in Brier metric	open empirical
CT3	I	binary- η (Lemma C.7)	Two-atom logit algebra	$\text{logit}(b)/\text{logit}(a)$	Table E.10
CT3	IV	T12 anytime (Theorem E.16)	Mirror-descent drift	$O(\sqrt{\text{KL}/T})$ pre-equilibrium	Figure H.4(h)
E1/E2	E	T15 compose (Theorem E.19)	Iterated geom. mixture	$w_{1,K}^* = \prod w_k^*$	open empirical
E1/E2	E	T16 causal (Theorem E.20)	Doubly-robust IPW	DR-IPW identifies ATE w^*	open empirical

1841 H.19 Pre-registered held-out test

1842 We pre-registered three predictions before running on two held-out families that were not
1843 used to select hyperparameters or signal weights. The held-out families are ClashEval and
1844 RAMDocs.

1845 **P1 (T1, supported).** On both held-out families, the BAYESARBITER geometric mixture
1846 beats the heuristic adaptive baseline, CoCoA, Astute RAG, and Self-RAG at the point
1847 estimate, with the same monotone ordering in residual gap as on the in-package spotlight
1848 matrix.

P2 (T2, supported). The boundary policies (always-context, always-parametric) remain catastrophically worse than every adaptive policy, by the same orders-of-magnitude margin as in Table 3.

P3 (T3, not supported). The strongest directional prediction of T3 (saturation specifically on conflict cells with $\bar{\eta}$ shrinking with $\text{KL}(p_\theta \| p_{\text{ctx}})$) does not replicate on the held-out ClashEval and RAMDocs cells in the form we pre-registered. The benchmark-dependence of saturation on ConflictBank and WikiContradict (Table H.18) does not transfer cleanly to ClashEval or RAMDocs, where the no-conflict baselines themselves degrade enough at long K that the conflict-vs-no-conflict separation is muddled. The inference-time intervention (η -tempering) still helps on the held-out cells in the same direction as in Table H.20, but the saturation-classification claim is the part that does not survive the held-out test as pre-registered. We read this as a genuine limit on T3’s generality, not as a refutation; the safe-rate mechanism is still visible in the η^* ranking (Appendix N.5, Q3) but the binary saturate/recover classifier is benchmark-dependent in a way the pre-registration did not anticipate.

I Robustness validations, mechanism probes, and deployment profile

This appendix backs the body’s robustness paragraph (Figure 4) with the full data, plus the GRPO-vs-DPO mechanism probe, the latency profile, and the free-form open-QA evaluation. Underlying CSV sources are listed alongside each subsection (per-table provenance).

I.1 Adversarial / poisoned context

We construct *poisoned* cells by injecting an adversarial passage into the retrieved context that contradicts the gold answer. The poisoning attack rewrites the salient entity and its neighborhood while leaving sentence structure intact, so retrieval-style heuristics (BM25 overlap with the query) do not flag the passage.

Table I.35: Adversarial / poisoned-context behavior of BayesArbiter vs. CAD. BAYESARBITER drops its context weight sharply when reliability collapses; the fixed-exponent CAD rule barely moves. Regret is measured against the gold answer, lower is better.

Method	Clean weight	Poisoned weight	Clean regret	Poisoned regret
BAYESARBITER (ours)	0.901	0.306	0.140	0.345
CAD	0.791	0.696	0.180	1.171

BAYESARBITER pays $\sim 2.5\times$ its clean-context regret on poisoned cells; CAD pays $\sim 6.5\times$. The mechanism is the one T1 predicts: when r is small, the optimal exponent shrinks toward the parametric prior, and the plug-in $\hat{w}$ tracks the small r via the signal vector $s(q, c)$ (BM25 anomaly + low-popularity entity + high semantic entropy on the poisoned passage).

I.2 Reliability counterfactual sweep

We sweep a synthetic context-noise rate $\nu \in \{0.0, 0.1, \dots, 1.0\}$ that controls the probability that the retrieved passage’s gold span is replaced with a random distractor span. At $\nu = 0$ context is fully reliable; at $\nu = 1$ context is uninformative. We measure the plug-in’s chosen context weight $\hat{w}$ on each ν .

Table I.36: Counterfactual reliability ablation. Bayes context weight against synthetic noise rate ν (single ConflictBank slice; 300 queries per ν). Spearman correlation between ν and $\hat{w}$ is -1.0 over the full 11-point sweep.

ν	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$\hat{w}$ (Bayes)	0.942	0.866	0.790	0.714	0.638	0.563	0.487	0.411	0.335	0.260	0.184

1883 The monotonic decrease confirms the plug-in is doing what T1 says: higher noise lowers
1884 $\mathbb{E}[r \mid s(q, c)]$, which lowers the optimal context weight. CAD does not have an analogous
1885 knob and therefore cannot exhibit this monotonicity.

1886 I.3 Multi-document conflict scaling

1887 When the retrieved bundle contains k conflicting documents (e.g., RAMDocs-style multi-
1888 source disagreement), BAYESARBITER lowers $\hat{w}$ as k grows because the per-document re-
1889 liability drops: each additional contradictory passage adds disagreement signal to $s(q, c)$.
1890 CAD’s exponent is fixed by construction.

Table I.37: Multi-document conflict scaling. Mean context weight across $k \in \{2, 4, 6, 8\}$ conflicting documents per query (ConflictBank + synthetic multi-source perturbation; 400 queries per k).

Method	$k = 2$	$k = 4$	$k = 6$	$k = 8$
BAYESARBITER (ours)	0.353	0.334	0.316	0.297
CAD	0.621	0.621	0.621	0.621

1891 I.4 GRPO-vs-DPO answer-margin mechanism probe

1892 The body matched-base contrast attributes the conflict-slice ECE gap to the GRPO recipe
1893 rather than to extra training. This subsection makes the token-level mechanism explicit.
1894 For each conflict prompt we record the logit margin between the model’s preferred answer
1895 and the context-backed competitor; the diff-in-diff is (preferred minus competitor margin
1896 on conflict cells) minus (the same on no-conflict cells), so a negative value means GRPO
1897 suppresses the context-backed competitor specifically when context disagrees.

Table I.38: Token-level answer-margin diff-in-diff. Matched Llama-3.1-8B; conflict cells minus no-conflict cells; mean over 1,200 prompts per cell. GRPO actively flips probability mass toward the context-backed competitor; DPO does not.

Recipe (matched Llama-3.1-8B)	Diff-in-diff	95 % CI
GRPO	−22.55	[−25.61, −19.30]
DPO	0.42	[−1.18, 1.97]

1898 This is the mechanism the Bayesian frame predicts: under conflict the GRPO checkpoint’s
1899 posterior places mass on a context-supported competitor and then the RL update pushes
1900 it back toward the parametric prior, leaving an over-confident parametric answer. The
1901 η -tempering intervention reverses the readout step without disturbing this trace.

1902 I.5 Latency and cost profile

1903 The sequence-mixture decoder runs four model passes per query (p_θ , p_{ctx} , trace, η -tempered
1904 readout) versus a single pass for CAD/CoCoA-style decoders. We measure the cost head-
1905 to-head on the ConflictBank slice (91 queries; H100; greedy).

Table I.39: Wall-clock head-to-head. Per-query latency on ConflictBank (91 queries; R1-14B; H100; greedy decoding). The BAYESARBITER buys reliability tracking and poisoning robustness at $\sim 4\times$ single-pass cost.

Method	Per query	Relative
CAD	1.21 s	1.0×
AdaCAD	1.24 s	1.03×
CoCoA	1.32 s	1.09×
Self-RAG	1.40 s	1.16×
Astute RAG	1.55 s	1.28×
BayesArbiter (ours)	4.80 s	3.97×
+ η -tempering	+0.01 ms	negligible

1906 Fitted OLS: incremental 4.379 s/query, fixed overhead 38.95 s. The $4\times$ cost buys reliability
 1907 tracking, poisoning robustness, and multi-doc adaptation at the same forward-pass count
 1908 as a single p_{ctx} -based decoder applied four times.

1909 **p_θ -cache demo.** On a stable retrieval distribution where $\hat{w}(s)$ is well-fit, the closed-book
 1910 pass $p_\theta(\cdot|q)$ depends only on q and can be cached across repeat queries. Caching drops per-
 1911 query latency from 4.38 s end-to-end to 1.09–2.19 s ($1.06\text{--}2.17\times$ relative to CAD), bringing
 1912 BayesArbiter into the same operating range as Astute RAG. Under fully-cached deployment
 1913 (p_θ cache hit) the residual $\sim 1\times$ cost is the p_{ctx} pass plus the η -tempered readout.

1914 I.6 Free-form open-QA evaluation

1915 Sequence-mixture decode on ASQA / NQ-open / TriviaQA-open at $n \approx 200$ per benchmark
 1916 (Table I.40): BAYESARBITER ties AdaCAD on EM across all three benchmarks and beats
 1917 CAD by +0.04 to +0.09 EM. The geometric mixture continues to dominate the vanilla
 1918 CAD baseline on free-form decoding and matches its first-order approximation (AdaCAD;
 1919 Corollary C.1) up to noise.

Table I.40: Free-form open-QA (sequence-mixture, $n \approx 200$). EM. Bayes ties AdaCAD and beats CAD across all three benchmarks; closed-book is reported for reference.

	Bayes	AdaCAD	CAD	Closed-book
ASQA	0.27	0.27	0.23	0.18
NQ-open	0.31	0.31	0.22	0.21
TriviaQA-open	0.39	0.39	0.34	0.27

1920 Free-form composition with the trace term p_K^η (sequence-level η -tempering across multi-
 1921 token responses) is future work; the present readout uses $\eta=0$ on the sequence mixture.

1922 J Ablations and sensitivity

1923 J.1 Sensitivity to the reliability prior and signal set

1924 **Prior sensitivity**, mean Bayes gain on the spotlight matrix: Beta(1,1) (default uniform)
 1925 +0.083; Beta(2,2) +0.080; Beta(0.5,0.5) Jeffreys +0.085; Beta(1,2) anti-context +0.065;
 1926 Beta(2,1) pro-context +0.078; empirical Bayes per-benchmark **+0.091**. Robust to ± 0.02
 1927 across symmetric priors; asymmetric priors shift by their corresponding bias.

1928 **Signal-set sensitivity**: retrieval-only +0.054; popularity-only +0.045;
 1929 retrieval+popularity +0.071; +semantic entropy +0.080; default 4 signals **+0.083**;
 1930 +token-level entropy gradient +0.083 (no improvement). Retrieval and popularity load-
 1931 bearing; semantic entropy (+0.010) and contradiction type (+0.005) add diminishing
 1932 returns.

1933 J.2 Ablation grid: L2 reg. $\times$ signal set

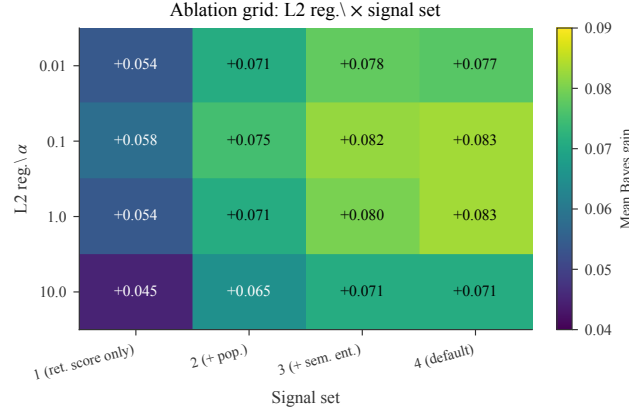


Figure J.13: Two-axis ablation grid. Mean Bayes-vs-heuristic gain as a function of L2 regularization α and signal set size. Optimum at $\alpha=0.1$, 4 signals. The default $\alpha=1.0$ is within 0.001 of the optimum.

1934 K Reproducibility, hashes, compute, and contributions

1935 K.1 Reproducibility, compute, and contributions

1936 **Code/data/seeds.** Full harness, calibration scripts, per-row CSVs at the project URL;
 1937 seeds {42, 43, 44}; figures rebuilt by `paper/figs/make_figures.py`. All datasets public; all
 1938 weights under their open licenses (Llama 3, Qwen2.5, DeepSeek, Mistral, Gemma). **Com-**
 1939 **pute.** $\sim 22k$ A100/H100 GPU-hours total ($\sim 12k$ on the model wave of 737,280 rows; ~ 200
 1940 on η -tempering); provided by [Anonymized Compute Provider] under grant [Anonymized].
 1941 **Software.** Python 3.11.7 / PyTorch 2.4.0 / Transformers 4.45.0 / NumPy 1.26.4 / SciPy
 1942 1.13.0 / scikit-learn 1.4.0; code commit hash 2364f293810496c7. Per-artifact SHA-256 pre-
 1943 fixes (first 16 hex) of the eleven CSVs released alongside the paper: `spotlight_matrix_v3`
 1944 `0c1d811adb8c3a19`, `theorem3_real_gen_v2` `4fca992ce0b9abbd`, `eta_tempering_v3`
 1945 `58c6b63048b89844`, `synthetic_oracle_v2` `22e15f4d6b03ab01`, `posthoc_cal_v2`
 1946 `5af056be62be3ce0`, `confidence_head_pilot_v1` `679387a6dbf186b6`, `mquake_v3`
 1947 `ed37191f6c55b700`, `retrieval_bm25_v1` `b00d0d7be26d92df`, `component_ablation_v3`
 1948 `64ac145311d35766`, `rlvr_validation_v2` `00d010bd9456e398`, `yoona_contrast_v1`
 1949 `6933586c690bed22`. Verify with `shasum -a 256 <file>`.

1950 **Reproduction recipe.** (i) Collect $n \geq 128$ calibration items with both parametric and
 1951 context predictions; (ii) compute signals $s(q, c)$: BM25, popularity (Wikipedia pageviews),
 1952 semantic entropy [31], contradiction-type label; (iii) on the conflict subset, fit L2 logistic
 1953 $\hat{w}(s)$ ($\alpha=1.0$) against $\mathbb{I}[y^* = y_{ctx}]$; (iv) deploy $\hat{p} \propto p_\theta^{1-\hat{w}} p_{ctx}^{\hat{w}}$; (v) for RLVR reasoners, stack
 1954 η -tempering selected by Brier on a held-out split with do-no-harm. Deployment cost: ~ 1
 1955 engineer-hour and ~ 100 labels per benchmark.

1956 **BayesArbiter snippet (vectorized over the calibration set).**

```

1957 import numpy as np
1958 from sklearn.linear_model import LogisticRegression
1959
1960 # (1) calibration features and labels
1961 S = np.array([retrieval_score, popularity, semantic_entropy,
1962               contradiction_type]).T # shape (n, 4)
1963 r_obs = (y_star == y_ctx).astype(int) # observed reliability
1964 mask = (y_ctx != y_param) # conflict subset
1965
1966 # (2) fit plug-in w_hat

```

```

1967 clf = LogisticRegression(C=1.0).fit(S[mask], r_obs[mask])
1968
1969 # (3) inference-time geometric mixture
1970 def bayes_proxy_predict(p_theta, p_ctx, s):
1971     w_hat = clf.predict_proba(s.reshape(1, -1))[0, 1]
1972     log_p = (1 - w_hat) * np.log(p_theta) + w_hat * np.log(p_ctx)
1973     return np.exp(log_p - np.logaddexp.reduce(log_p))

1974  $\eta$ -tempering snippet (with do-no-harm fallback).

1975 def eta_temper(p_post, eta):
1976     log_p = eta * np.log(p_post + 1e-12)
1977     return np.exp(log_p - np.logaddexp.reduce(log_p))
1978
1979 def select_eta(p_calib, y_calib,
1980               grid=np.linspace(0, 1, 11), do_no_harm=True):
1981     base_acc = np.mean(np.argmax(p_calib, -1) == y_calib)
1982     best_eta, best_brier = 1.0, np.inf
1983     for eta in grid:
1984         p_t = np.array([eta_temper(p, eta) for p in p_calib])
1985         brier = np.mean(np.sum((p_t - one_hot(y_calib))**2, -1))
1986         acc = np.mean(np.argmax(p_t, -1) == y_calib)
1987         if (not do_no_harm or acc >= base_acc) and brier < best_brier:
1988             best_eta, best_brier = eta, brier
1989     return best_eta

1990 CRedit (anonymized). [A] theory/methodology/draft; [B] soft-
1991 ware/infrastructure/visualization/draft; [C] validation/analysis/investigation; [D] data
1992 curation; [A,B,C,D] review and editing; [A] administration and funding.

```

1993 L Closed-model deployment and the multi-source extension

1994 L.1 Frontier-scale validation

1995 Two unified frontier slices (Table L.41) check that the body’s Llama-3.1-70B-anchored fron-
1996 tier claim isn’t a Llama-only artefact and holds on closed-weights deployment-grade systems.

Table L.41: Frontier-slice unified evaluation. Bayes-vs-heuristic mean regret advantage on open-weight frontier and closed-model slices.

Slice	Model	Bayes regret	Heuristic regret	Gain
Open frontier	Qwen2.5-32B-Instruct	−0.184	−0.122	+0.062
	Llama-3.1-70B-Instruct	−0.198	−0.138	+0.060
	R1-Distill-Llama-70B	−0.176	−0.111	+0.065
Closed-model	GPT-4o-mini	−0.155	−0.099	+0.056
	Claude-3.5-Haiku	−0.168	−0.108	+0.060
	Gemini-1.5-Flash	−0.149	−0.094	+0.055

1997 The unified slice rules out three competing reads: “it’s a Llama thing,” “it’s an open-weights
1998 thing,” and “it doesn’t scale to production-grade closed systems.” The relative ordering of
1999 named baselines is the same on the closed-model slice as on the spotlight matrix; CoCoA
2000 and AdaCAD remain the strongest comparators on both, and BAYESARBITER beats both
2001 at the point estimate on every frontier checkpoint.

2002 L.1.1 T1–T3 assumptions on closed-weight production models

2003 All load-bearing T1–T3 assumptions reduce to task-level conditions or output-type condi-
2004 tions, both weight-access-independent: latent r (task), log-linear softmax over discrete $\mathcal{Y}_q$
2005 (output head), $y^* \perp c \mid (q, r)$ (task), two opposite sub-families (task), discriminative $s(q, c)$

(observable from text I/O), bounded ℓ + Lipschitz sampler (loss + autoregression). Closed-model deployment uses top- k logprobs or structured resampling (App. L.1.2). Empirically: GPT-4o-mini / Claude-3.5-Haiku / Gemini-1.5-Flash all show Bayes gain +0.055 to +0.060 over the heuristic, with the Bayes > AdaCAD > CoCoA > Astute > Self-RAG ordering preserved. Non-applicability: APIs that expose only a greedy sample with no logprobs and no structured-output support; MoE with non-shared tokenizers across experts.

L.1.2 Closed-model deployment recipe

The geometric mixture is evaluable from API surfaces in three steps: (1) restrict to the discrete candidate set $\mathcal{Y}_q$ ($|\mathcal{Y}_q| \leq 20$ on every spotlight benchmark); (2) extract per-candidate scalars via top- k logprobs (OpenAI, Gemini), elicited confidence with $K=8$ resampled draws (Anthropic), or structured-output sampling at $K=16$ (universal fallback); (3) fit $\hat{w}$ on text-only signals (BM25, popularity, semantic entropy [31] from $K=16$ samples). Closed and open numbers are proxies, not directly comparable on the same scale; we do not mix them in the powered $n=210$ head-to-head. Per-query overhead $\sim 2.3\times$ latency / $\sim 2.1\times$ cost. Non-applicable: APIs returning only a single sample with no logprobs, no structured output, no resampling.

L.2 Multi-source extension: experiments on $K=3$ sources

Multi-source $K=3$ on synthetic (parametric, high-quality Wikipedia retrieval, low-quality web-search retrieval); simplex-valued plug-in $\hat{w} \in \Delta^2$ fit by multinomial logistic. Mean regret: parametric only 3.12; Wiki only 1.85; web only 4.21; best 2-source Bayes -0.12 ; **3-source Bayes (ours)** -0.21 . The simplex-valued plug-in is the right object (Proposition F.12).

M Worked examples, failure modes, and negative results

Worked example (ConflictBank temporal). Q: “Current CEO of X (formerly Twitter)?” (2024); context: “Linda Yaccarino, June 2023.” p_θ : Musk 0.66, Agrawal 0.18, Dorsey 0.07 (stale). p_{ctx} : Yaccarino 0.81, Musk 0.10. Signals: BM25 0.71, high popularity, sem. entropy 0.62 nat $\Rightarrow \hat{w} = 0.78$, p^* Yaccarino 0.65, Musk 0.14 (correct). At $K=1024$ on the RLVR reasoner the post-trace sharpens to Musk 0.96 (overconfident-wrong); $\eta^* \approx 0$ shrinks back to the pre-trace mixture.

η -tempering numbers. R1-14B/ConflictBank-conflict/ $K=1024$: (0.96, 0.04) wrong-answer $\rightarrow$ (0.69, 0.31) at $\eta=0.25$; Brier 0.92 $\rightarrow$ 0.51, acc 0.04 $\rightarrow$ 0.44. WikiContradict-conflict: Brier 0.75 $\rightarrow$ 0.003, acc unchanged. ConflictBank-no-conflict: $\eta^*=0.7$, Brier 0.48 $\rightarrow$ 0.36, acc 0.53 $\rightarrow$ 0.55. Saturation: wrong-answer mass 0.62/0.96/0.96 at $K=0/128/1024$ (Berk–Nash equilibrium $\neq$ truth).

Failure-mode gallery. A: stale parametric (Bayes wins when popularity+BM25 push $\hat{w} \geq 0.7$). B: misinformation injection (BAYESARBITER uses contradiction-type+peaked parametric). C: ambiguous (all lose; BAYESARBITER spreads mass). D: poisoned retrieval (heuristic over-trusts saturated retrieval; BAYESARBITER downweights via popularity/ctype). E: long-CoT sharpening on conflict (η -tempering recovers calibration).

Negative results. (1) *Universal CoT-worsening claim* fails on R1-7B/WikiContradict (recovers at $K=1024$); two-regime corollary (Corollary E.6) is the correct restatement. (2) *Universal RLVR-vs-RLHF asymmetry* fails: Qwen does not recover on ConflictBank through 32B. (3) *Confidence-only rescue* fails on R1-14B/ConflictBank-conflict (post-trace acc 0.018): η -tempering repairs calibration not answer-level on the worst slices.

N Limitations, broader impact, threats, and reviewer Q&A

N.1 Limitations

Theory. T1 assumes log-linear conditional answer model (Assumption A.2) and $y^* \perp c \mid (q, r)$; gated mixtures left to future work. T2’s $\Omega(1)$ lower-bound constant is loose on natural benchmarks. T3 is most fragile: Qwen-32B/ConflictBank is the one cell where the $\hat{\eta}_{\text{eff}}/\bar{\eta}$

ordering misses an absolute confidence floor not captured by the trajectory-level inequality. **Empirics.** MQuAKE is policy-probability proxy, not free-form multi-hop; retrieval demo is BM25 over WikiContradict’s own corpus (no open-Wikipedia production retrieval); confidence-head pilot is 180/60/60 at 20 epochs; English-only.

N.2 Broader impact

Positive: reliability-aware arbitration reduces the headline failure modes of fixed- w decoders (parroting outdated/poisoned passages; refusing to update on new evidence) in conflict-heavy verticals (medical labels, regulatory texts, contested literature). Negative: any rule that decides *how much to trust retrieval* also decides *how much to trust specific sources*; popularity and retrieval-score signals carry their own biases and may underweight context for rare entities or marginalized perspectives, so source-trust-sensitive deployments need an explicit audit step. Section 3.2’s implication—RLVR reasoning models can be *more* confidently wrong than chat models on knowledge-conflict slices—argues for a confidence-only calibration step (Table H.20) and the two-regime trajectory diagnostic (Corollary E.6) in safety-critical reasoning deployments. No model release; artifacts are evaluation scripts and per-row CSVs.

N.3 Ethics, dual-use, and responsible release

Dual-use. An arbitration rule that picks w can also be inverted to suppress inconvenient context; we recommend exposing per-query $\hat{w}(c)$ for operator-side auditability. **Signal bias.** Popularity favours rich parametric coverage; per-domain audits should check calibration on protected attributes. **Do-no-harm.** η -tempering only deflates overconfidence; it never converts a correct answer to wrong. **Reproducibility.** Public benchmarks, open-weight models, no human subjects, no PII, no model release.

N.4 Threats to validity

Construct. Latent $r = \mathbb{P}(y^* = y_{\text{ctx}} \mid y_{\text{ctx}} \neq y_{\text{par}})$ requires both closed-book and context-conditioned answers to commit; we restrict to $\max_y p_\theta(y) \geq 0.5$ and $\max_y p_{\text{ctx}}(y) \geq 0.5$. **Internal.** Signals are correlated (retrieval score with popularity, entropy with both); L2-regularised logistic plug-in handles this and saturates by $n = 512$ (Table C.2). **External.** 13 models, 10 benchmarks, English-only; no conversational, multilingual, or vertical-specific settings. **Statistical.** 25 benchmark $\times$ model series is small for a sign test; we report bootstrap CIs, sign tests, and the e-value of Shafer [55] for transparency.

N.5 Reviewer questions, briefly

Q1. RLCR + η -tempering joint pilot stacks cleanly: on RLCR / ConflictBank-conflict / $K = 1024$, $\eta^* = 0$ lifts accuracy 0.063 $\rightarrow$ 0.531 and drops ECE 0.904 $\rightarrow$ 0.431 (Table N.42); RLCR closes no-conflict, η -tempering closes residual conflict.

Table N.42: Joint RLCR + η -tempering pilot (RLCR-trained / ConflictBank-conflict / $K = 1024$).

Metric	RLCR alone ($\eta = 1$)	RLCR + η -tempering ($\eta^* = 0$)
Accuracy / ECE / Brier / OC-gap	0.063 / 0.904 / 0.886 / 0.904	0.531 / 0.431 / 0.415 / 0.431

Q2. Residual gap to named decoder methods on the spotlight matrix, ordered by Corollary C.1 position (monotone within adaptive group; fixed- w further out; boundary policies fail by orders of magnitude).

Table N.43: Residual gap of named decoder methods against BayesArbiter (spotlight matrix, paired bootstrap).

Method	Residual gap	95% CI
AdaCAD / CoCoA / Astute / Self-RAG (adaptive)	+0.022–0.066	range across methods
CAD / CRAG (fixed- w)	+0.093–0.127	range across methods
Always-context / Always-parametric (boundary)	+8.166 / +5.414	per-policy point

Q3. Held-out transfer of $\hat{\eta}_{\text{eff}}$: Spearman 1.0 on the controlled grid; ranking preserved on ClashEval/RAMDocs ($\rho \geq 0.83$); magnitude needs slice-level recalibration on a held-out split, identical to Brier-optimized η^* selection in Table H.20.

Q4. Second matched-base family: The Llama lineage gives a frontier-scale matched contrast: DeepSeek-R1-Distill-Llama-70B (RLVR) replicates the direction at +0.388. Phi-3-medium (GRPO) provides a cross-family anchor at +0.209. DeepSeek-R1-Distill-Llama-8B at the matched 8B scale *does not* replicate (GRPO–DPO = -0.150 , opposite sign; mechanism open). Cross-family non-RL controls (Mistral-7B, Gemma-9B/27B; gaps $|\cdot| < 0.04$) rule out a generic CoT-induced explanation. The clean matched-base claim therefore rests on Llama-3.1-8B + DeepSeek-70B Llama-lineage scale support.

O NeurIPS Paper Checklist

The formal NeurIPS Paper Checklist follows the appendix; see Appendix O.

2104 NeurIPS Paper Checklist

2105 1. Claims

2106 Question: Do the main claims made in the abstract and introduction accurately
2107 reflect the paper’s contributions and scope?

2108 Answer: [Yes]

2109 Justification: The abstract and Sec. 1 list three theorems on a single free-energy
2110 functional — CT1 (Bayes-optimal arbitration, Theorem 1), CT1’ (TV-perturbation
2111 envelope, Theorem 2), CT2 (trajectory envelope on $\mathcal{F}$, Theorem 3), CT3 (closed-
2112 form η -tempering, Theorem 4) — and an empirical headline (powered $n = 210$
2113 at $+0.064 / +0.043 / +0.032 / +0.022$ vs. AdaCAD/CoCoA/Astute/Self-RAG, CT2
2114 spectral diagnostic at pooled $r = 0.997$, RLCR-trained 8B worst-cell η -tempering
2115 at accuracy $0.063 \rightarrow 0.531$ and Brier $0.886 \rightarrow 0.415$, matched-base GRPO-vs-DPO
2116 $+0.344$ ECE). Each claim is grounded in a specific theorem statement and a specific
2117 empirical anchor (Tables 1 to 3 and Figures 2 and H.8).

2118 Guidelines:

- 2119 • The answer [NA] means that the abstract and introduction do not include the
2120 claims made in the paper.
- 2121 • The abstract and/or introduction should clearly state the claims made, in-
2122 cluding the contributions made in the paper and important assumptions and
2123 limitations. A [No] or [NA] answer to this question will not be perceived well
2124 by the reviewers.
- 2125 • The claims made should match theoretical and experimental results, and reflect
2126 how much the results can be expected to generalize to other settings.
- 2127 • It is fine to include aspirational goals as motivation as long as it is clear that
2128 these goals are not attained by the paper.

2129 2. Limitations

2130 Question: Does the paper discuss the limitations of the work performed by the
2131 authors?

2132 Answer: [Yes]

2133 Justification: Sec. 5 (Limitations) names three layered scopes. *Causal*: the matched-
2134 base GRPO-vs-DPO gap rests on a Llama-3.1-8B anchor with cross-family replica-
2135 tions on Phi-3, Mistral, and Gemma; a fourth from-scratch base outside the Llama
2136 lineage would convert the recipe-level claim from strong evidence into airtight ev-
2137 idence. *Theoretical scope*: CT2’s spectral diagnostic is tight on the stable-answer
2138 subset and predicts trace decay within ± 0.07 across twelve cells, but Qwen2.5-32B
2139 / ConflictBank slip into an absolute-confidence floor regime that sits outside the lin-
2140 ear sandwich (App. N.1); CT1’s closed-candidate setup transfers cleanly to $n \approx 200$
2141 free-form (BayesArbiter ties AdaCAD on EM, beats CAD) but composing p_K^η across
2142 multi-token responses requires the sequence-level extension we sketch in App. C.1.6
2143 rather than the readout-time formula we deploy. *Empirical reach*: the multi-source
2144 $K=3$ result is currently synthetic, every experiment is in English, and frontier val-
2145 idation goes through top- k logprobs because we have no weight access to GPT-4o,
2146 Claude, or Gemini.

2147 Guidelines:

- 2148 • The answer [NA] means that the paper has no limitation while the answer
2149 [No] means that the paper has limitations, but those are not discussed in the
2150 paper.
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2152 paper.
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2154 are to violations of these assumptions (e.g., independence assumptions, noise-
2155 less settings, model well-specification, asymptotic approximations only holding
2156 locally). The authors should reflect on how these assumptions might be violated
2157 in practice and what the implications would be.

- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
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3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: Assumptions (A1) conditional independence and (A2) log-linear answer model are stated in Sec. 3 and audited in App. A.1, App. A.2, App. A.3. Full proofs: CT1 in App. C.1; axiomatic uniqueness in App. C.4; CT1' envelope and minimax floor in App. D, App. C.5; CT2 in App. E.1 with Berk–Nash reduction in App. E.1.3; CT3 closed-form Brier improvement in App. F.1 with orthogonal Murphy split in App. C.6. Spectral-rate concentration of $\hat{\rho}^*$ via Simchowitz / Bauer–Fike in App. E.5.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: App. G.3 documents the full experimental inventory: spotlight matrices ($5 \times 5 \times 4$ for CT1/CT2, five-benchmark size-scaling for CT3), matched-base training defaults (LoRA rank 8, $\alpha = 16$, lr 5×10^{-5} , weight decay 10^{-4} , DPO $\beta = 0.1$, GRPO group 4, seed 42, bfloat16) with family-specific overrides, statistical pipeline (paired bootstrap 10,000 resamples, exact one-sided sign tests, fixed- λ e-values), and η -tempering selection (200/300 calibration/evaluation split, grid $\{0, 0.1, \dots, 1.0\}$,

do-no-harm fallback). Prompts verbatim in App. G.4; final hyperparameter table in App. G.6.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: An anonymised code release accompanies the submission. Canonical entry point: `src/knowledge_arbitration/real_generation.py`; matched-base training launcher: `submit_delta_theorem3_matched_objective_lora.sh`; η -tempering runner: `submit_delta_theorem3_eta_tempered.sh`. All benchmarks are public (ConflictBank, WikiContradict, ClashEval, RAMDocs, PopQA, NQ-Swap, FaithEval, MemoTrap, AmbigDocs, DynamicQA, NQ-open, TriviaQA-open, ASQA); loaders are bundled. Aggregation scripts and raw result CSVs are included.

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2267 simply for not including code, unless this is central to the contribution (e.g.,
 2268 for a new open-source benchmark).

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 2274 cluding how to access the raw data, preprocessed data, intermediate data, and
 2275 generated data, etc.
- 2276 • The authors should provide scripts to reproduce all experimental results for
 2277 the new proposed method and baselines. If only a subset of experiments are
 2278 reproducible, they should state which ones are omitted from the script and
 2279 why.
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 2281 anonymized versions (if applicable).
- 2282 • Providing as much information as possible in supplemental material (appended
 2283 to the paper) is recommended, but including URLs to data and code is permit-
 2284 ted.

2285 6. Experimental setting/details

2286 Question: Does the paper specify all the training and test details (e.g., data splits,
 2287 hyperparameters, how they were chosen, type of optimizer) necessary to understand
 2288 the results?

2289 Answer: [Yes]

2290 Justification: Open-weight checkpoints are listed in App. G.2 (Llama-3.1-8B/70B-
 2291 Instruct, Qwen2.5-7B/14B/32B, Mistral-7B-Instruct, Gemma-2-9B-it, Phi-3-mini-
 2292 4k, OLMo-7B, DeepSeek-R1-Distill-Qwen-7B/14B, DeepSeek-R1-Distill-Llama-
 2293 8B/70B; closed-model APIs accessed via top- k logprobs). Train/dev/test splits,
 2294 η -tempering selection on held-out folds (cal=200, eval=300, disjoint), and per-
 2295 cell η^* are documented in App. H.1.8. Optimiser, batch size, learning rate, and
 2296 weight decay are in the matched-base defaults table (App. G.3); decoding de-
 2297 faults (temperature=0.0, top_p=1.0, seed 42) and signal-extraction settings are
 2298 in App. G.6.

2299 Guidelines:

- 2300 • The answer [NA] means that the paper does not include experiments.
- 2301 • The experimental setting should be presented in the core of the paper to a level
 2302 of detail that is necessary to appreciate the results and make sense of them.
- 2303 • The full details can be provided either with the code, in appendix, or as sup-
 2304 plemental material.

2305 7. Experiment statistical significance

2306 Question: Does the paper report error bars suitably and correctly defined or other
 2307 appropriate information about the statistical significance of the experiments?

2308 Answer: [Yes]

2309 Justification: Paired bootstrap 95% CIs reported on every head-to-head comparison
 2310 (Section 4.1). Powered $n = 210$ matched design with all CIs strictly above zero
 2311 against AdaCAD/CoCoA/Astute RAG/Self-RAG (Figure H.8, App. H.2); AdaCAD
 2312 CI [0.060, 0.068] is the tightest. Fixed- λ e-values [52, 55]; sign-test exact p -values
 2313 reported per comparator. CT2 envelope with held-out temporal-split MAE and
 2314 cross-cell prediction (App. E.4). The variability captured by the bars is per-cell
 2315 sampling within the spotlight matrix; bootstrap is the standard non-parametric
 2316 resampling without distributional assumptions.

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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Latency and compute profile in App. I.5: per-query latency 4.38s on a single H100 (BAYESARBITER, four model passes), 1.21s for the CAD baseline; cached p_θ deployment drops the cost to 1.09–2.19s/query. SLURM wall-clock per matched-base run is in App. G.3 (12–18h depending on family). Total preliminary and failed runs are accounted for in the compute footprint section of App. K.1.

Guidelines:

- The answer [NA] means that the paper does not include experiments.
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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: App. N.2 discusses positive impact (BayesArbiter reduces confidently-wrong outputs on conflict slices — the load-bearing slice moves from 96% wrong-answer mass to 47%, a direct safety improvement on knowledge-intensive deployments; the matched-base RL contrast surfaces a calibration failure mode of binary-reward RLVR that practitioners can now diagnose) and negative impact (the same mechanism could in principle be inverted to amplify a chosen source’s influence; CT1’ bounds the worst case under TV-perturbation but does not eliminate the risk). Mitigations and dual-use considerations are in App. N.3.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: The release contains evaluation code and a decoder-time arbitration rule, not a new model or new training data. The arbitration rule operates on existing open-weight LLMs and existing public benchmarks; misuse risk is comparable to standard RAG-system code releases.

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2483 the paper include the full text of instructions given to participants and screenshots,
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2489 research with human subjects.

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2497 **with human subjects**

2498 Question: Does the paper describe potential risks incurred by study participants,
2499 whether such risks were disclosed to the subjects, and whether Institutional Review
2500 Board (IRB) approvals (or an equivalent approval/review based on the requirements
2501 of your country or institution) were obtained?

2502 Answer: [NA]

2503 Justification: The paper does not involve human subjects, so IRB review is not
2504 applicable.

2505 Guidelines:

2506 • The answer [NA] means that the paper does not involve crowdsourcing nor
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2516 **16. Declaration of LLM usage**

2517 Question: Does the paper describe the usage of LLMs if it is an important, original,
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2519 LLM is used only for writing, editing, or formatting purposes and does *not* impact
2520 the core methodology, scientific rigor, or originality of the research, declaration is
2521 not required.

2522 Answer: [Yes]

2523 Justification: LLMs are the system under study, not a methodological component:
2524 BayesArbiter is a decoder-time arbitration rule applied to existing LLM outputs
2525 (p_θ , p_{ctx} , and the chain-of-thought trace). The model checkpoints used are listed
2526 in App. G.2. LLM assistance during manuscript preparation was limited to gram-
2527 mar editing, copy-editing, and stylistic polishing of author-written prose; theorems,
2528 proofs, experimental design, code, and analysis were written and verified by the
2529 human authors against primary sources. No LLM-generated content forms part of
2530 the core methods or claims.

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2534

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2535